How to succeed

- Read the Chapter ahead of lectures
- Come to class
- **Start paper early**
- Study groups
- Prepare exams
- Don’t cheat, plagiarize, or otherwise participate in un-ethical behavior
- **Learn with Internet: Wikipedia, google(baidu)…**
- Ask questions and Think skeptically

Last Week

- **Viscoelasticity**

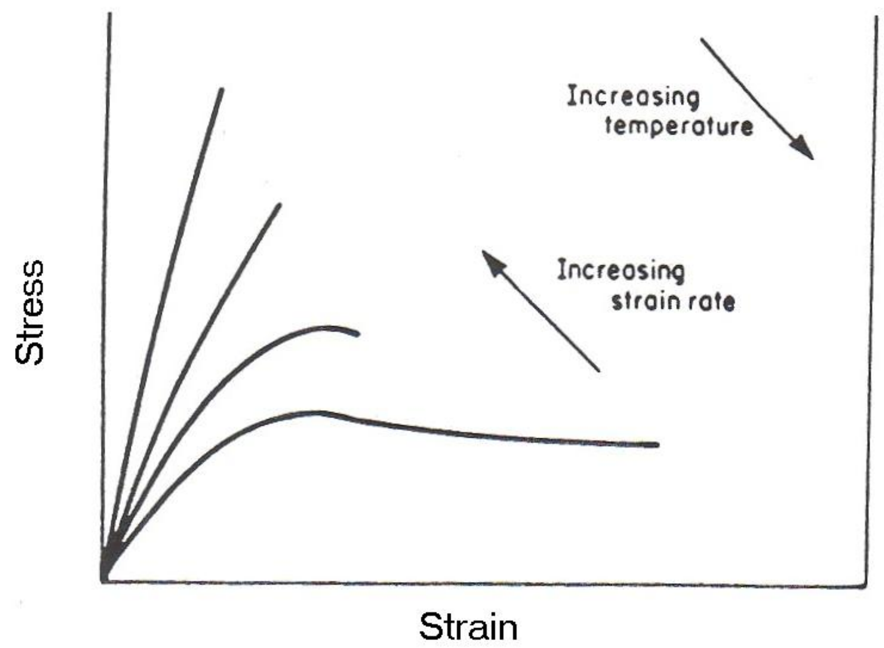
Lecture 8:
Degradation & Fracture of Polymers

SUN Dazhi

Mechanisms leading to degradation of polymers

Physical	Chemical
Sorption	Thermolysis
Swelling	Radical scission
Softening	Depolymerization
Dissolution	Oxidation
Mineralization	Chemical
Extraction	Thermooxidative
Crystallization	Solvolysis
Decrystallization	Hydrolysis
Stress cracking	Alcoholysis
Fatigue fracture	Aminolysis, etc.
Impact fracture	Photolysis
	Visible
	Ultraviolet
	Radiolysis
	Gamma rays
	X-rays
	Electron beam
	Fracture-induced radical reactions

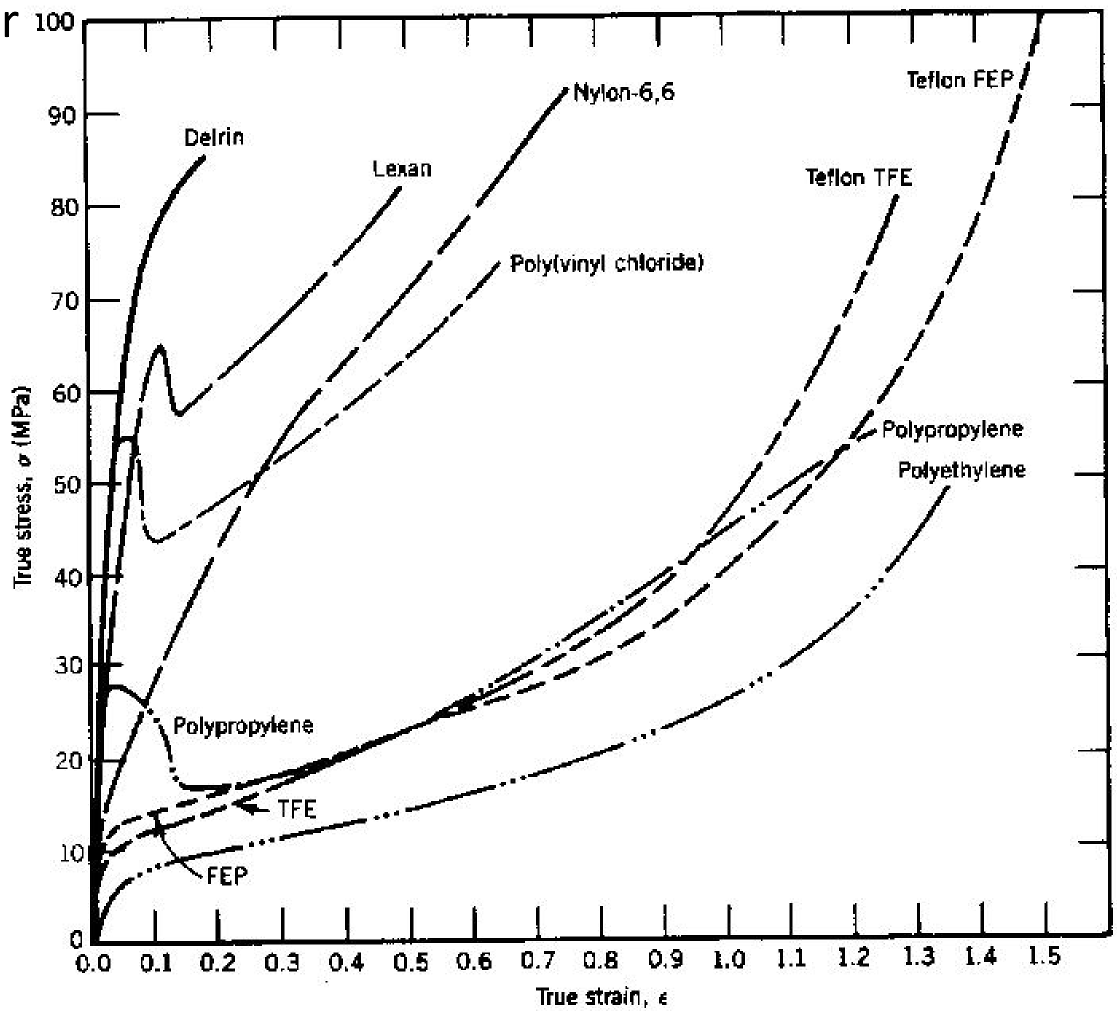
Deformation behavior of polymers is time and temperature dependent



Fast	**Loading Rate**	**Slow**
Low	**Temperature**	**High**
Brittle/Rigid	Deformation Behavior	**Tough/Ductile**

High-Speed Static Tensile Testing

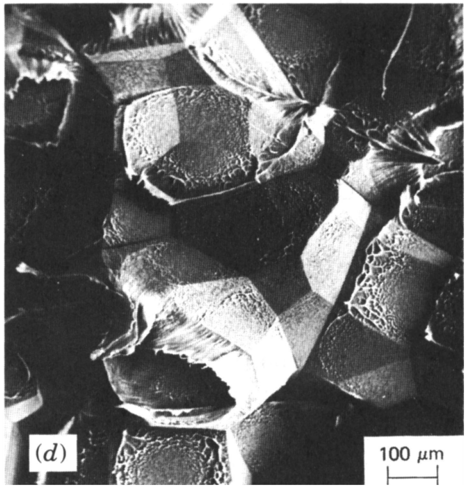
- True stress-vs-true strain behavior of polymers (taking necking into account) measured at high strain rate (typically 100 s^{-1}).
  - Note differences in yielding behavior
- Delrin is an acetal resin
- Lexan a polycarbonate
- Teflon TFE is the tetrafluoroethylene homopolymer
- Teflon FEP is a copolymer of tetrafluoroethylene and hexafluoropropylene



Polymer Failure Modes

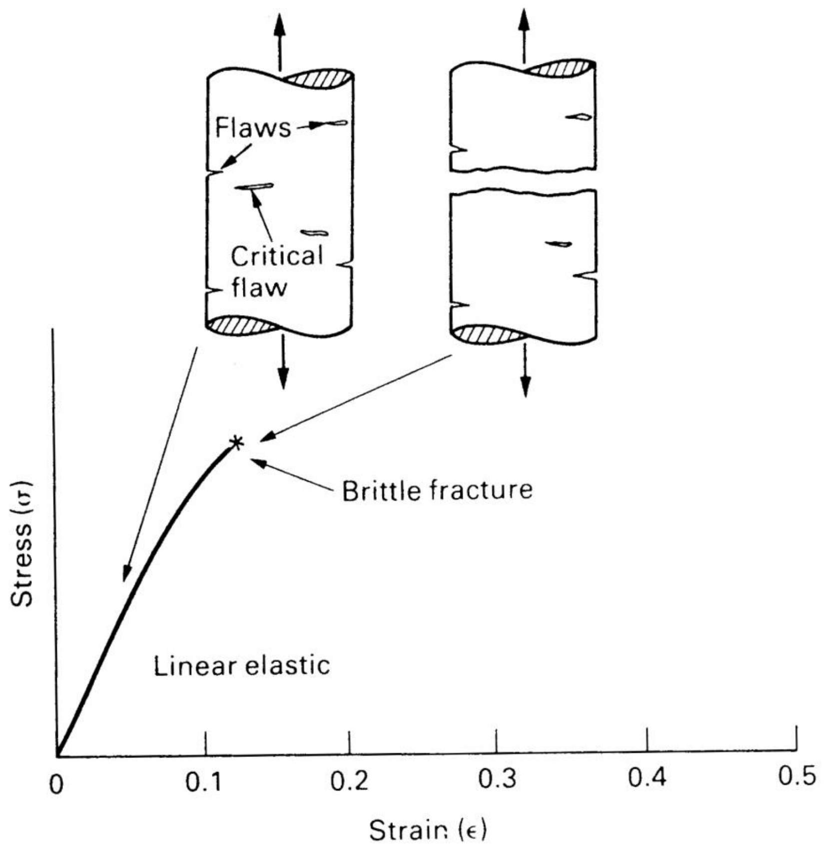
- **Rupture**
  - Ductile-large deformation, one piece
  - Brittle-small deformation, many pieces
- **Fatigue**
- **Creep**- slow increase in deformation under stress

Polypropylene



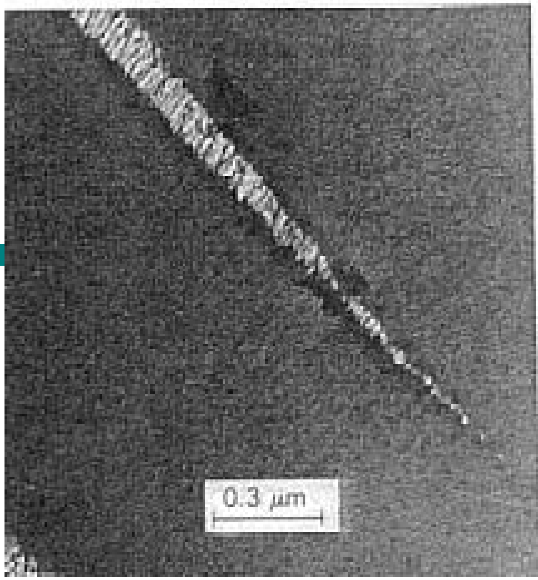
Brittle Fracture

- Very brittle behaviour may occur **below the glass transition temperature** since plastic deformation by chain movement is difficult. The strength is high, but can be reduced significantly by notches and surface damage.



Crazing

- At higher temperatures, below the glass transition temperature, crazing is observed in some thermoplastics. Voids, bridged by polymer chain ligaments develop before brittle fracture.



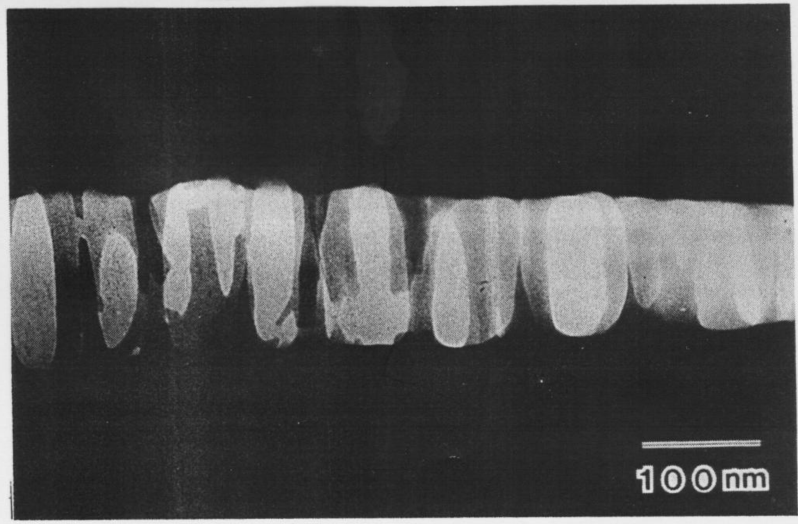
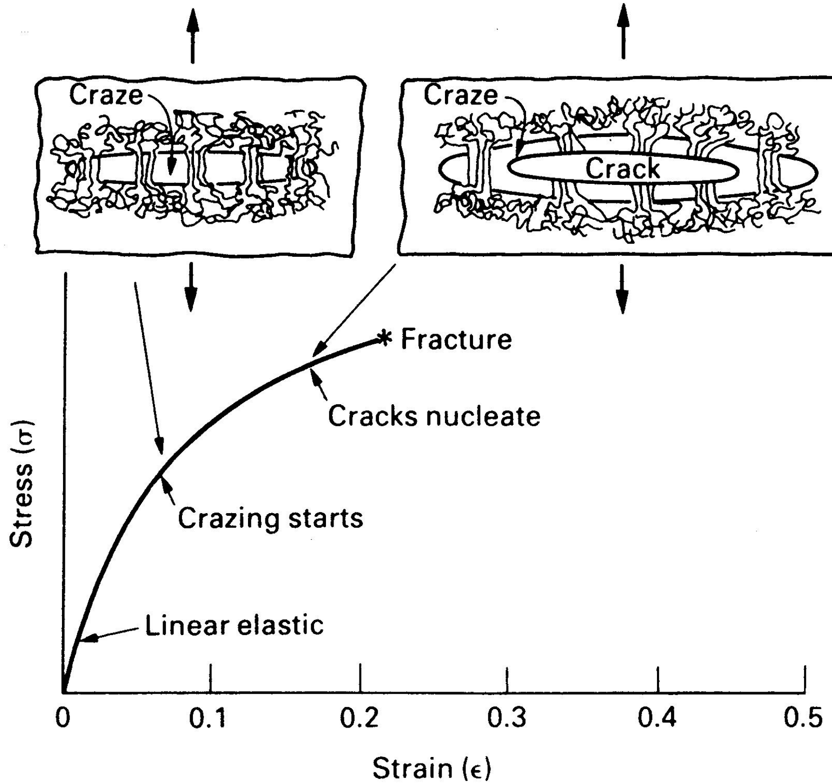
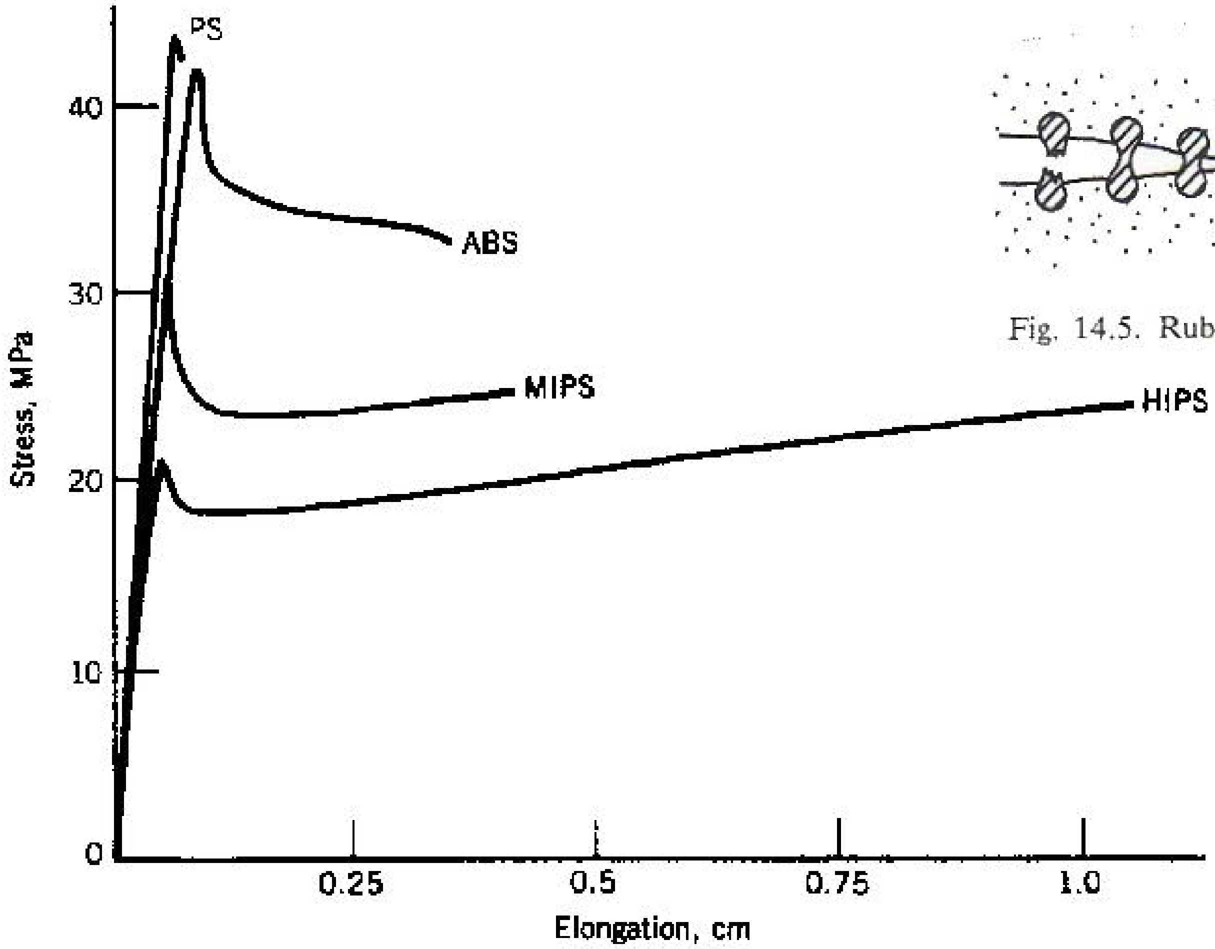


Fig. 1. Electron micrograph of a single craze in polystyrene. Thin fibrils, which are ~20 nm in diameter, are stretched across the craze boundaries. The applied strain is ~1%. Some of the fibrils have fractured; the remaining ones will break on further extension. This will lead to catastrophic crack development.



Rubber Toughening - Crazing

- Typical uniaxial tensile stress-strain behavior of polystyrene (PS), medium-impact PS (MIPS), high-impact PS (HIPS), and poly(acrylonitrile-co-styrene-graft-butadiene) ABS.



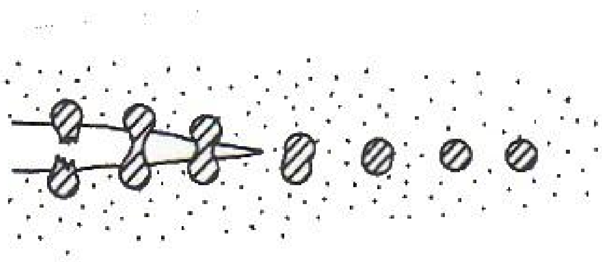
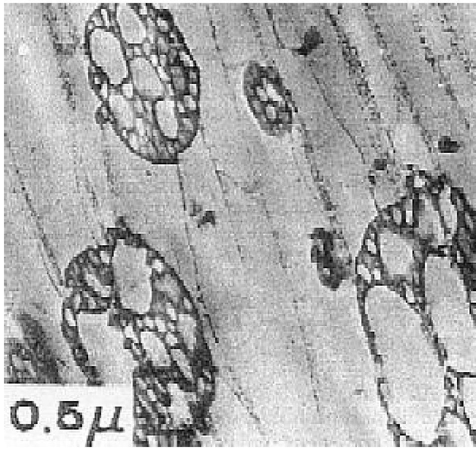


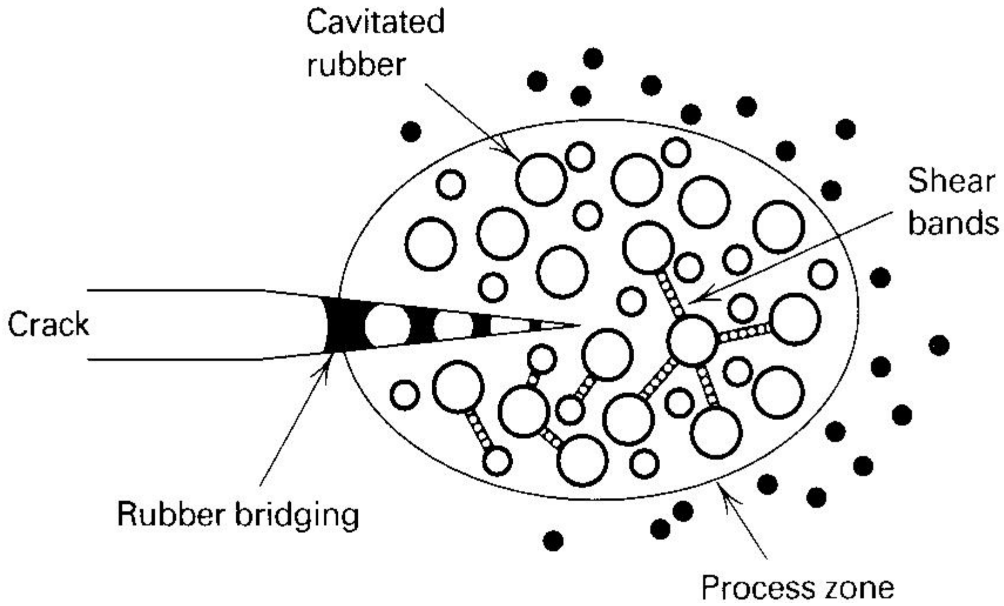
Fig. 14.5. Rubber-toughened polymers.



Toughening of Polymers

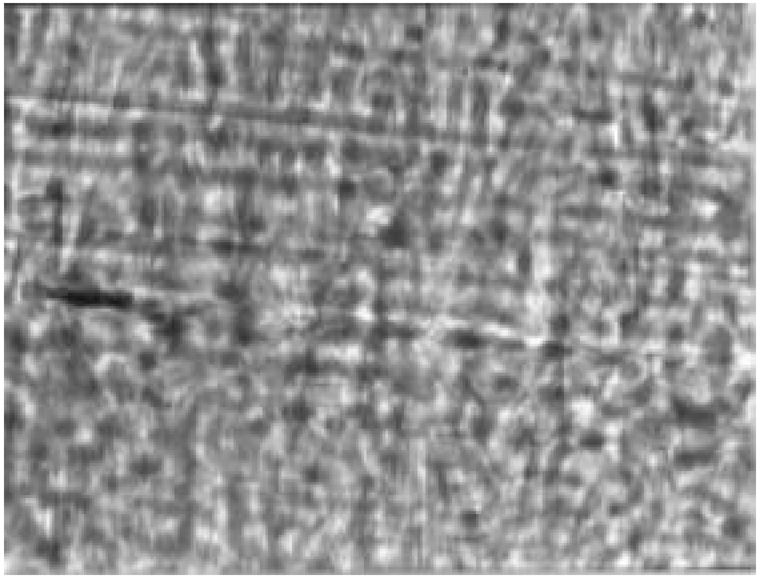
Several toughening mechanisms may develop

- **Rubber Bridging**
- **Crazing**
- **Shear Bands**



Fatigue: Cyclic Stress on Materials

Polypropylene lid of a Tic Tacs box, with a living hinge

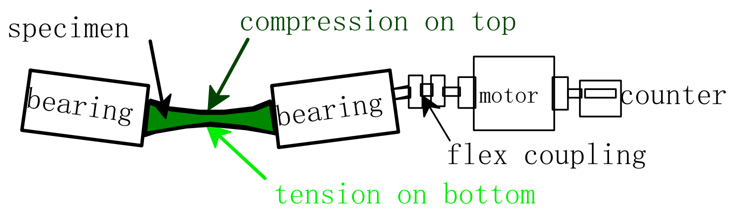


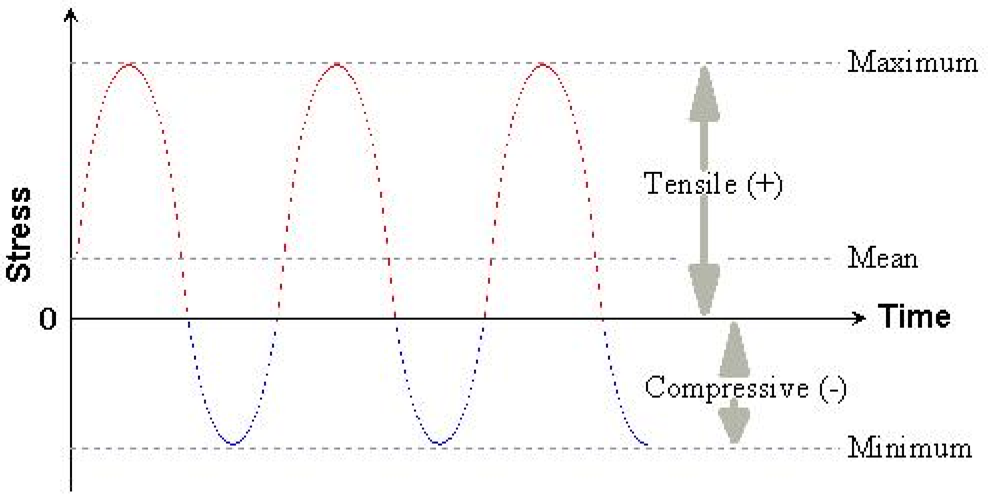
Micrograph of fatigue stressed hinge

Polypropylene & polyethylene are excellent materials for hinges

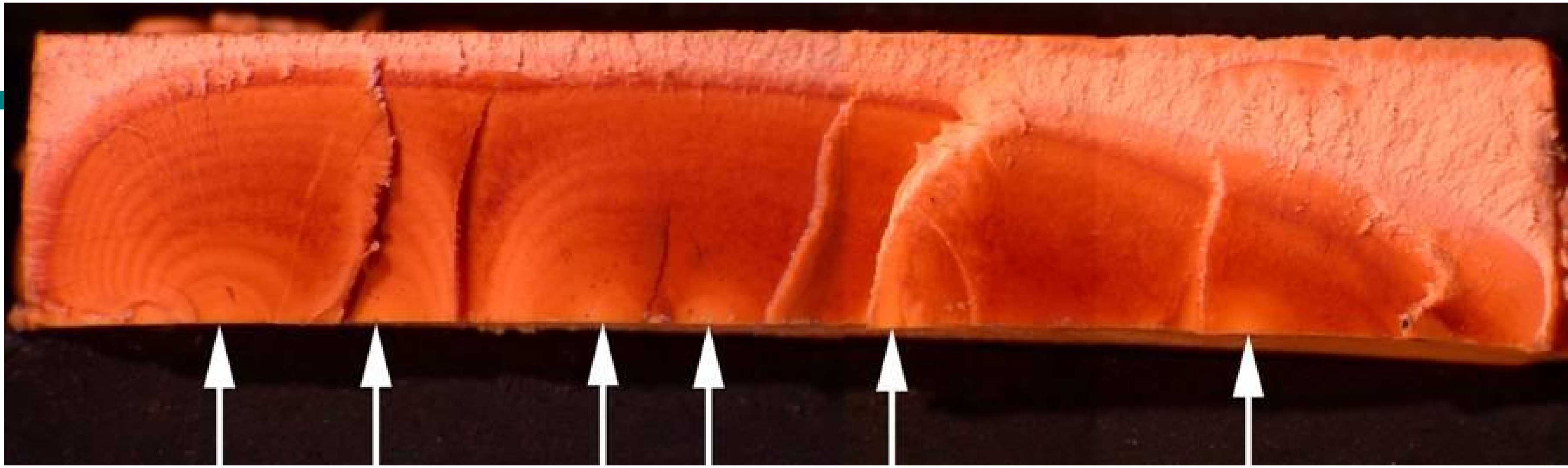
Fatigue failure

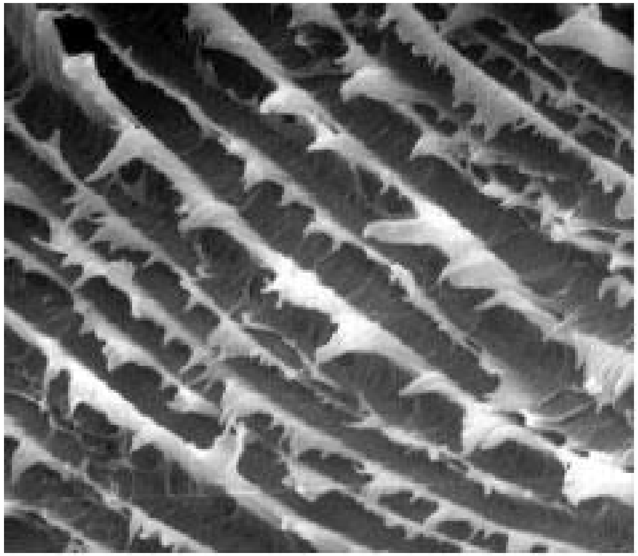
- Fatigue is a failure process →which a crack grows as a result of cyclic loading.
- This type of loading involves → stresses that alternate between high and low values over time.
- The stress values may be entirely positive (tensile), entirely negative (compressive), or a combination of the two (see diagram).

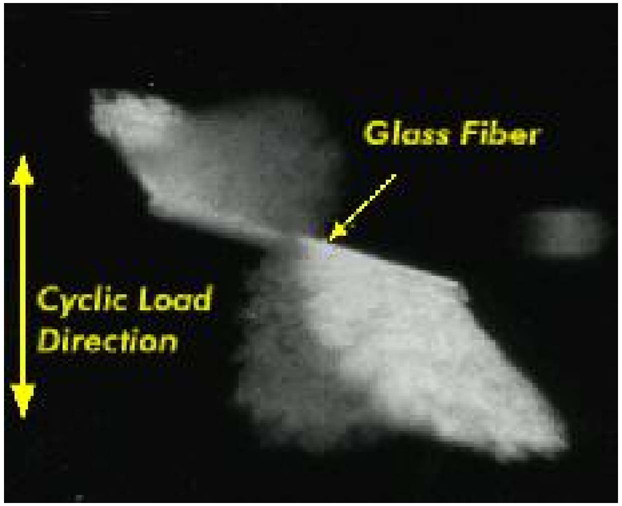




Cyclic stress that gives rise to fatigue in materials

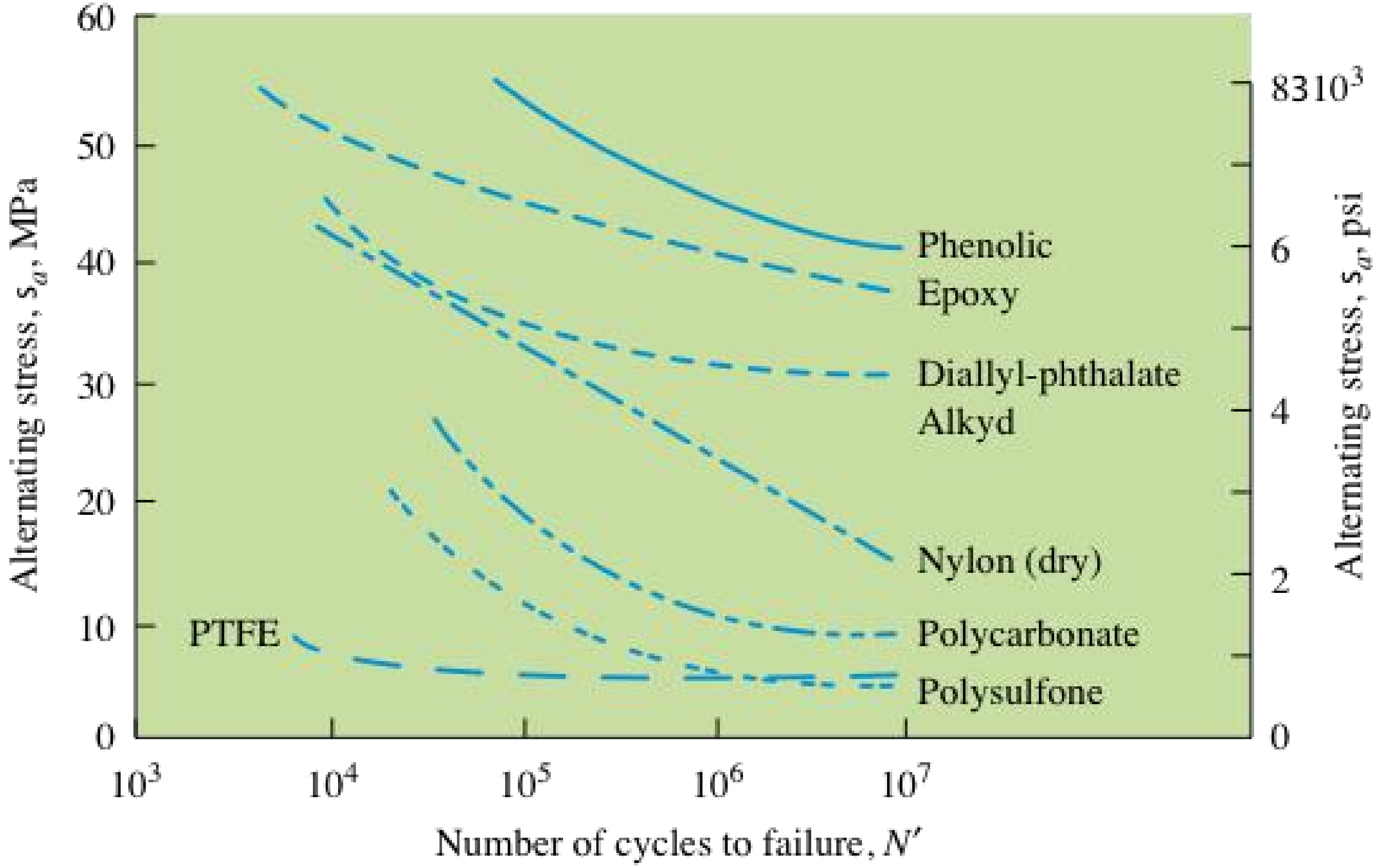




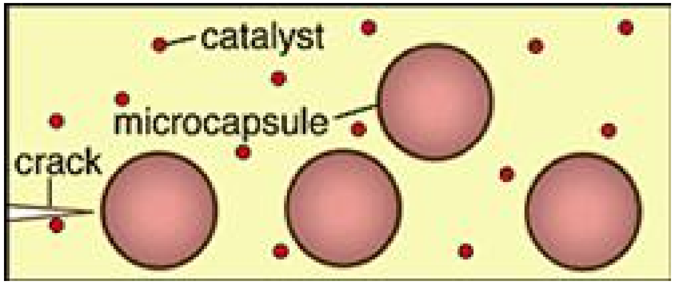

Glass Fiber
Cyclic Load
Direction

Fatigue Testing: S/N plots

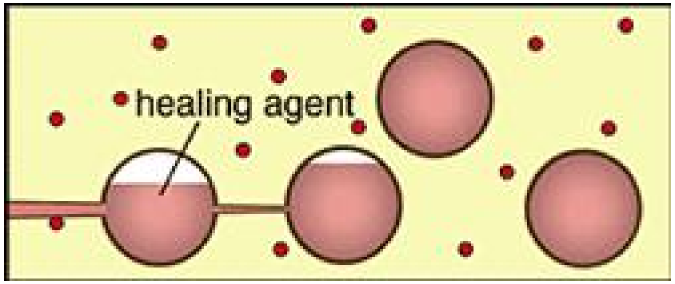
50-250 Hz cyclic stress using stress strain analyzer



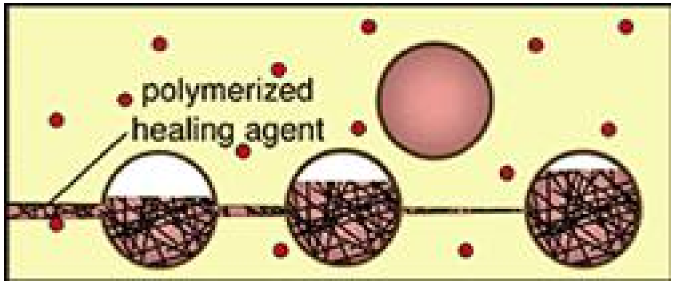
Self-Healing Polymers and Composites



To achieve "autonomic healing," this technology combines a micro-encapsulated healing agent and a chemical catalyst within an epoxy matrix.



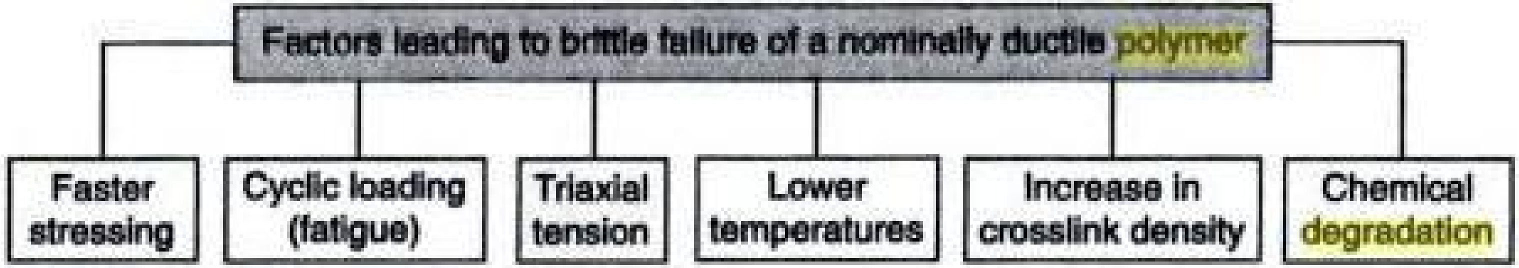
As cracks develop, they rupture the embedded microcapsules, releasing the healing agent into the crack plane through capillary action.



Polymerization of the healing agent is triggered by contact with the embedded catalyst, bonding the crack faces. The damage-induced triggering mechanism provides site-specific autonomic control of repair.

Table 12.1 Mechanical Achilles' heels of various polymers

Polymer	Achilles' heel(s)
Polyacetal	Low strain to break and notch-sensitivity
ABS	Stress embrittlement, poor creep resistance
Epoxies	Low impact strength
Polyethylene	Creep
PS, HIPS	Fatigue
PC	Poor notched-impact strength
PVC	Notch-sensitivity
Polysulfones	Poor notched-impact strength
'Kevlar'	Low compressive strength
PTFE	Low strain to break, creep
SBS	Dynamic fatigue
PPS	Brittleness/low impact strength
PEI	Loss of properties at elevated temperatures
PPO	Physical ageing

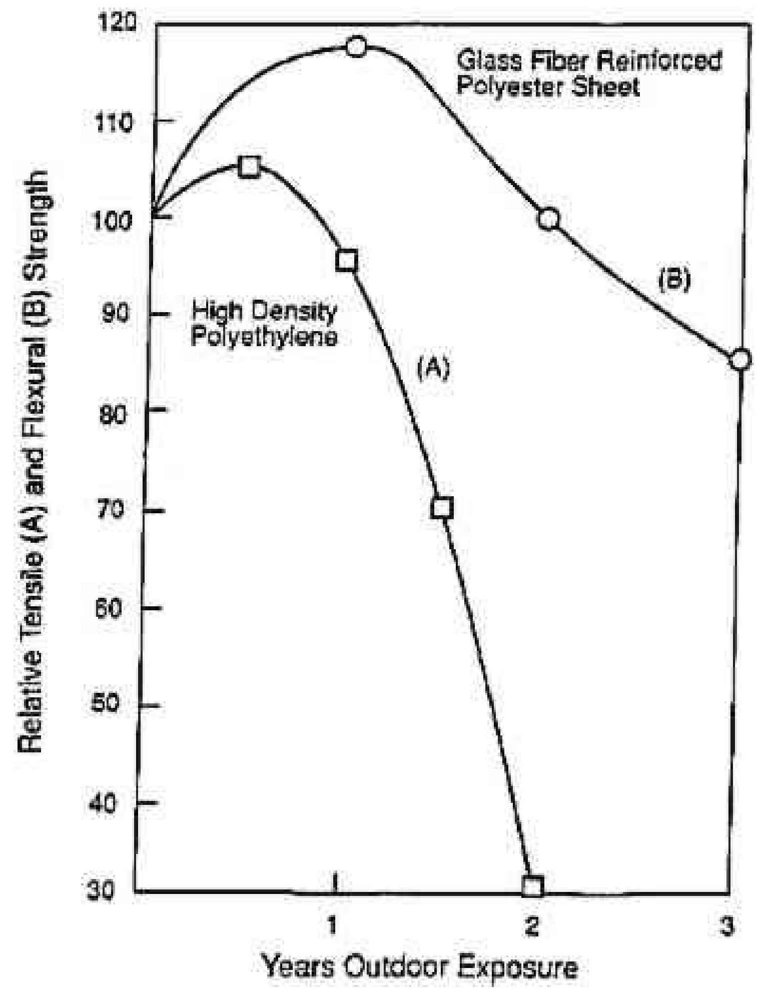


Stimuli – Response

	Environmental Stimuli	**Result**
Polymers	Time $h\nu$ Heat Oxygen Ozone Water Biologicals Mechanical	Loss of mech. Properties *embrittlement* Discoloration Permanent change in volume Cracking Deformation

Thermal Degradation & Ageing of Polymers

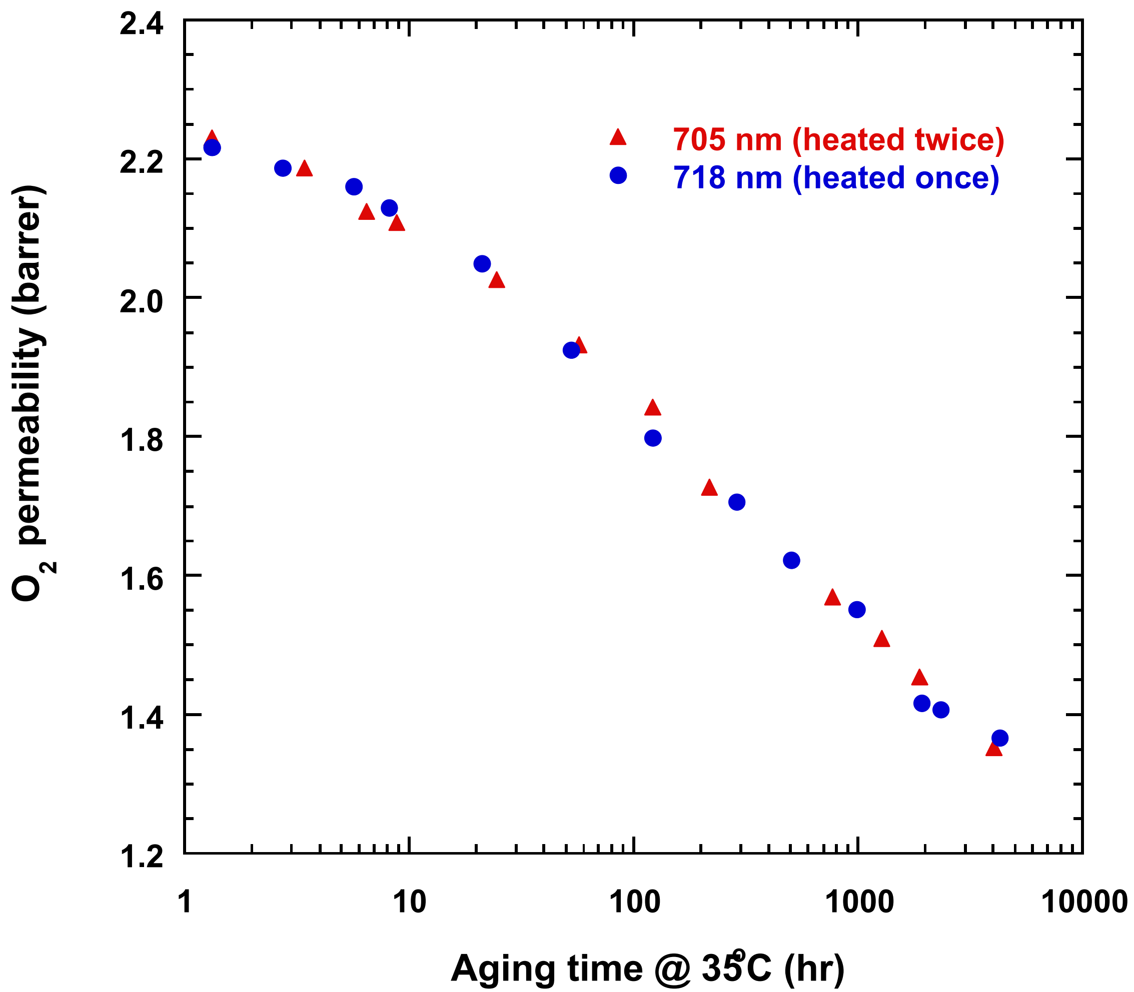
- Time dependent processes
- Reactive degradation
- Solvent damage
- Loss of additives



Time dependent processes

- Loss of free volume
- Creep-cohesive failure of adhesives

Aging: Collapse of Void volumes



Reactive Degradation

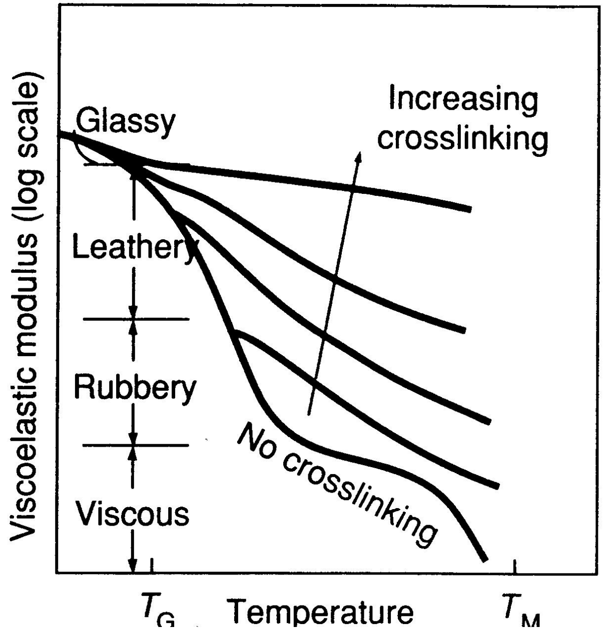
- **Reactions with air-*Oxidation***
- **Electrochemical oxidation**
- **Reactions with water-*Hydrolysis***
- **Radiation damage-*Photochemistry & Radiolysis***
- **Heat damage-*Thermal degradation***
- **Flora & Fauna-*biological degradation***

Effects of Degradation

- Reduced Ductility & embrittlement
- Chalking
- Color changes
- Cracking

Thermal, Radiation, Photochemistry & Oxidation

High Levels of Radicals: Crosslinking and formation of insoluble gels, brittle materials.



1,1-disubstitution

favors depolymerization

Low levels of Radicals: Chain Scission & Reduction in Molecular Weight

Polymer Oxidation

Homolysis

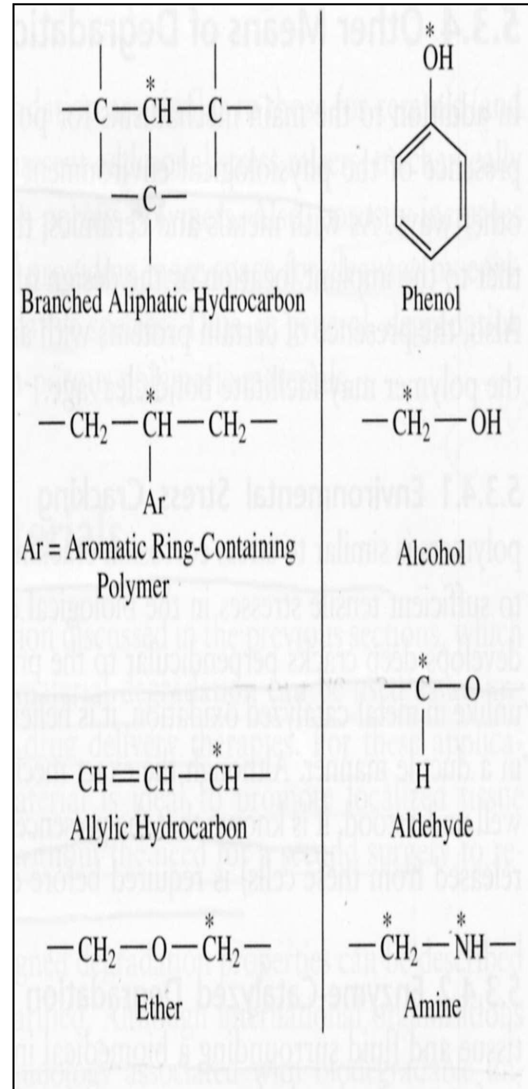
$$R{-}R \longrightarrow 2R\cdot$$

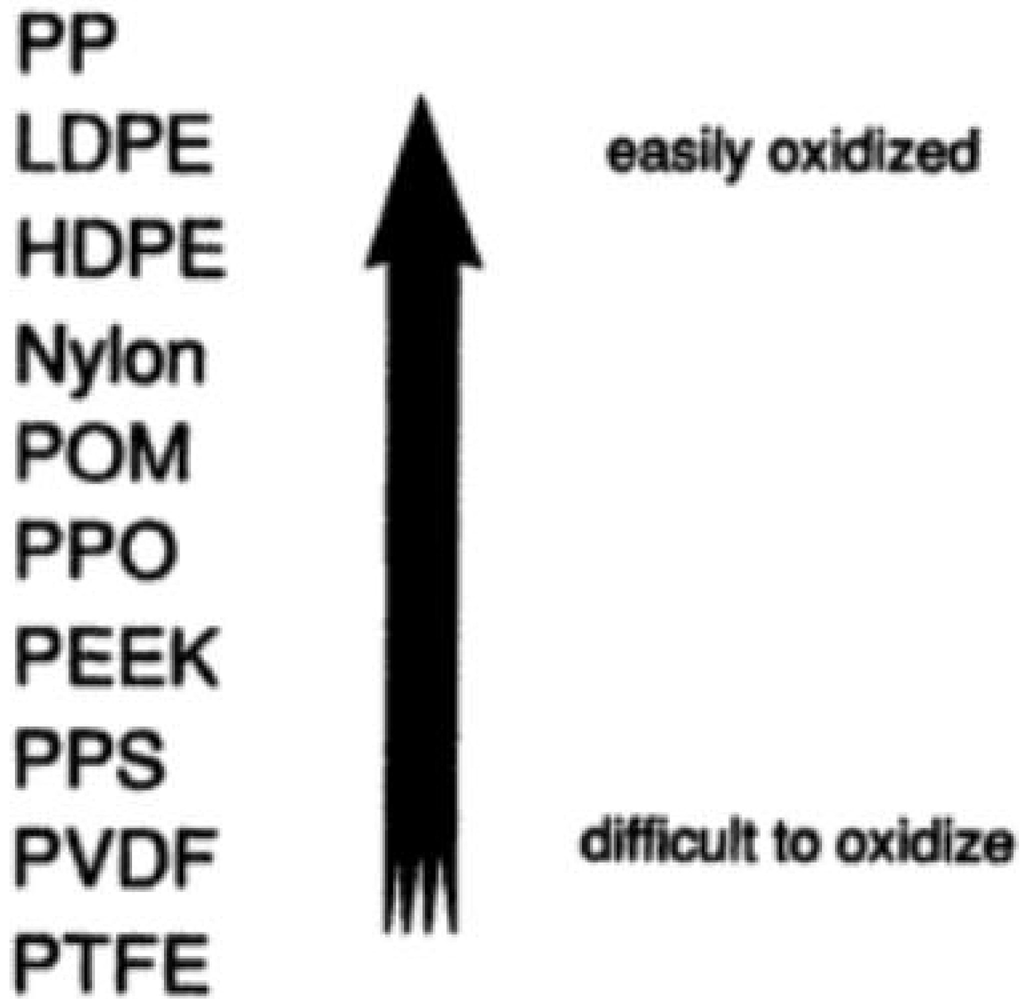
Heterolysis

$$R{-}R \longrightarrow R^{+} + R^{-}$$

Initiation steps in oxidation may be homolysis (top) or heterolysis (bottom)

Examples of chemical domains that are susceptible to oxidative degradation. * denotes sites of homolysis or heterolysis



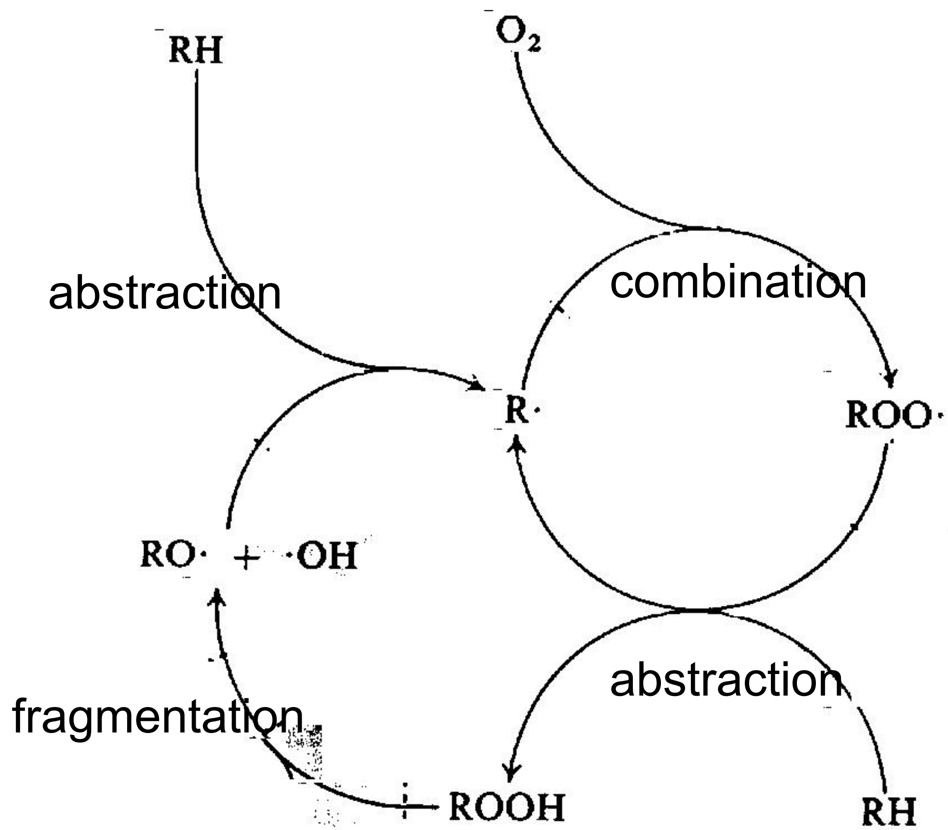

PP
LDPE
HDPE
Nylon
POM
PPO
PEEK
PPS
PVDF
PTFE
easily oxidized
difficult to oxidize

Oxidative Degradation

- Polymer degradation is almost always faster in the presence of oxygen (air), due primarily to the autoaccelerating nature of reactions between oxygen and carbon centred radicals.

- Interactions with oxygen lead to an increase in the concentration of polymer alkyl radicals (R•), and therefore to higher levels of scission and crosslinking products.

- Additionally, fragmentation reactions of oxygen-centred radicals (RO•) yield new species (oxidation products), not found in polymers processed under air-free conditions.



Oxidative Degradation:Susceptibility of Polyolefins

- Influence of polyolefin chain branching on oxidation rate (139° C).

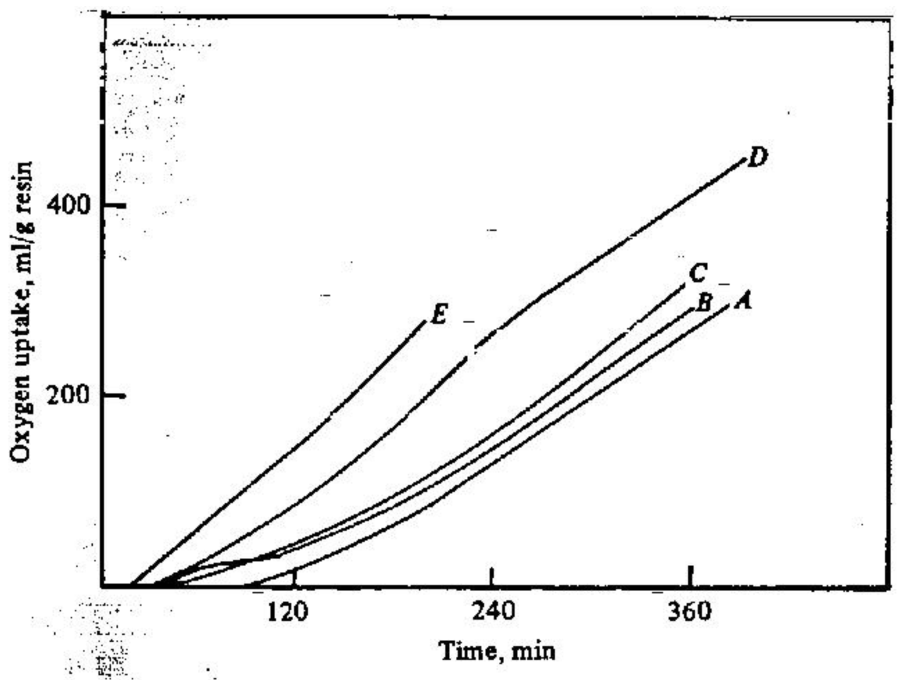
  A Linear polyethylene (1 methyl group per 1000 carbon atoms);

  B Ethylene propylene copolymer, EPM (10.7 Me / 1000 C);

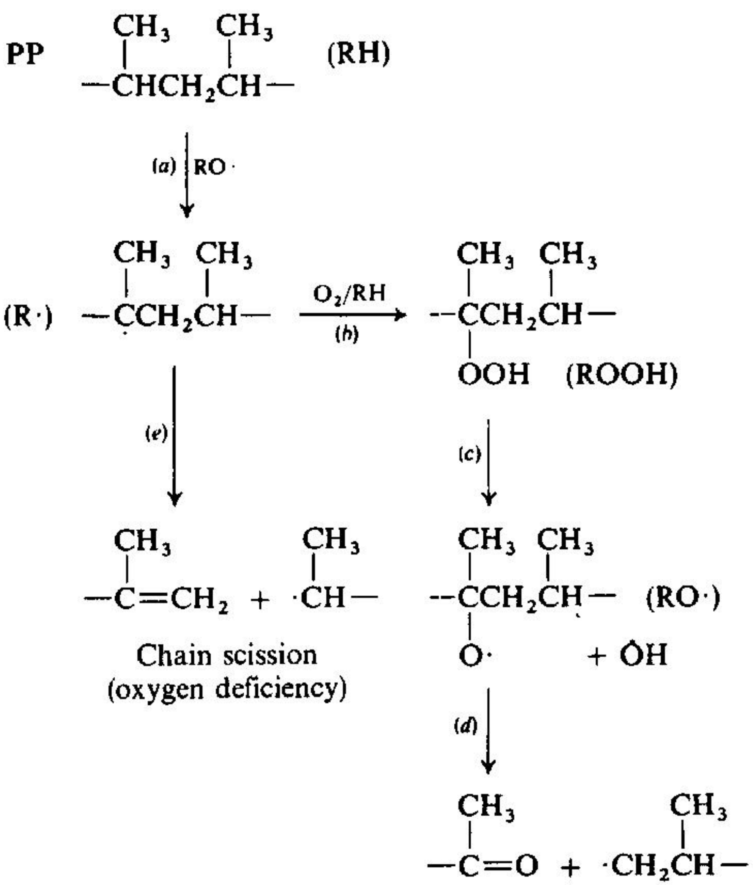
  C EPM (21.0 Me / 1000 C);

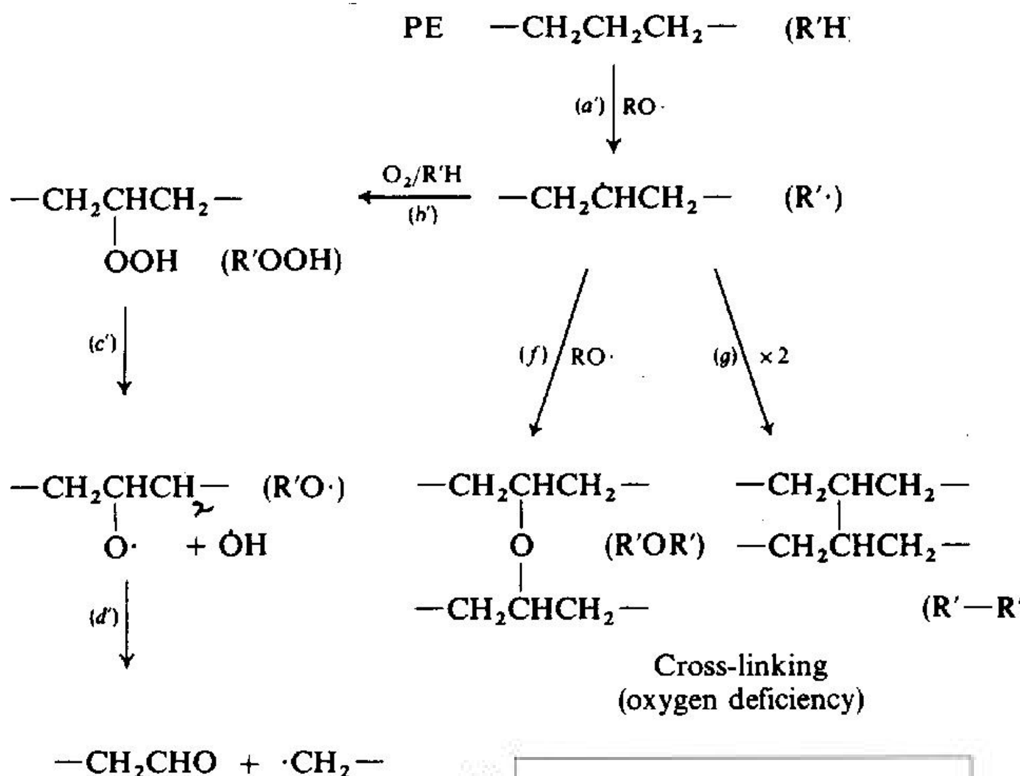
  D EPM (35.5 Me / 1000 C);

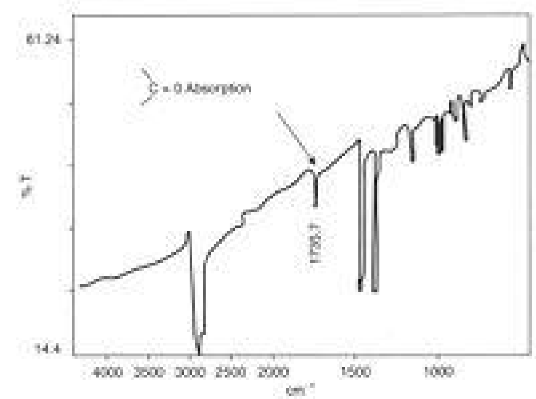
  E Polypropylene (333 Me / 1000 C).



Oxidative Degradation: Polyolefins

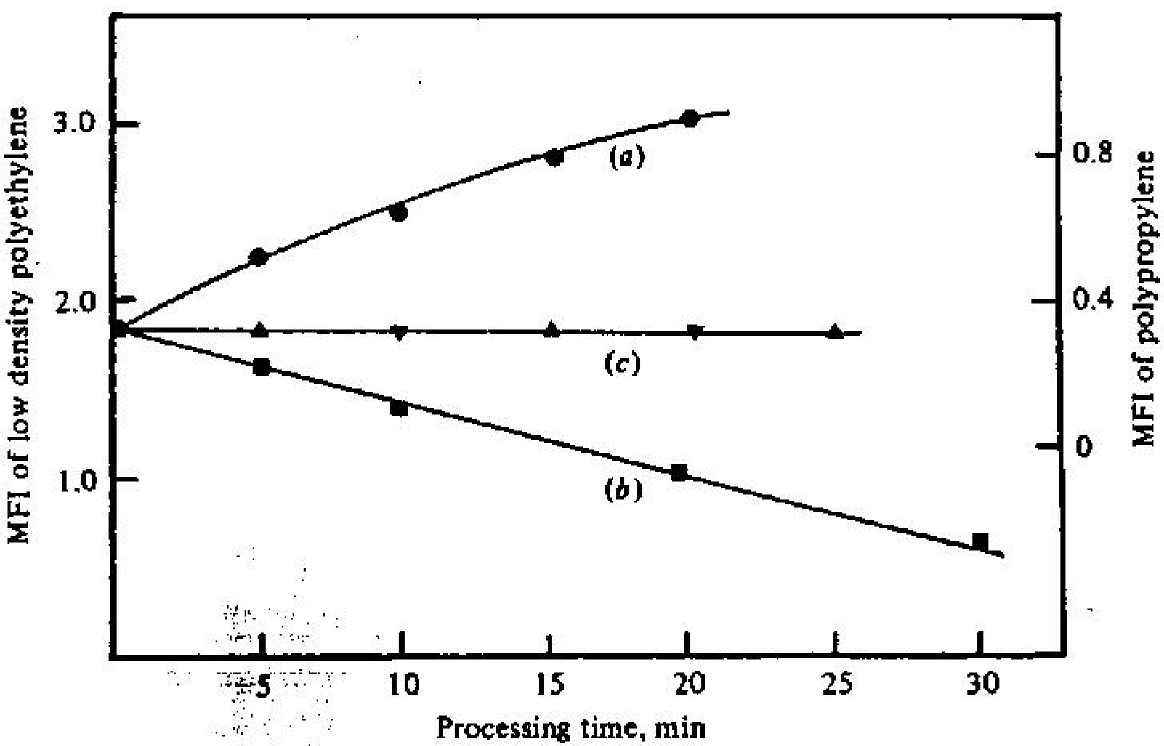






Oxidative Degradation: Polymer Processing

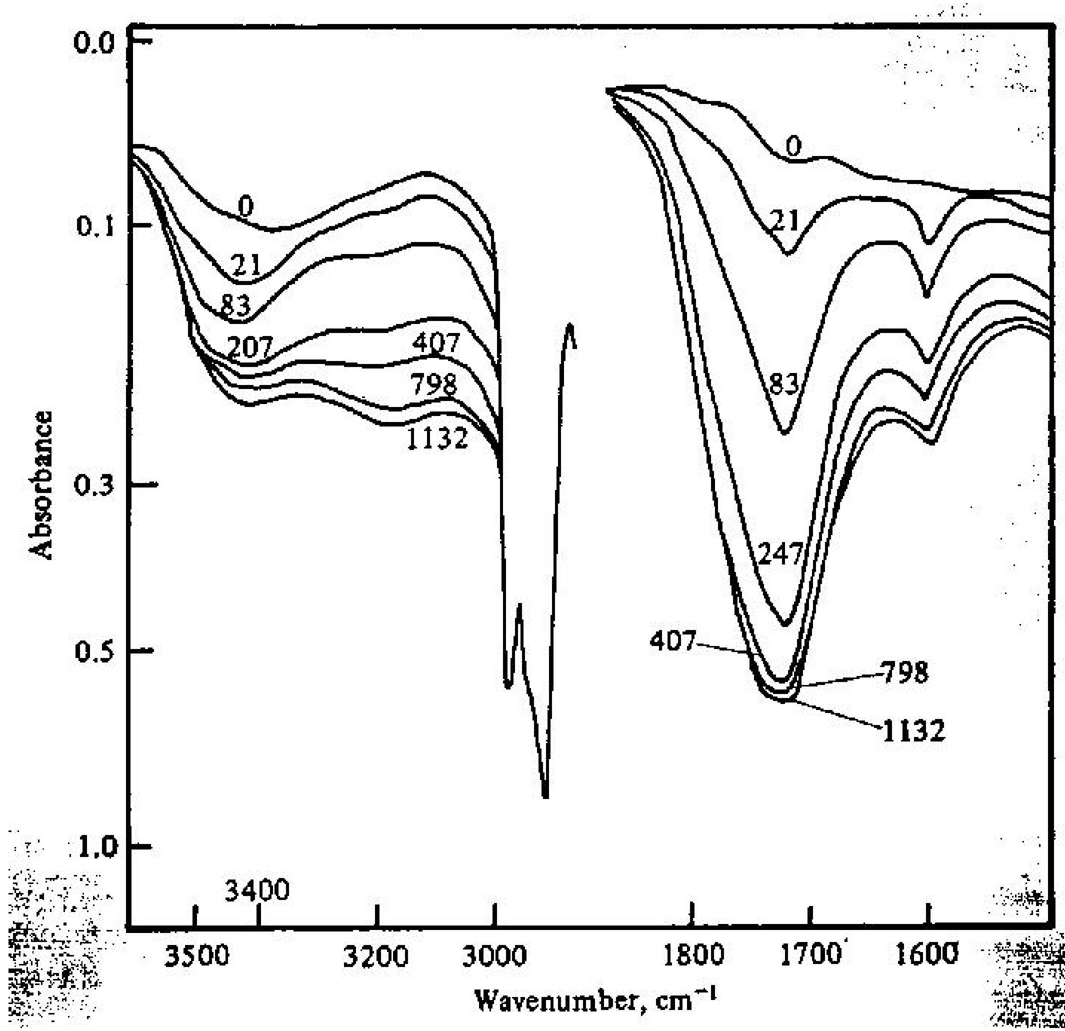
- Effect of processing of polyolefins in a shearing mixer.
  - (a) Polypropylene mixed at 180° C; ●
  - (b) Low density polyethylene mixed at 150° C; ■
  - (c) Polypropylene (♦) and low density polyethylene (♦) in a mixer purged with argon



Oxidative Degradation: Photoxidation

- Infra-red spectrum of polypropylene during photo-oxidation in the hydroxyl (3420 cm^{-1})
- and carbonyl (1720 cm^{-1}) regions.

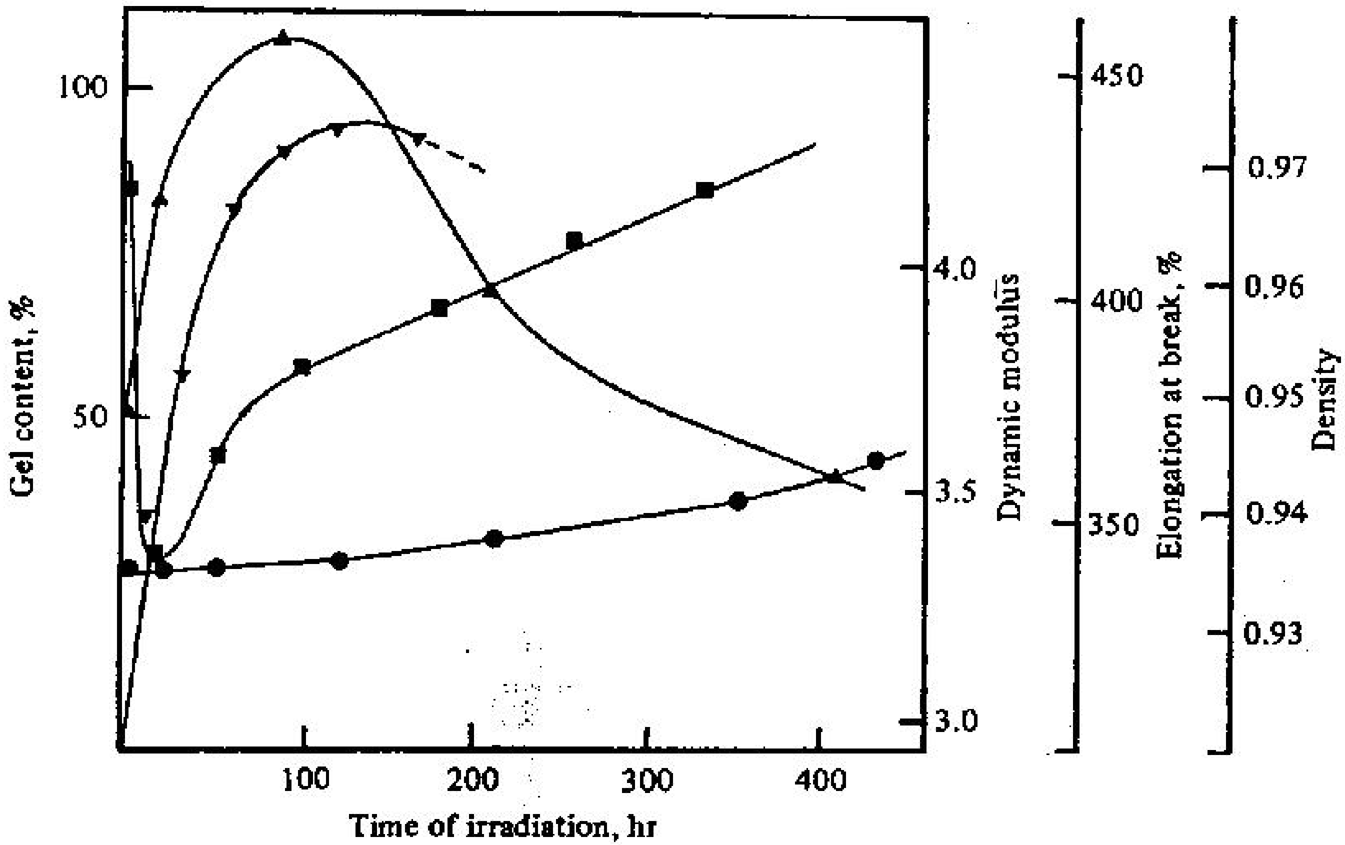
- Numbers on the curves represent UV irradiation times (hours).



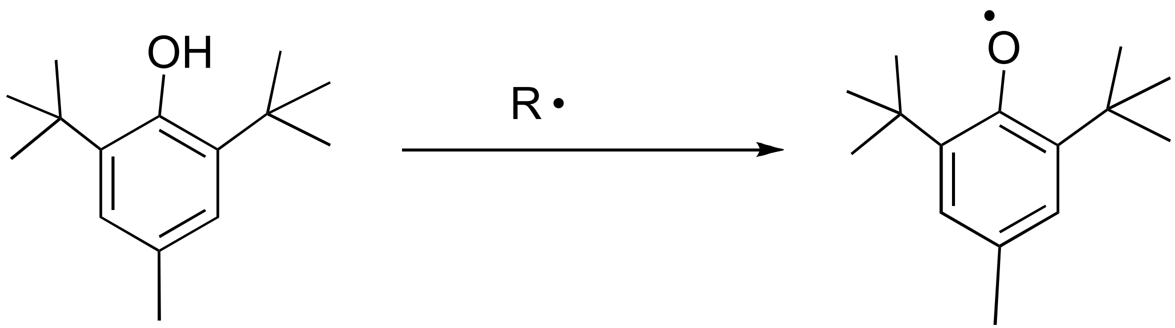
Oxidative Degradation: Photo-oxidation of LDPE

Effect of UV irradiation on the mechanical and physical properties of low-density polyethylene during the early stages of exposure.

- ■ Dynamic modulus
- ♦ Gel content
- ♦ Elongation at break
- ● Density



Antioxidants: Helping Healthy Bodies Grow

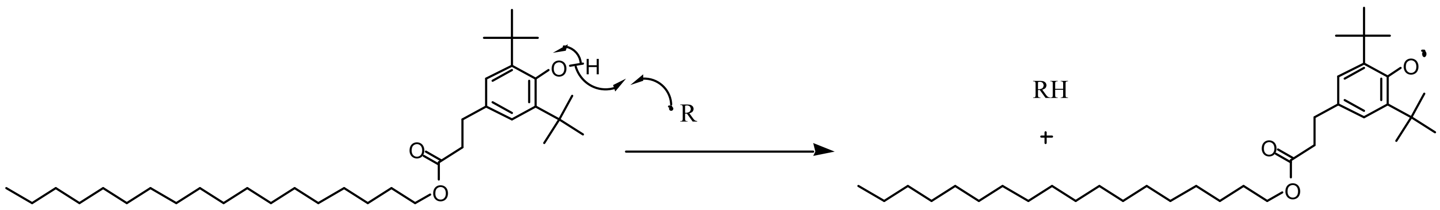


Molecules that retard oxidation by intercepting radicals and converting them into more stable, un-reactive radicals

Delaying the inevitable

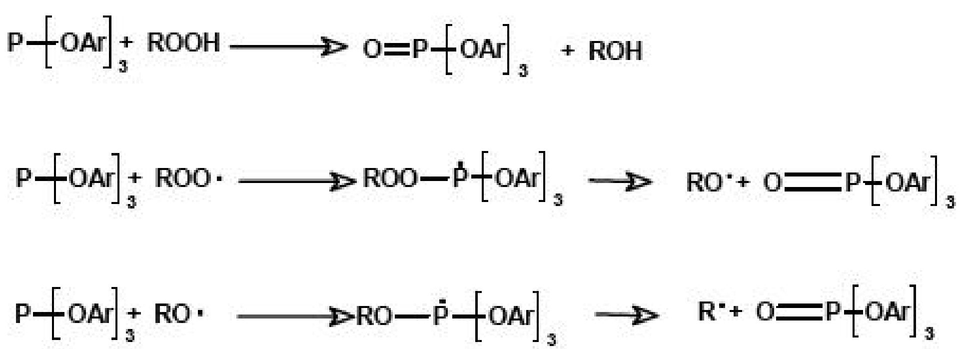
Functions of Antioxidants

- Primary antioxidants react with free radical species



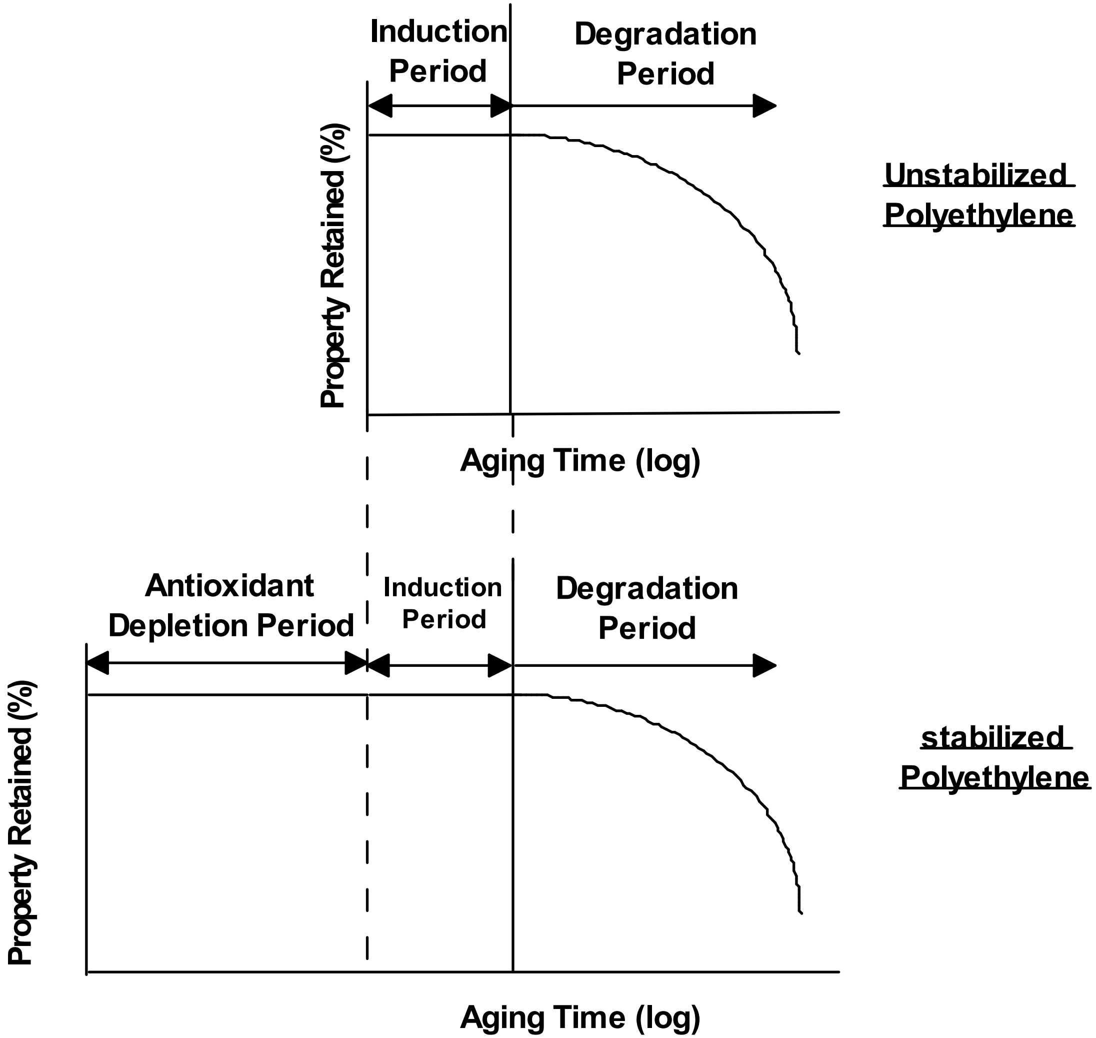
Irganox 1076

•Secondary antioxidants decompose *ROOH* to prevent formation of free radicals



Scheme 4: Hydroperoxide decomposition involving trivalent organo phosphorous compounds

Using antioxidants:Hollywood approach to ageing



Antioxidants: Mechanisms of Action

Antioxidants function by interfering with radical reactions that lead to polymer oxidation and, in turn, to degradation.

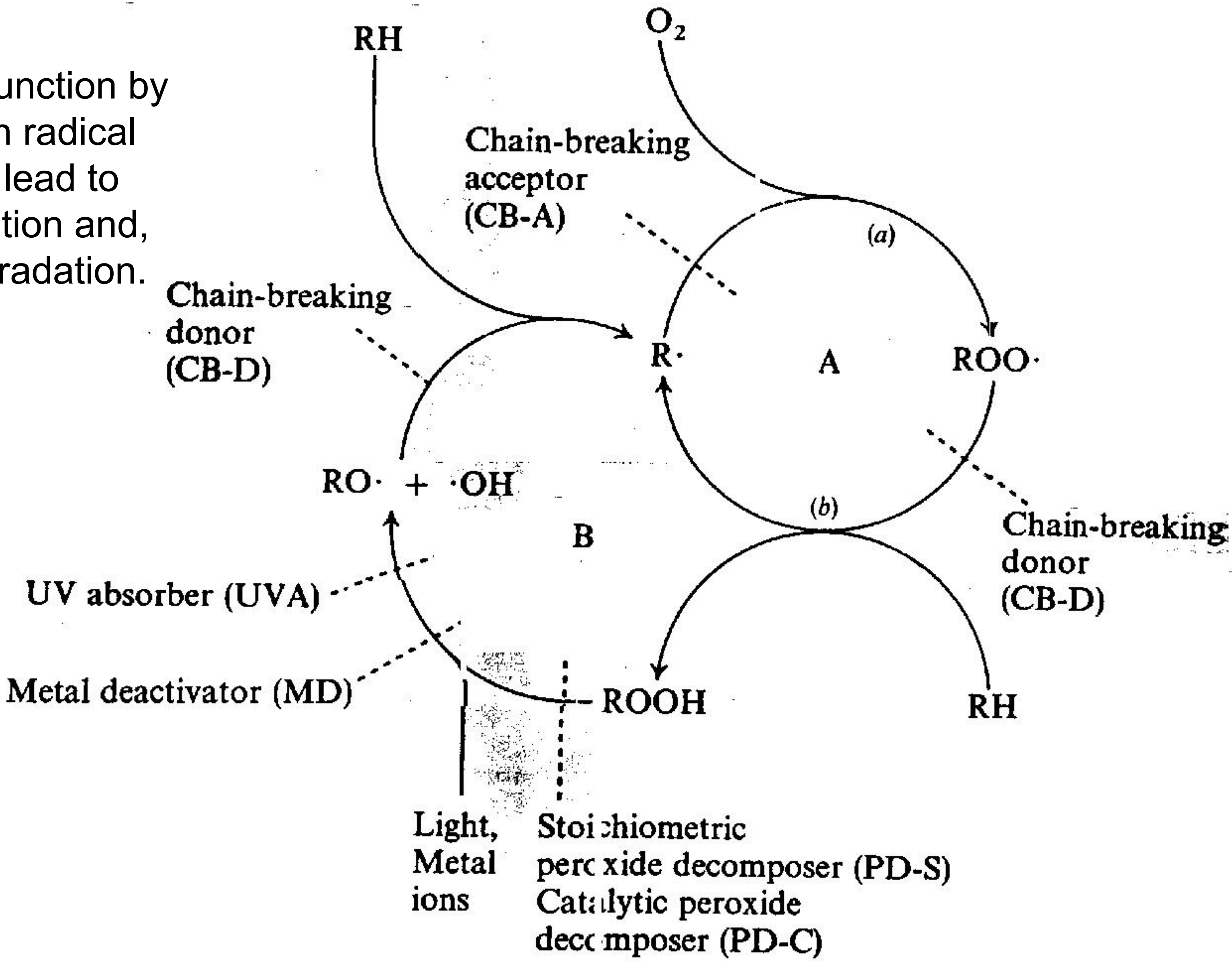


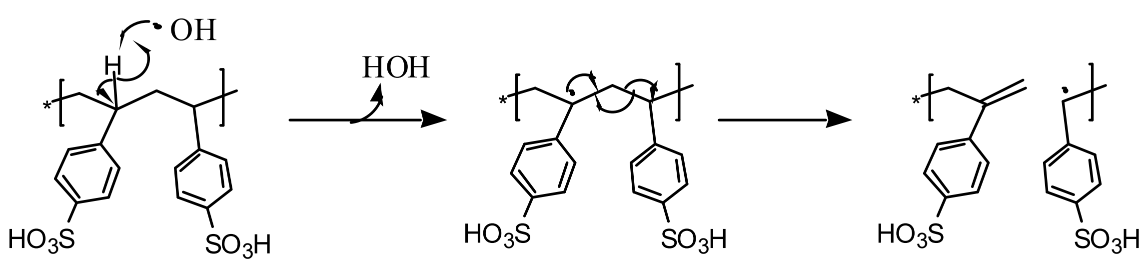
Table 1 UV Characteristics and Lifetimes of Common Polymers

Polymers	Absorption maximum (nm)[a]	Most efficient wavelength (nm)[b]	Cutoff (nm)[c]	Outdoor lifetime (yr)[d]
polyethylene	<150	300 310-340[e]	<180	0.5-1.0
polypropylene	<200	310, 370[f] 320-380[e]	<180	0.2
polyoxymethylene			<210	0.3
poly(vinyl chloride)	<210	310, 320[f] 310-370[e]	~220	0.5
poly(methyl methacrylate)	<240	290-315	~240	>20
polyamides			~240	3-4
polyamide			~240	~8
polyvinyl acetate	<250	<280		
polystyrene	<260	318	~270	~0.1
polycarbonate	260	295, 310[e] 345[f]	~280	0.5
poly(phenylene oxide)			~280	<0.5
polyurethane			~280	~2
poly(ethylene terephthalate)	~290	290-320	~310	~3
poly(m-phenylene isophthalamide)			~340	~0.2
poly(p-phenylene terepthalamide)			~350	~0.3
polyarylate		350-385[e]		
ABS		340-380[e]		
polysulfone		310-312[e]		
polyester		325[f]		

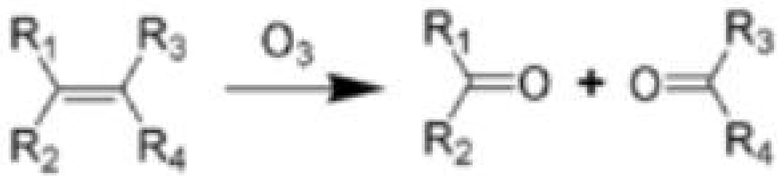
Cathodic Degradation in Fuel Cells

$$Pt + O_2 \longrightarrow Pt{-}O{-}O^{\bullet}$$

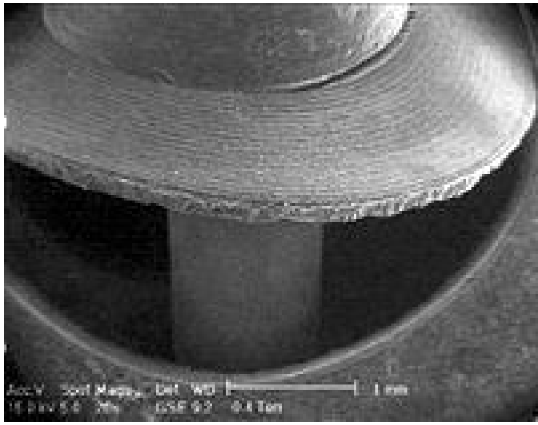
$$Pt{-}O{-}O^{\bullet} + HOH \longrightarrow Pt{-}O{-}OH + {}^{\bullet}OH$$

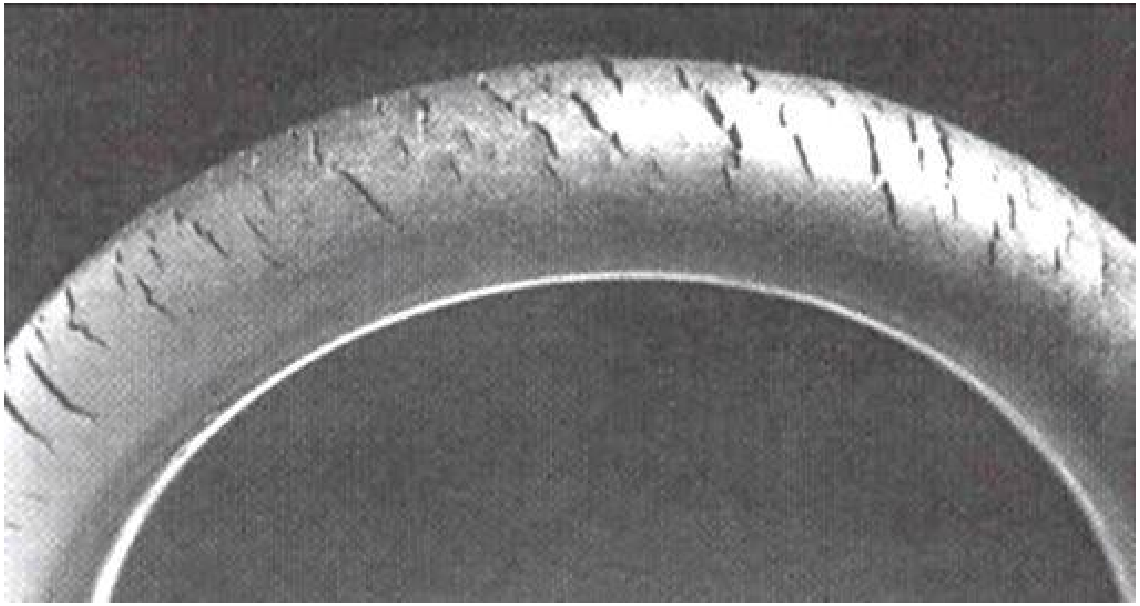


Ozone cleavage of polymers



Cuts through carbon-carbon double bonds
e.g. Polydiene elastomers





Chlorine (or bromine) attack

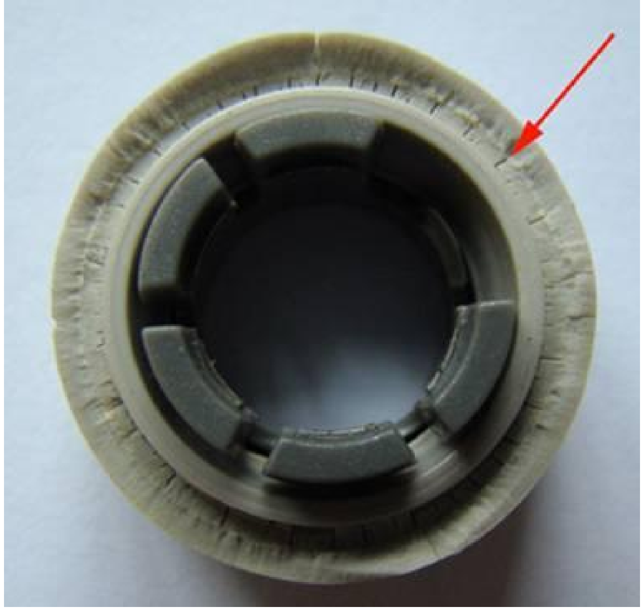
Pool and spas, water chlorination

acetal resin and polybutylene pipes

Stress corrosion cracking

Hypochlorite attack on Polyacetal

Chlorinated water



Delrin

Effect of metal ion oxidation potential on properties of poly(ether urethane)

Aqueous solution	Standard oxidation potential	Change in tensile strength (%)	Change in elongation (%)
$PtCl_2$	Ca + 1.2	− 87	− 77
$AgNO_3$	+ 0.799	− 54	− 42
$FeCl_3$	+ 0.771	− 79	− 10
Cu_2Cl_2	+ 0.521	− 6	+ 11
$Cu_2(OAc)_2$	+ 0.153	− 11	+ 22
$Ni(OAc)_2$	− 0.250	− 5	+ 13
$Co(OAc)_2$	− 0.277	+ 1	+ 13

Polyetherurethane	Polyether content	Change in tensile strength (%)	Change in elongativon (%)
Pellethane 2363-80A	High	− 54	− 42
Pellethane 2363-55D	Low	− 23	− 10
Model segmented polyurethane	None	+ 9	+ 3

Effect of ether content on poly(ether urethane) on susceptibility to metal ion-induced oxidation

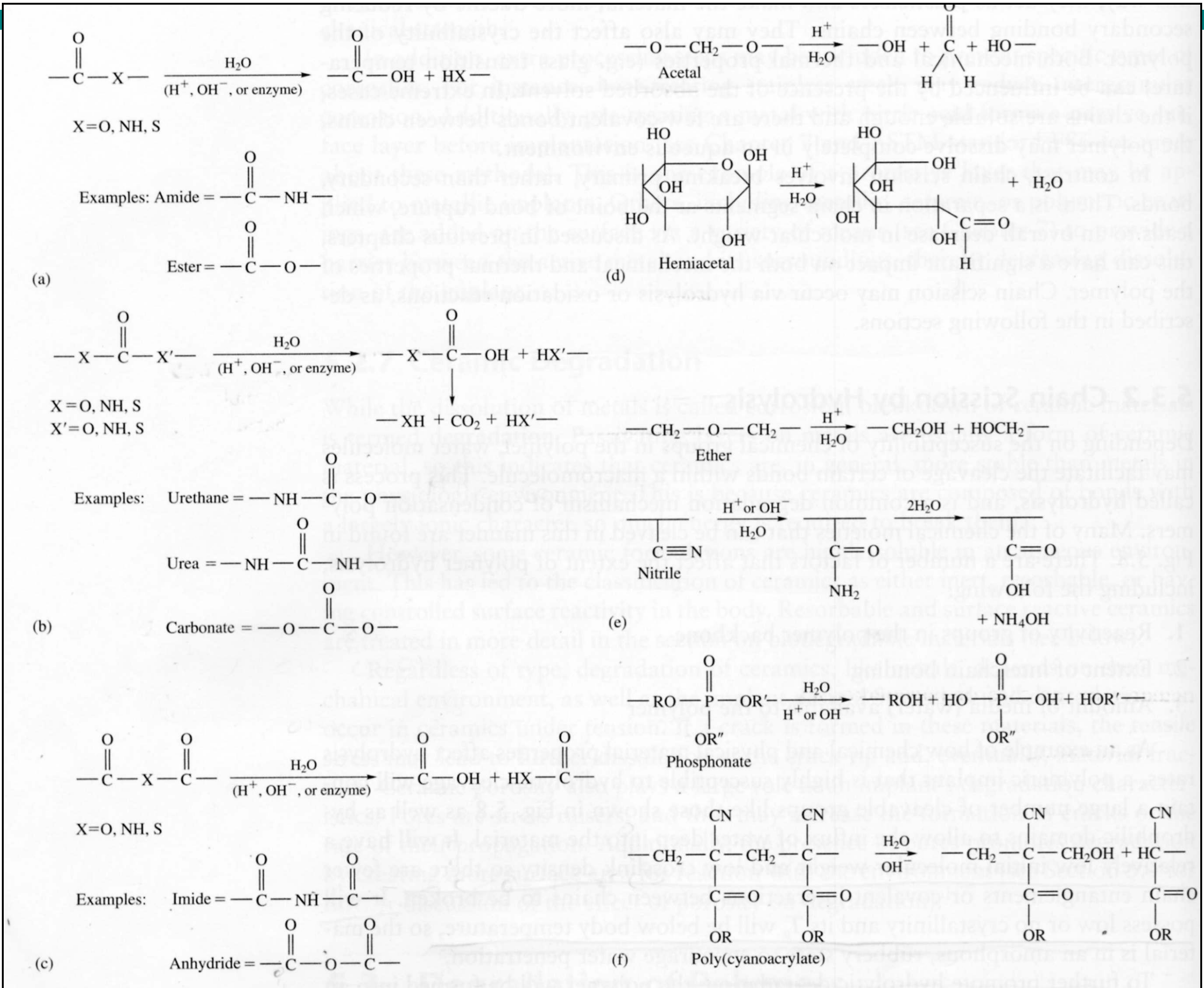
Table 16.5 Resistance to Degradation by Various Environments for Selected Elastomeric Materials[a]

Material	*Weather-Sunlight Aging*	*Oxidation*	*Ozone Cracking*	*Alkali Dilute/ Concentrated*	*Acid Dilute/ Concentrated*	*Chlorinated Hydrocarbons, Degreasers*	*Aliphatic Hydrocarbons, Kerosene, Etc.*	*Animal, Vegetable Oils*
Polyisoprene (natural)	D	B	NR	A/C-B	A/C-B	NR	NR	D-B
Polyisoprene (synthetic)	NR	B	NR	C-B/C-B	C-B/C-B	NR	NR	D-B
Butadiene	D	B	NR	C-B/C-B	C-B/C-B	NR	NR	D-B
Styrene-butadiene	D	C	NR	C-B/C-B	C-B/C-B	NR	NR	D-B
Neoprene	B	A	A	A/A	A/A	D	C	B
Nitrile (high)	D	B	C	B/B	B/B	C-B	A	B
Silicone (polysiloxane)	A	A	A	A/A	B/C	NR	D-C	A

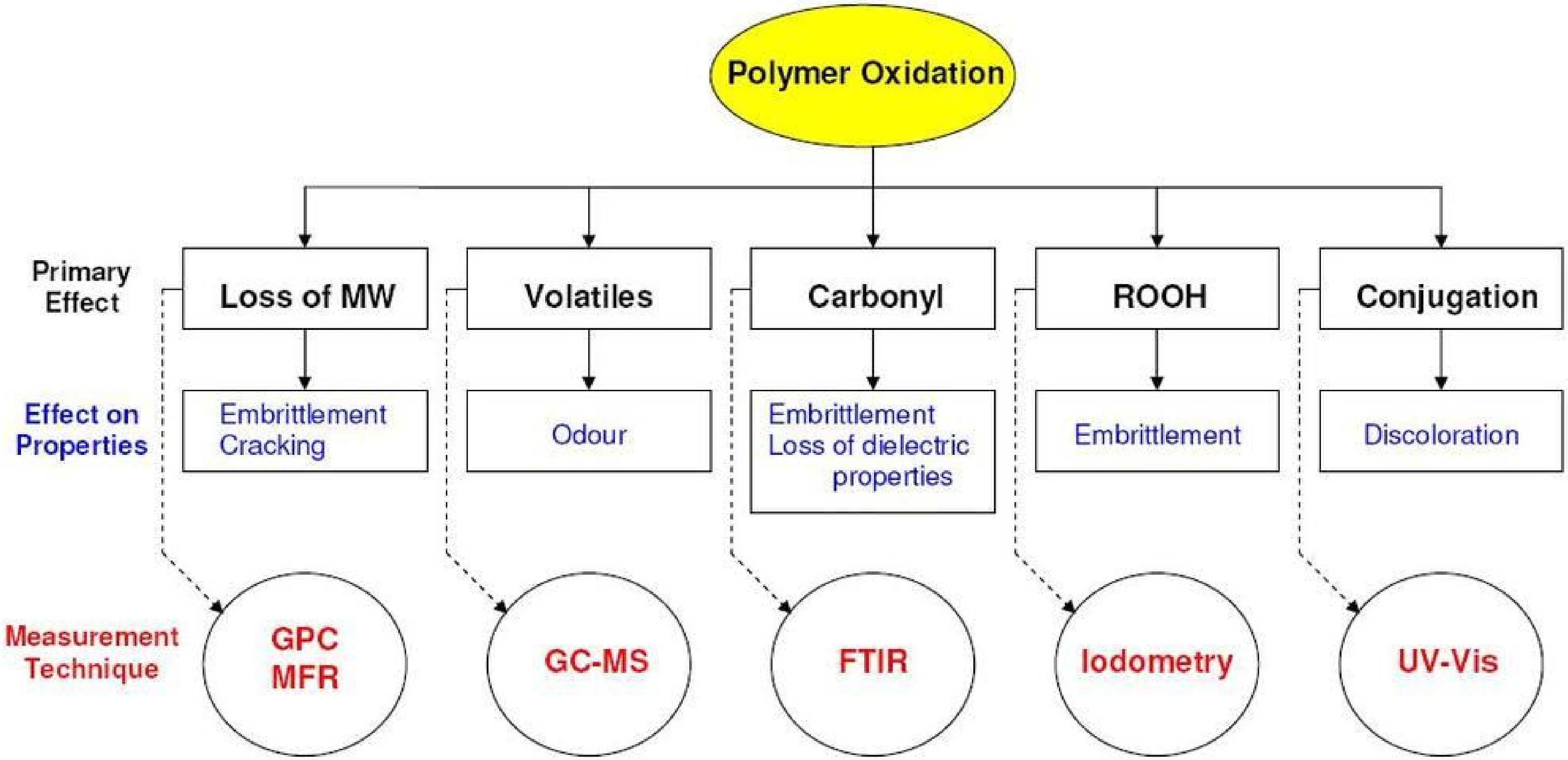
[a] A = excellent, B = good, C = fair, D = use with caution, NR = not recommended.

Source: *Compound Selection and Service Guide*, Seals Eastern, Inc., Red Bank, NJ, 1977.

Cleavage via hydrolysis



Polymer	Achilles' heel(s)
Polyacetal	Acids
PET	Base-catalysed hydrolysis
Nylon 6	Acid-catalysed hydrolysis
Polyurethanes	Hydrolysis
Polyolefins	Oxidizing acids
Polyetherimide	Alkalis
PC	Amines
PPS	Oxidizing agents (e.g. sodium hypochlorite)
PVDF	Primary amines


Polymer Oxidation
Primary Effect
Loss of MW
Volatiles
Carbonyl
ROOH
Conjugation
Effect on Properties
Embrittlement Cracking
Odour
Embrittlement Loss of dielectric properties
Embrittlement
Discoloration
Measurement Technique
GPC MFR
GC-MS
FTIR
Iodometry
UV-Vis

Thermal degradation

- **Depolymerization**
- **Fragmentation**
- **Cross-linking**
- **Thermal loss of additives**

Thermal Gravimetric Analysis:

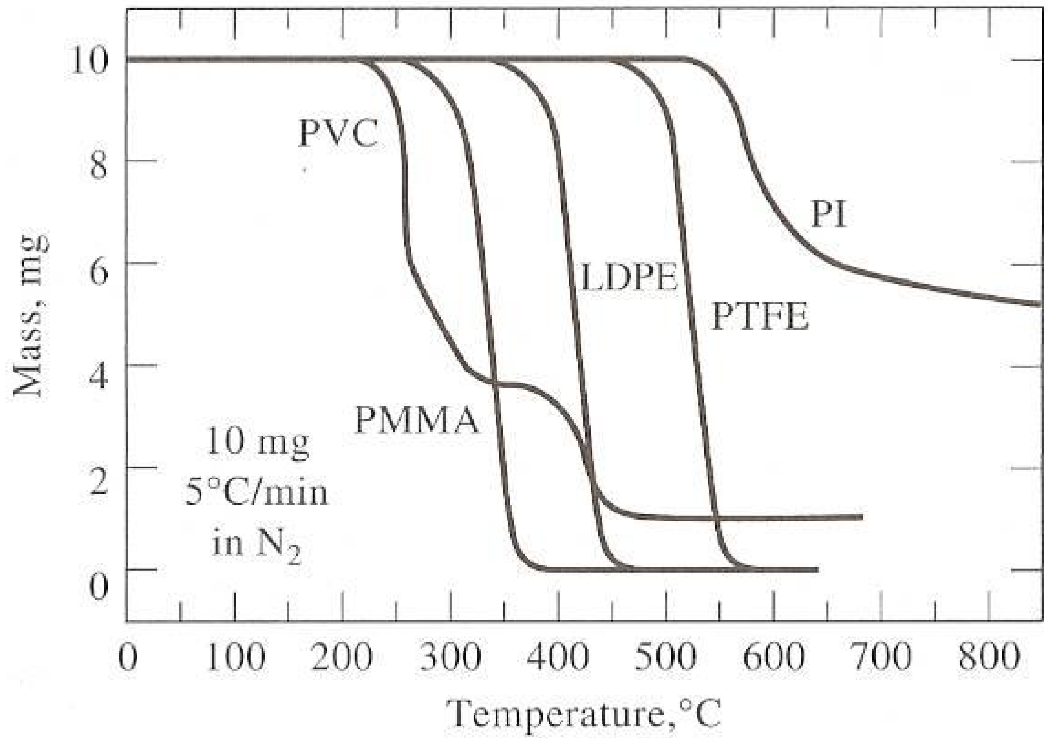
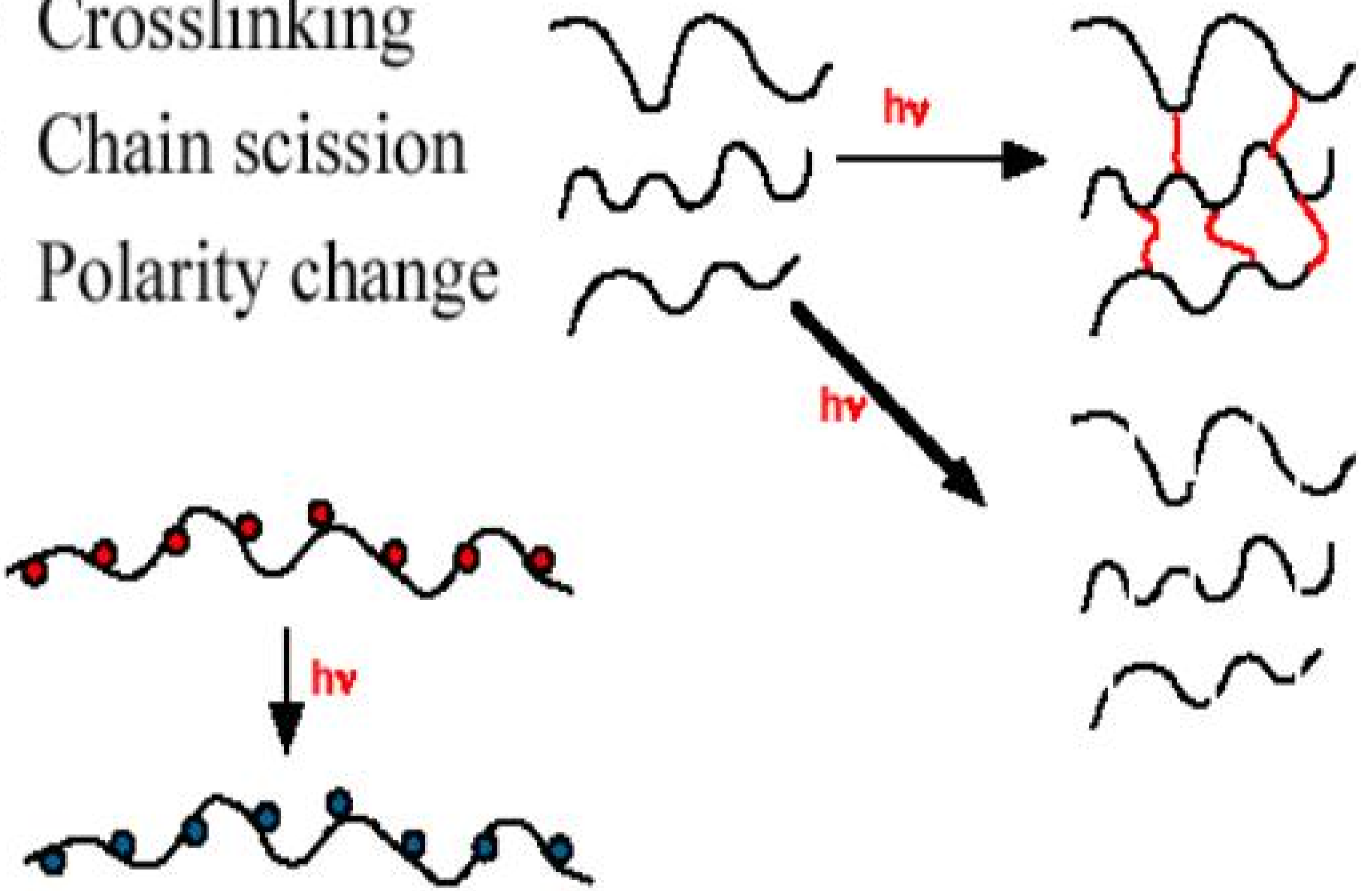


Figure 31-3 Thermograms for some common polymeric materials. PVC = polyvinyl chloride; PMMA = polymethyl methacrylate; LDPE = low density polyethylene; PTFE = polytetrafluoroethylene; PI = aromatic polypyromellitimide. *(From J. Chiu, in* Thermoanalysis of Fiber-Forming Polymers. *R. F. Schwenker, Ed., p. 26. New York: Interscience, 1966. Reprinted by permission of John Wiley & Sons, Inc.)*

Monomer yields from polymers.

Polymer	Monomer, %
Methyl methacrylate	100
styrene	42
α-deuterostyrene	70
β-deuterostyrene	42
α-methylstyrene	100
m-methylstyrene	52
ethylene	<1
methacrylonitrile	100
vinylidene cyanide	100
isobutene	32
propylene	2
methyl acrylate	trace
butadiene	1.5
isoprene	12
tetrafluoroethylene	100
chlorotrifluoroethylene	28

- Crosslinking
- Chain scission
- Polarity change



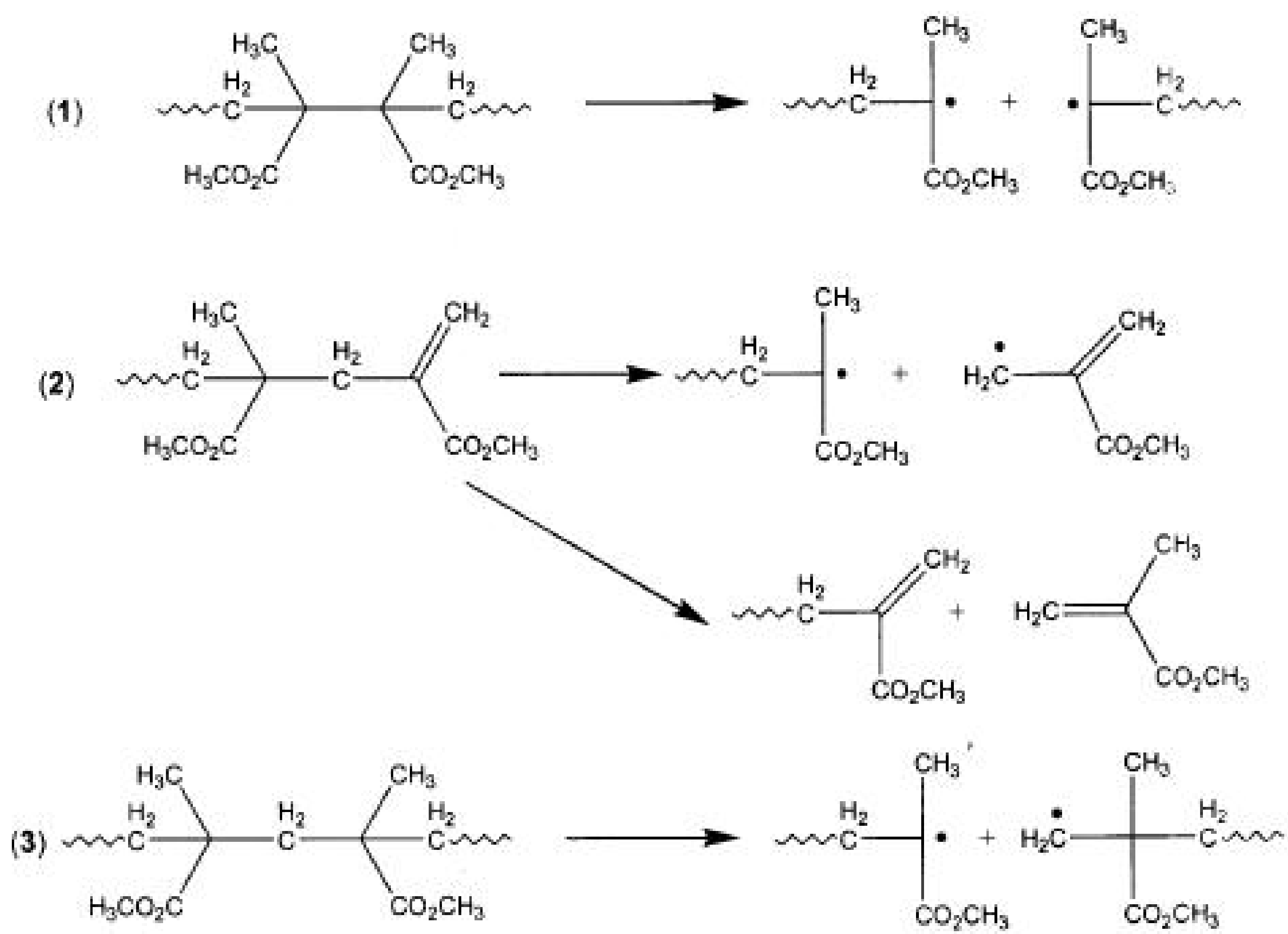
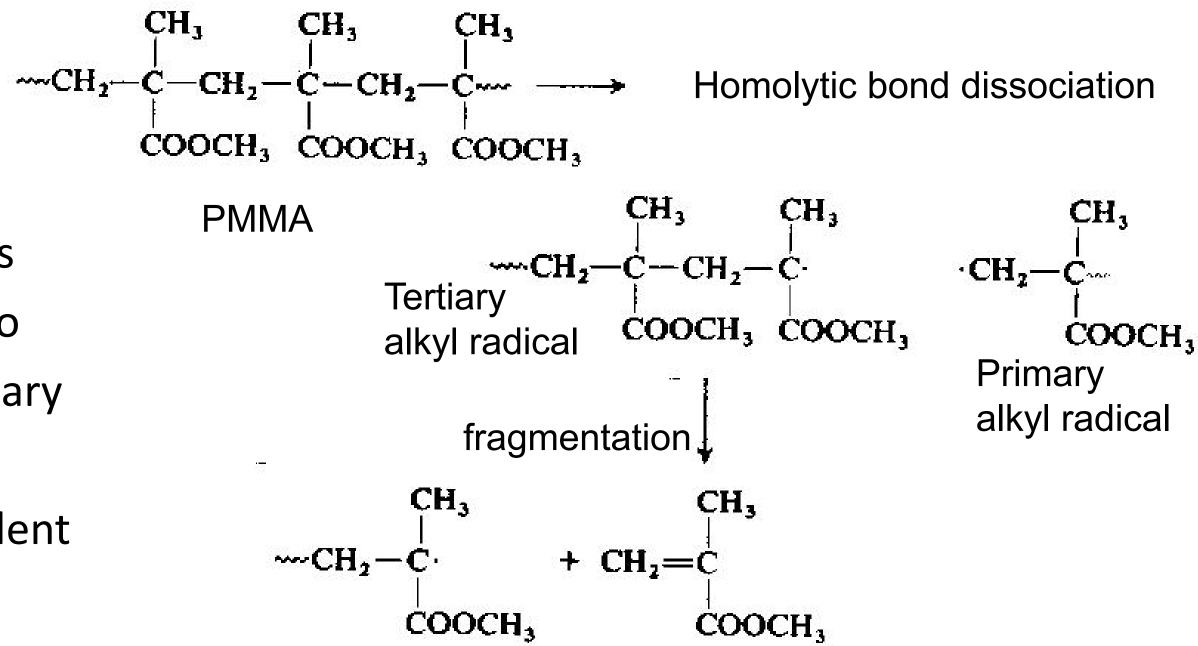


Fig. 1. Three chain scission steps leading to thermal degradation in acrylic polymers.

Thermal Degradation: Depolymerization

- At high temperature, polymers such as poly(methyl methacrylate) become thermally unstable, leading to degradation by depolymerization to yield a mixture of monomer and polymer.
  - PMMA depolymerization is favoured at 300° C. External sources of radicals and defects in chemical structure make the material more susceptible to this mode of degradation.

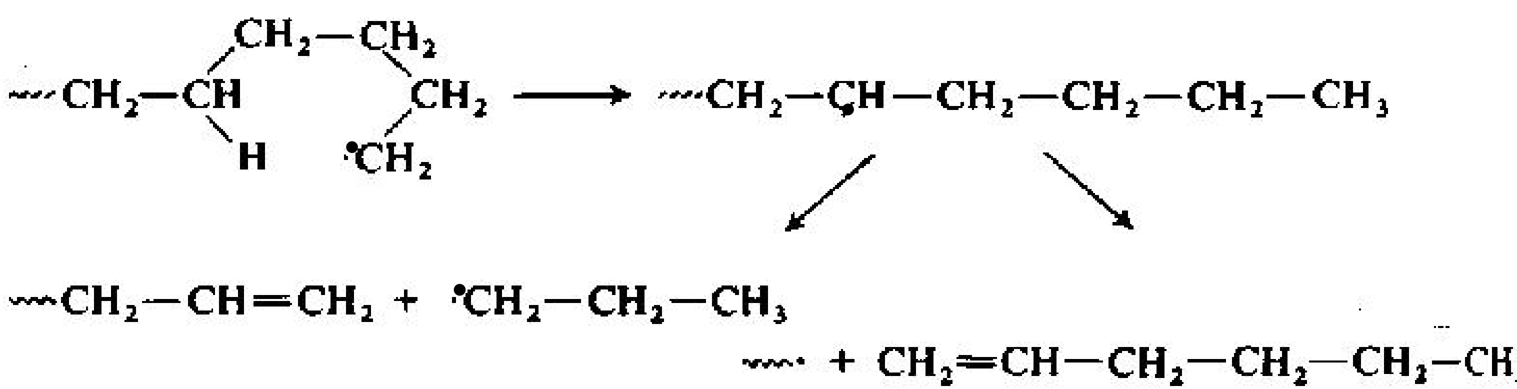


- Radical initiation is thermolytic, leading to fragmentation of tertiary radicals to yields monomer and equivalent tertiary radical.

Thermal Degradation: Fragmentation

- Degradation in many polymer systems leads not to the generation of monomer, but different low molecular weight products.
  - Chain transfer reactions wherein a hydrogen atom is abstracted from a new site is responsible

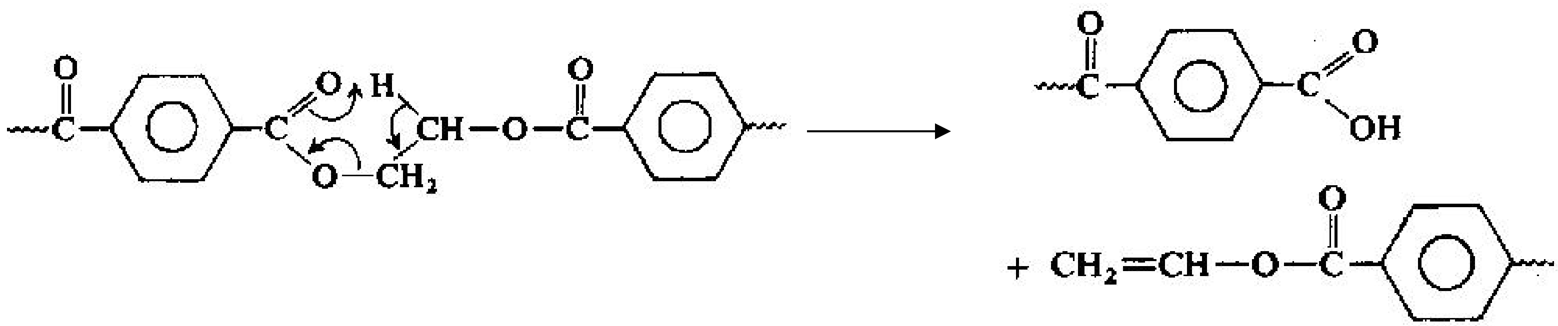
- Poly(ethylene) degradation is a good example, wherein an intramolecular chain transfer leads to 1-hexene, propylene and other low molecular weight compounds



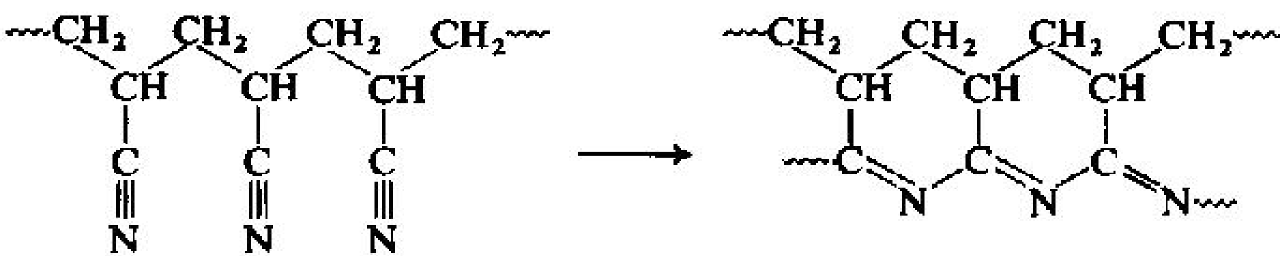
- Abstraction between polymer chains (intermolecular chain transfer) can lead to significant molecular weight losses.

Thermal Degradation: Non-Radical Processes

•Poly(ethylene terephlatate): Rearrangement lowers mol. weight.



•Poly(acrylonitrile): Cyclisiation leads to coloured degradation product

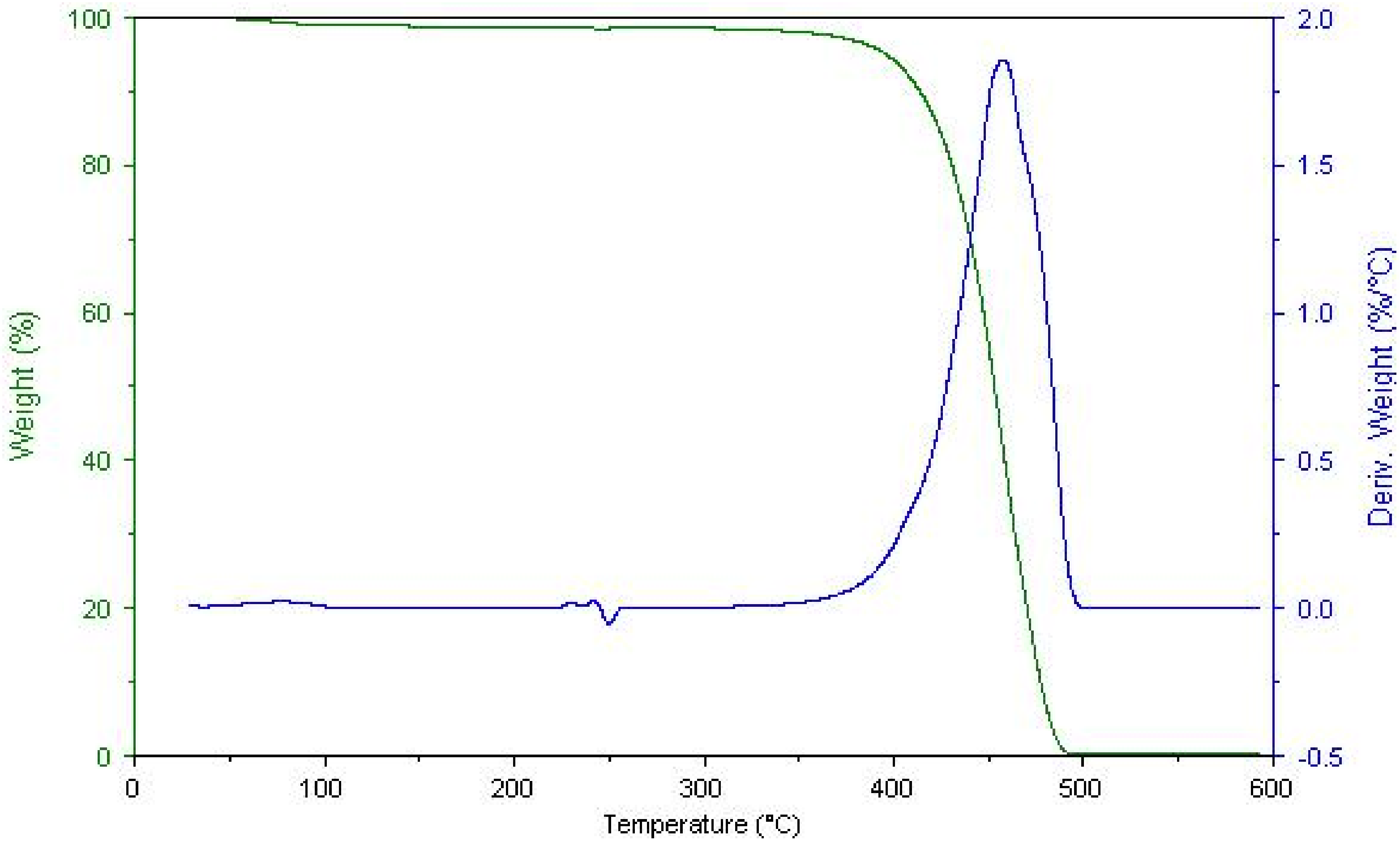


•Poly(vinylchloride):

$$\sim CH_2-CHCl-CH_2-CHCl\sim \;\rightarrow\; \sim CH=CH-CH=CH\sim \quad + HCl$$

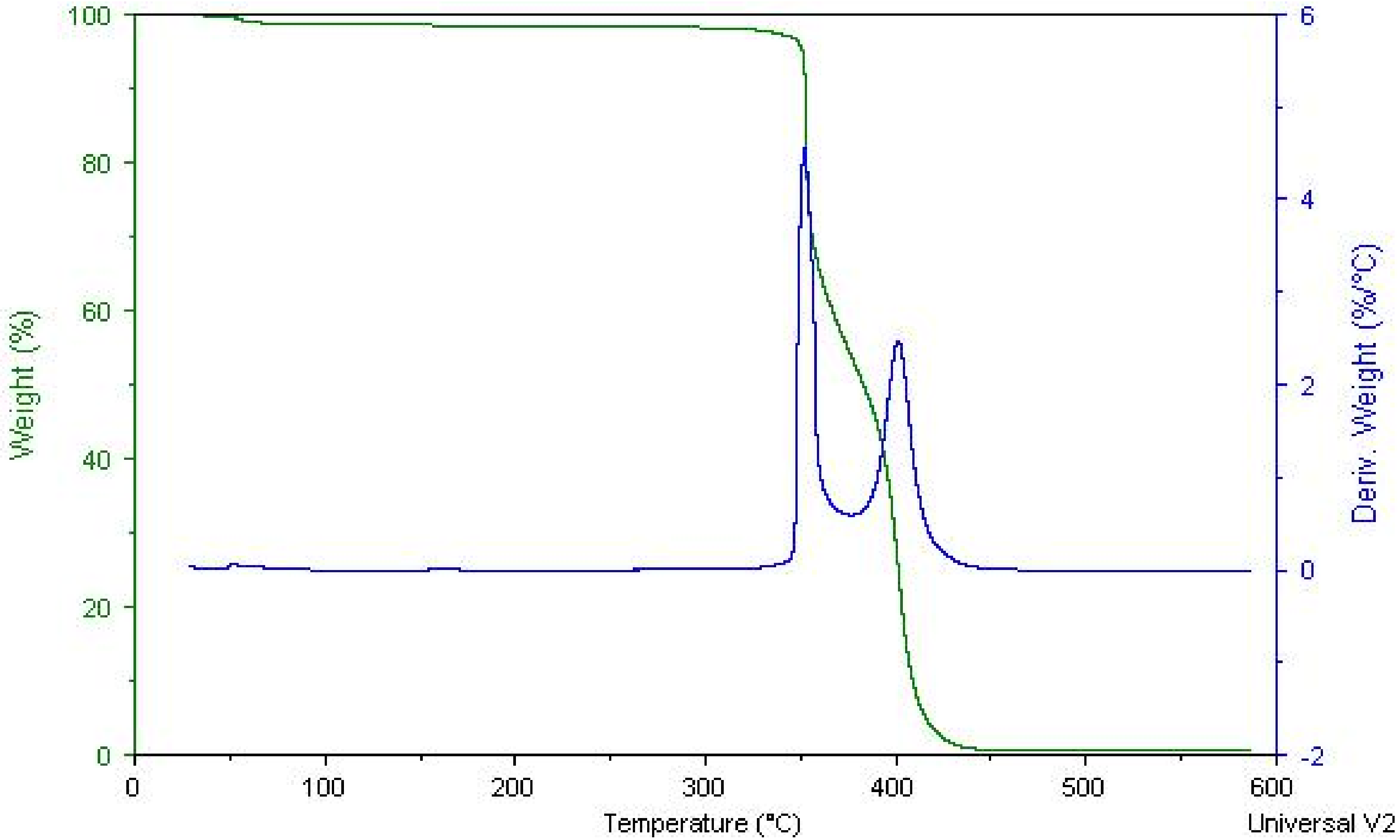
HCl elimination yields a coloured residue that is readily oxidized. Without stabilization, PVC would find little application.

Nylon-6

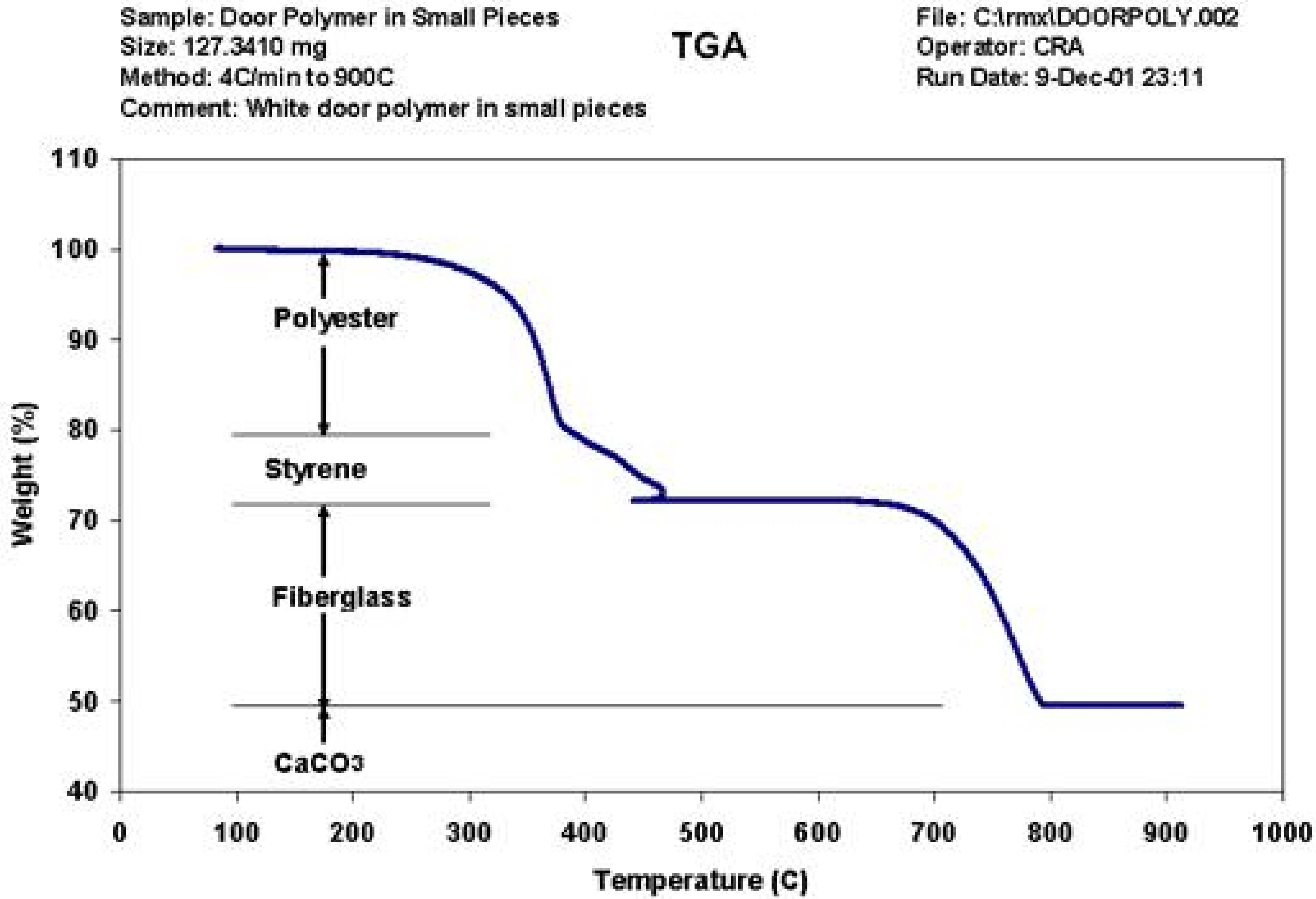


Weight (%)
Deriv. Weight (%/°C)
Temperature (°C)
100
80
60
40
20
0
2.0
1.5
1.0
0.5
0.0
-0.5
0
100
200
300
400
500
600

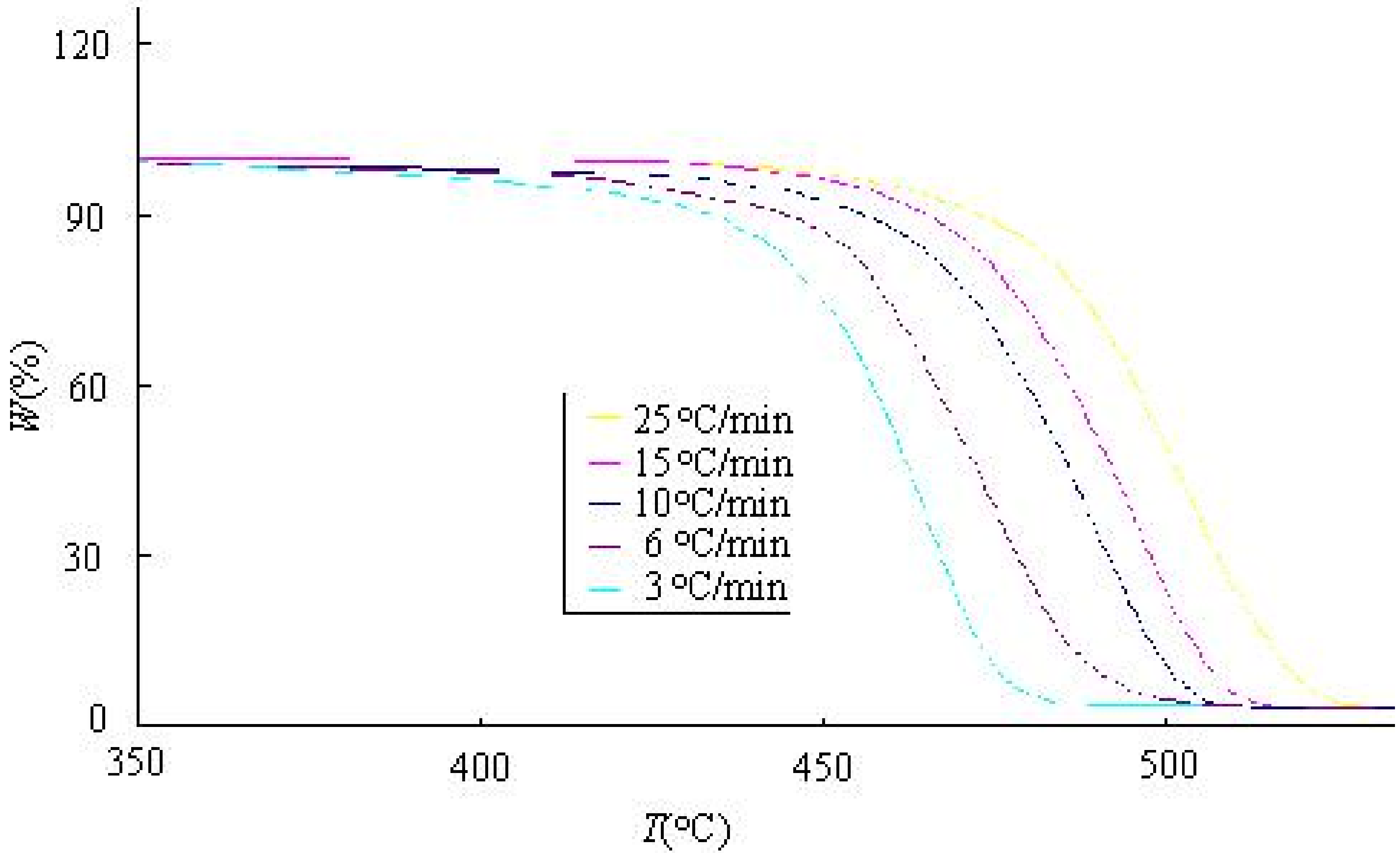
Nylon-6/LDPE



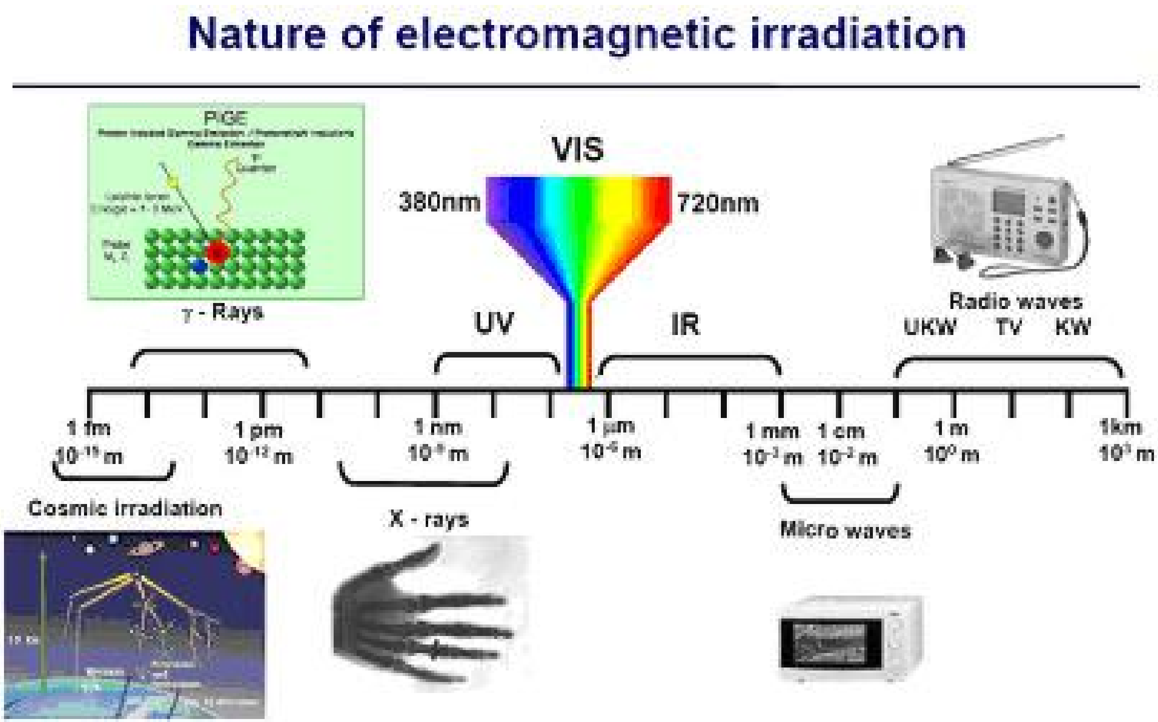
100
80
60
40
20
0
Weight (%)
6
4
2
0
-2
Deriv. Weight (%/°C)
0
100
200
300
400
500
600
Temperature (°C)
Universal V2


Sample: Door Polymer in Small Pieces
Size: 127.3410 mg
Method: 4C/min to 900C
Comment: White door polymer in small pieces
TGA
File: C:\rmx\DOORPOLY.002
Operator: CRA
Run Date: 9-Dec-01 23:11
Polyester
Styrene
Fiberglass
CaCO3
Weight (%)
Temperature (C)
110
100
90
80
70
60
50
40
0
100
200
300
400
500
600
700
800
900
1000

TGA: Rate of Heating

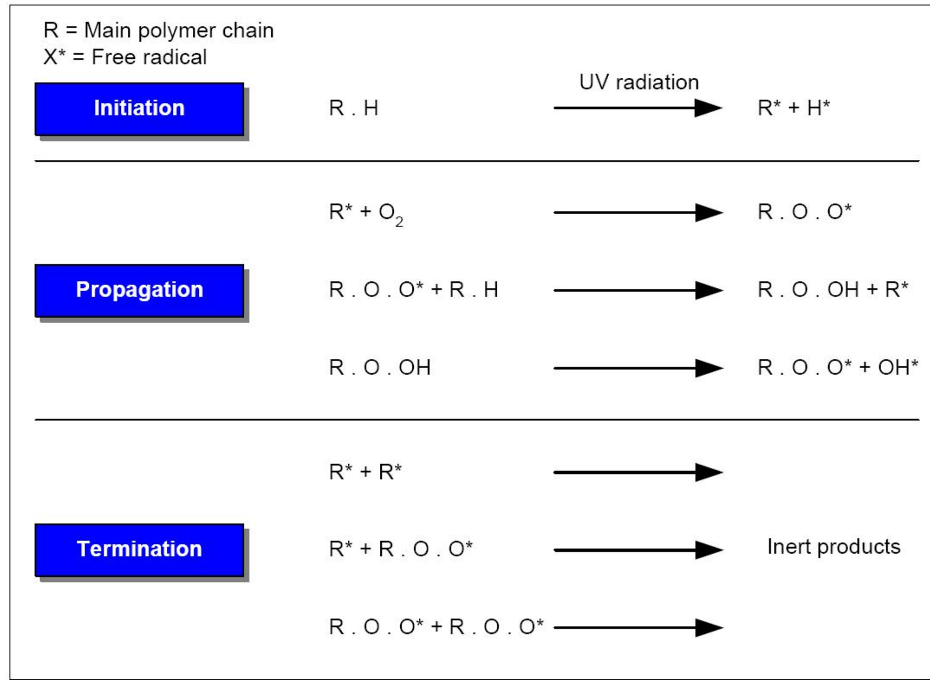


ULTRAVIOLET (RADIATION) DEGRADATION



Plastic degradation

- UV energy absorbed by plastics can excite photons, which then create free radicals.
- While many pure plastics cannot absorb UV radiation, the presence of catalyst residues and other impurities will often act as free radical receptors, and degradation occurs.



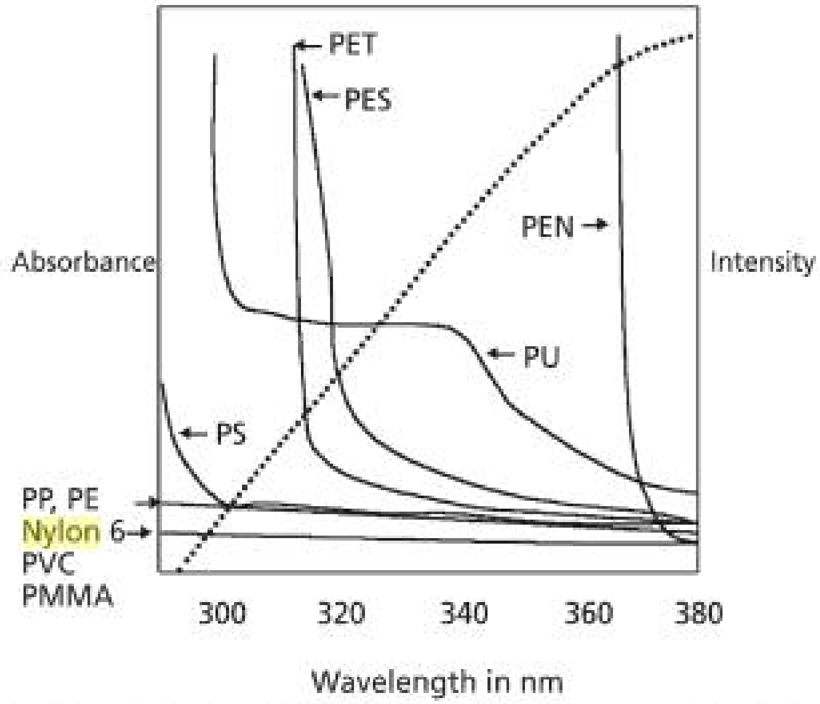


Figure 3.3 UV absorbance of various plastics compared with the intensity of natural sunlight (•••••)

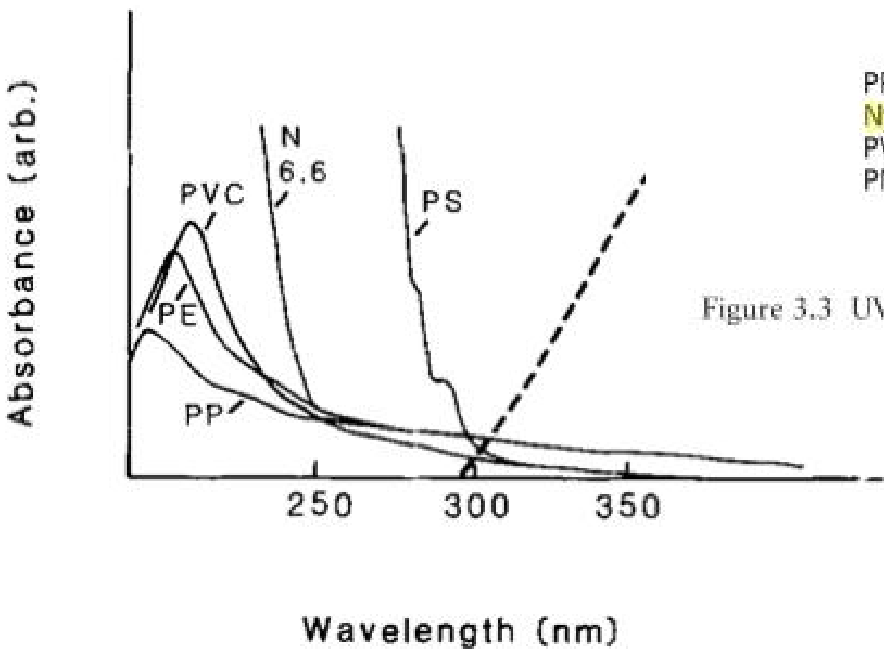
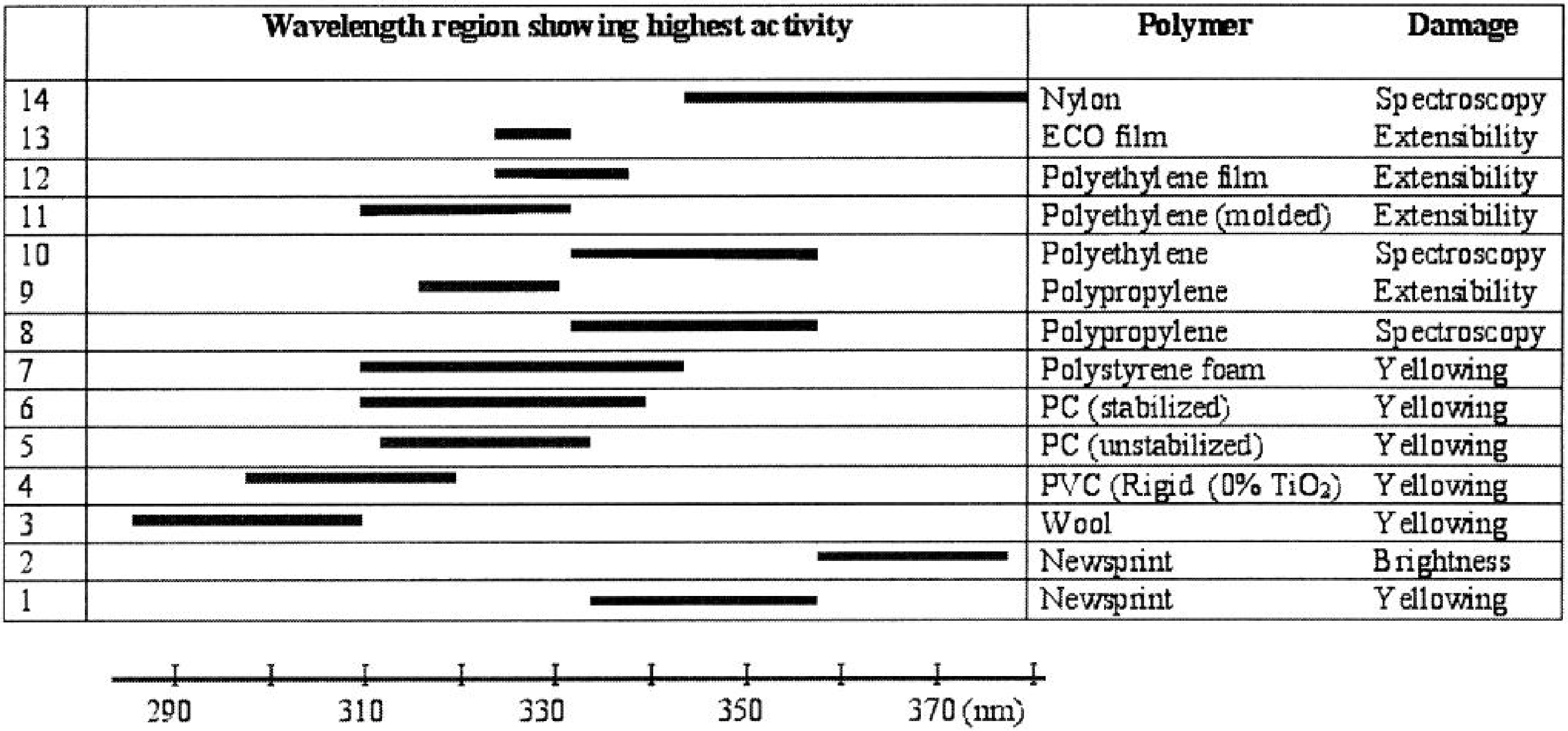
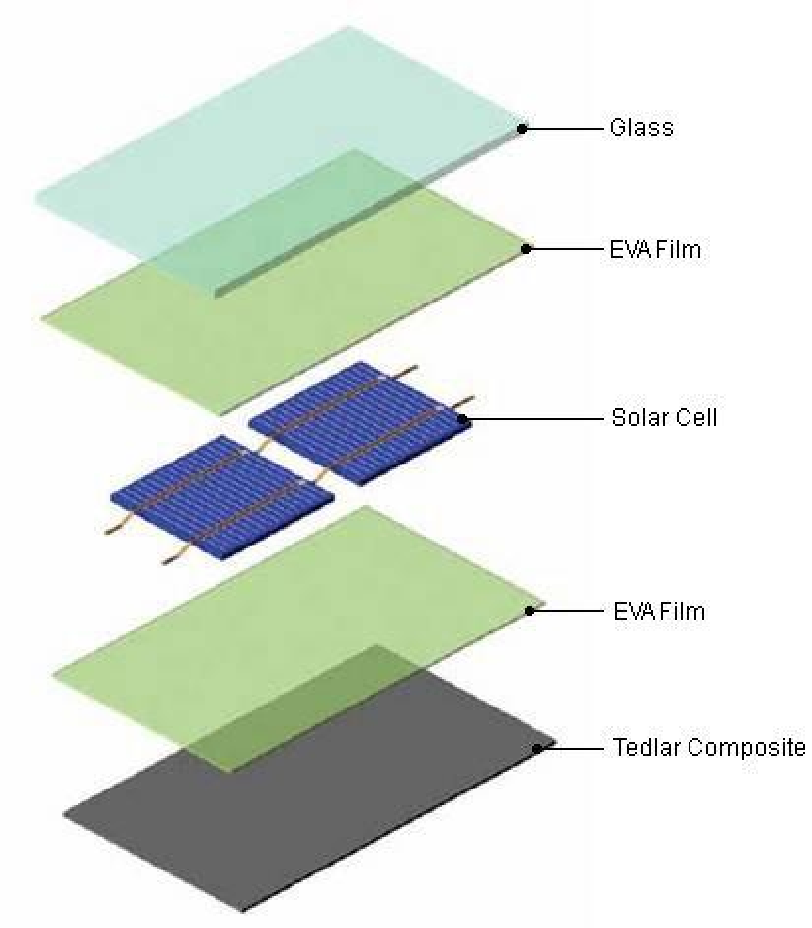


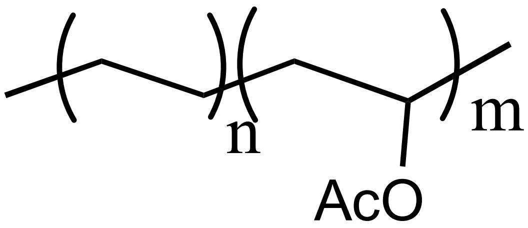
Fig. 2. UV absorption spectra of: PVC [poly(vinyl chloride)], PE (polyethylene), PP (polypropylene), PS (polystyrene) and N 6.6 (6,6-nylon).

Increment = $H(\lambda)\,(T_i(\lambda) - T_{i+1}(\lambda))$

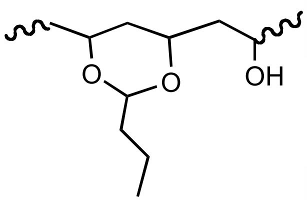


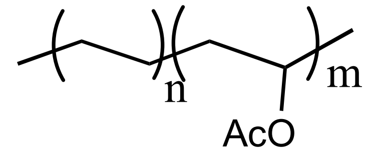
UV Degradation of Polyvinyl acetate-Polyethylene





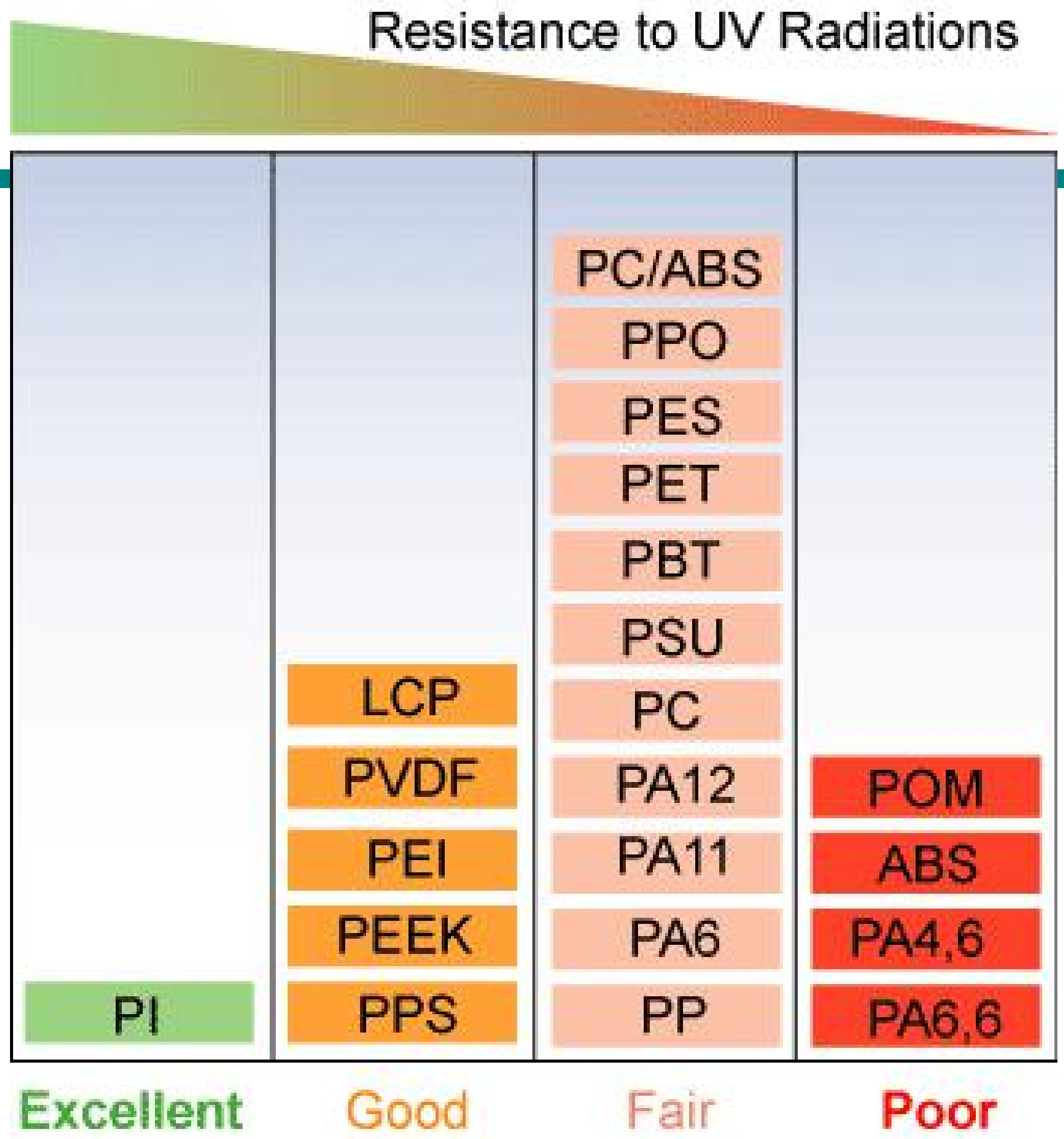
UV Degradation of Polyvinyl acetate-Polyethylene





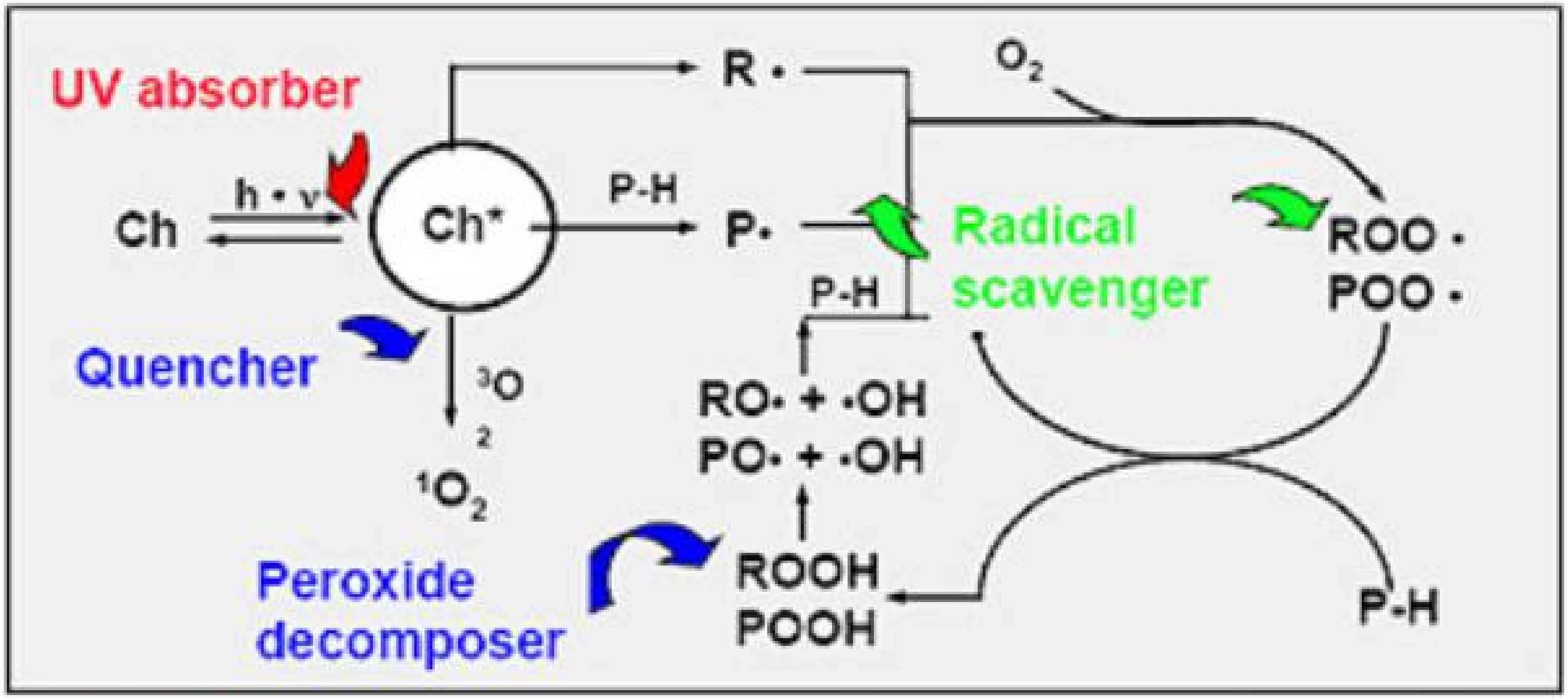
polyvinyl butyral

EVA

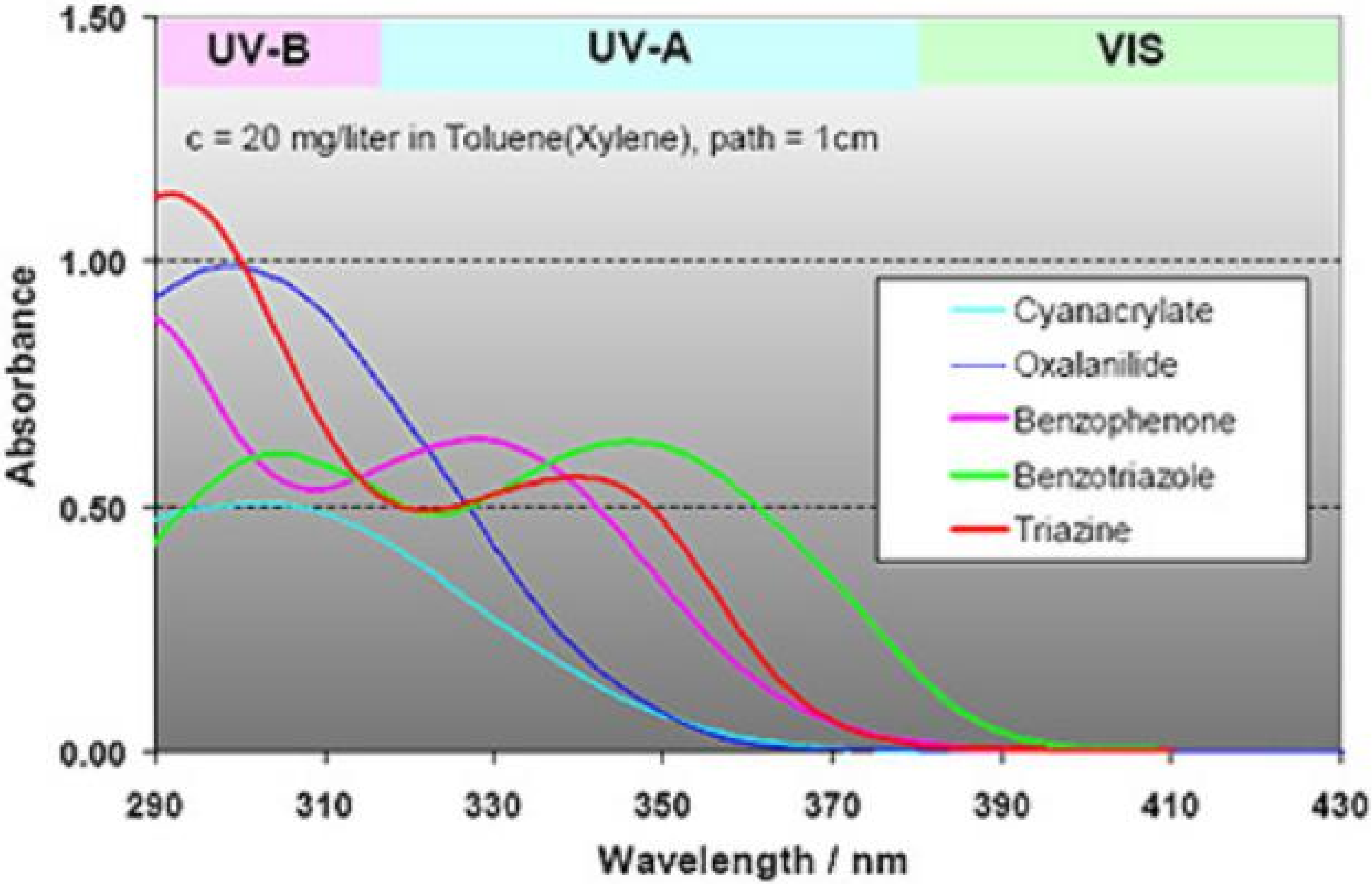

Resistance to UV Radiations
PI
LCP
PVDF
PEI
PEEK
PPS
PC/ABS
PPO
PES
PET
PBT
PSU
PC
PA12
PA11
PA6
PP
POM
ABS
PA4,6
PA6,6
Excellent
Good
Fair
Poor

PMMA

excellent resistance to UV

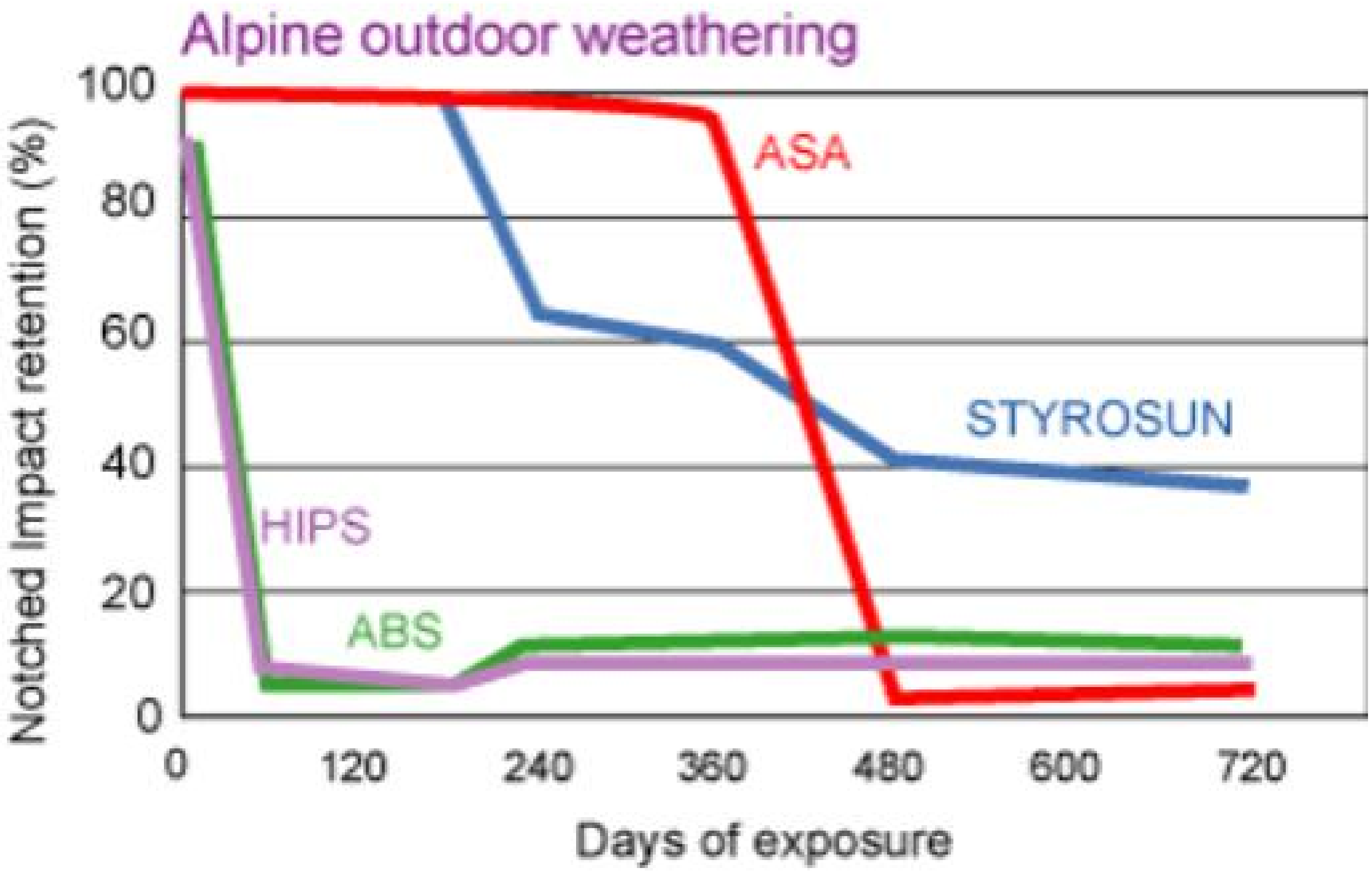


- **UV absorber:** Absorption (filter effect) of the damaging light before CH* can be formed
- **Radical scavenger:** trapping of radicals before subsequent reactions (polymer degradation) take place
- **Peroxide decomposer:** decomposition of peroxides (e.g. phosphites)
- **Quenchers:** Excited state (Ch*) quenching (ESQ) through suitable acceptors (e.g. Ni – compounds)



UV absorption spectrum

UV Stabilizers



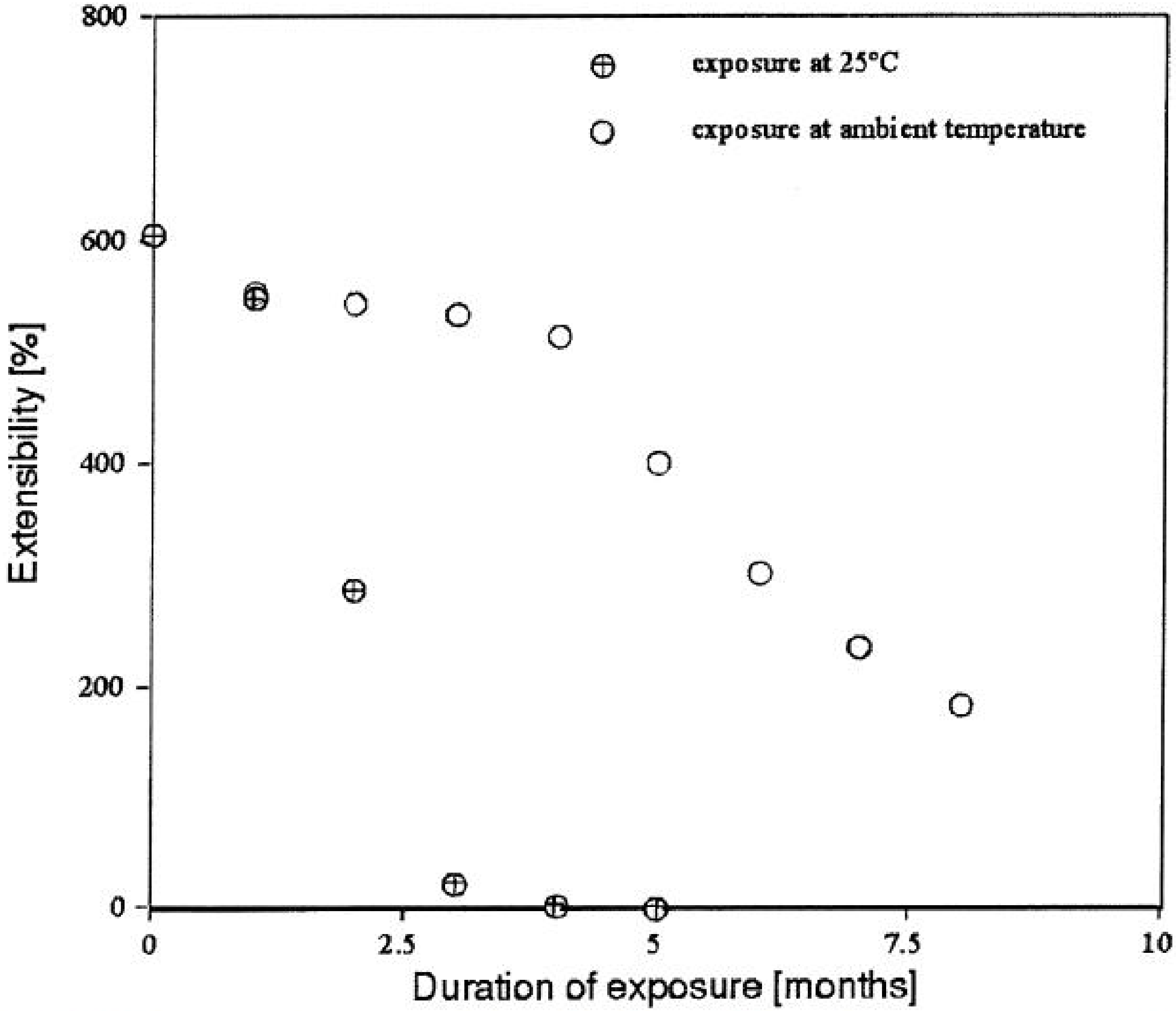
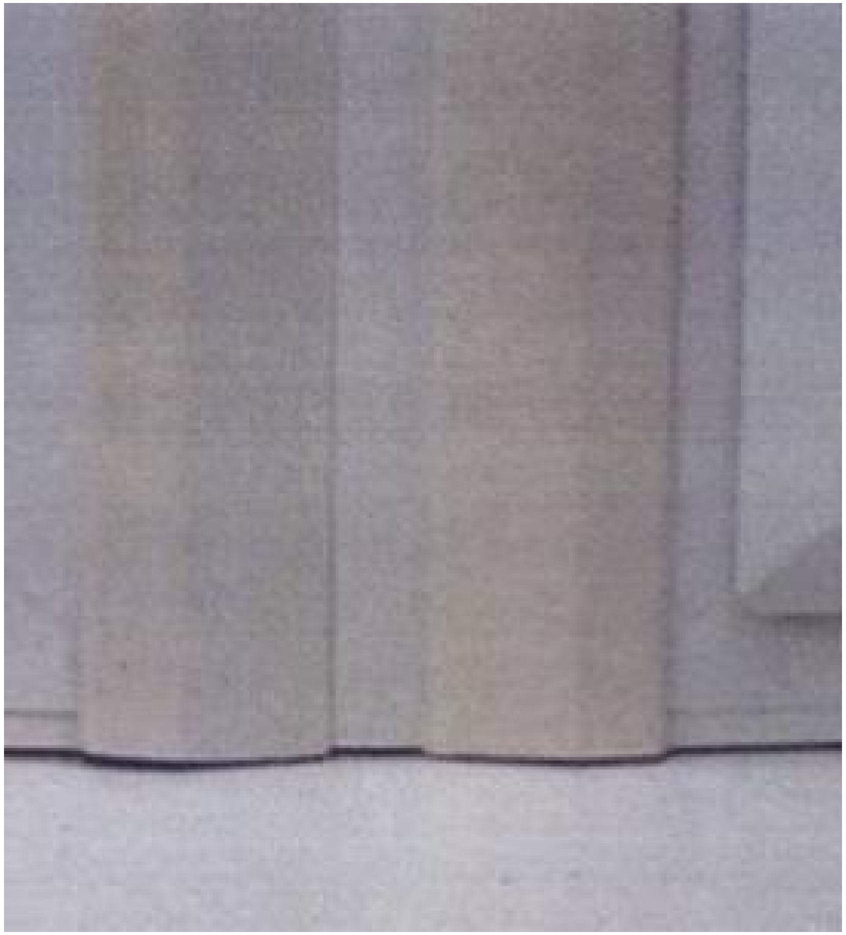


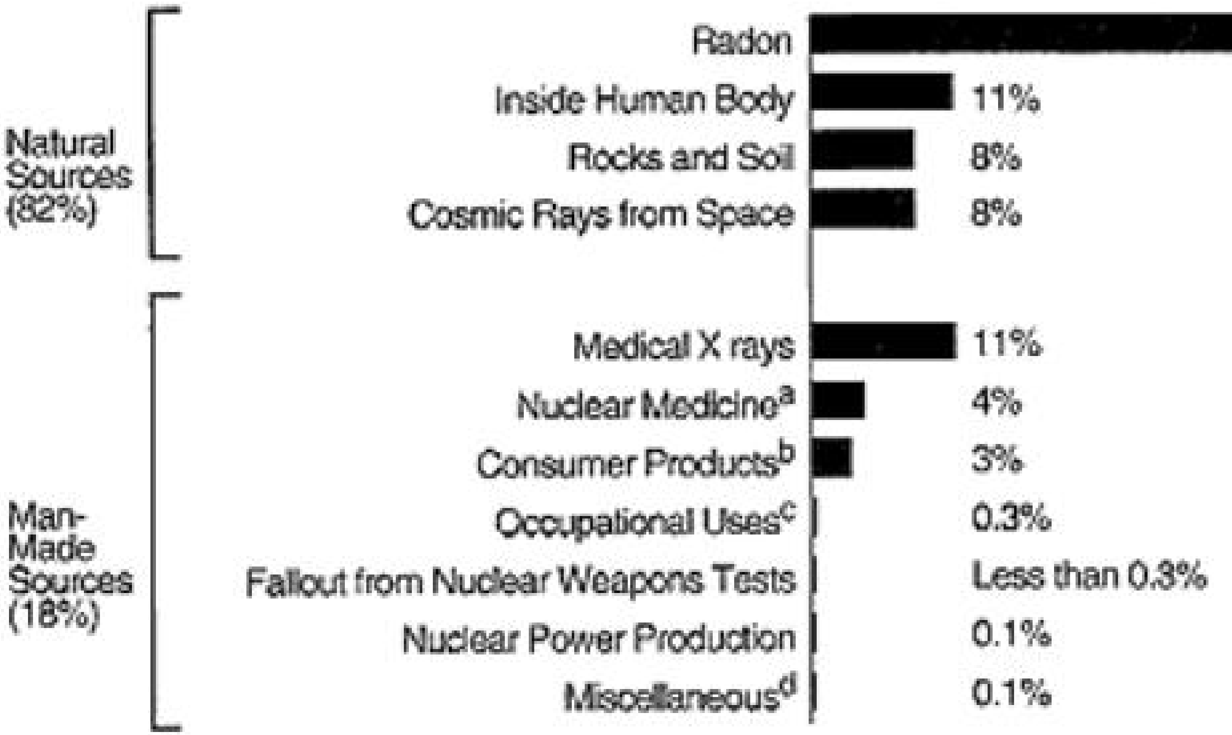
Fig. 1. Change in the extensibility of polyethylene film samples exposed in Dhahran, Saudi Arabia. The open symbols are for samples maintained at 45°C during the exposure, and the filled symbols are samples exposed under ambient conditions [12].

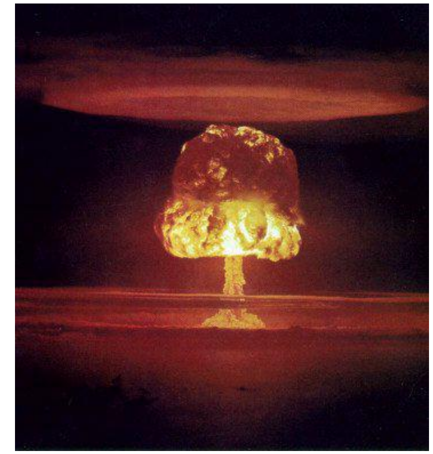
UV & Polyvinyl Chloride-Titania Composites



5 micron deep penetration

Sources of Radiation Exposure





[a] Involves the use of radioactive materials in diagnosing and treating patients with cancer and other diseases.

[b] Building materials, tobacco, mining and agricultural products, water supplies, etc.

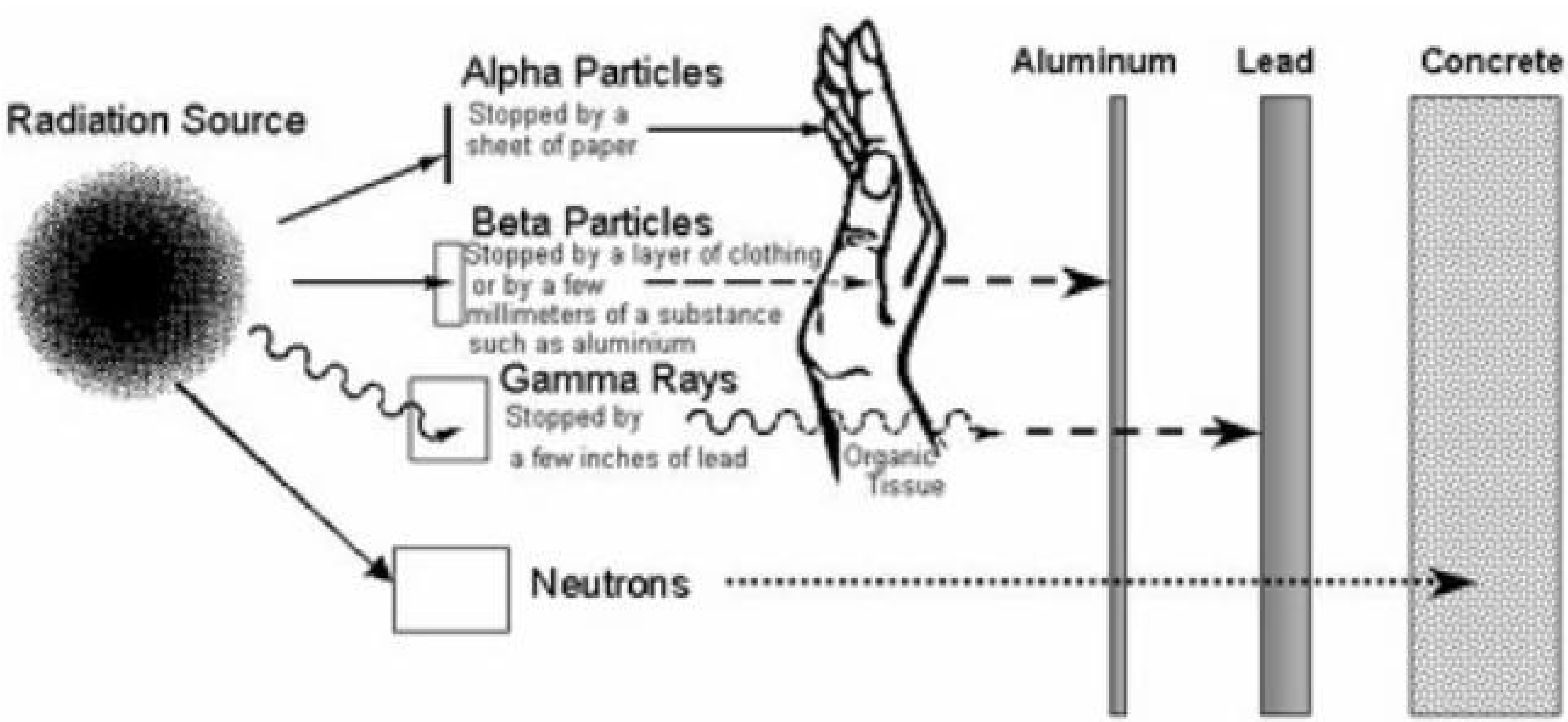
[c] Uranium mines, industrial and medical users, etc.

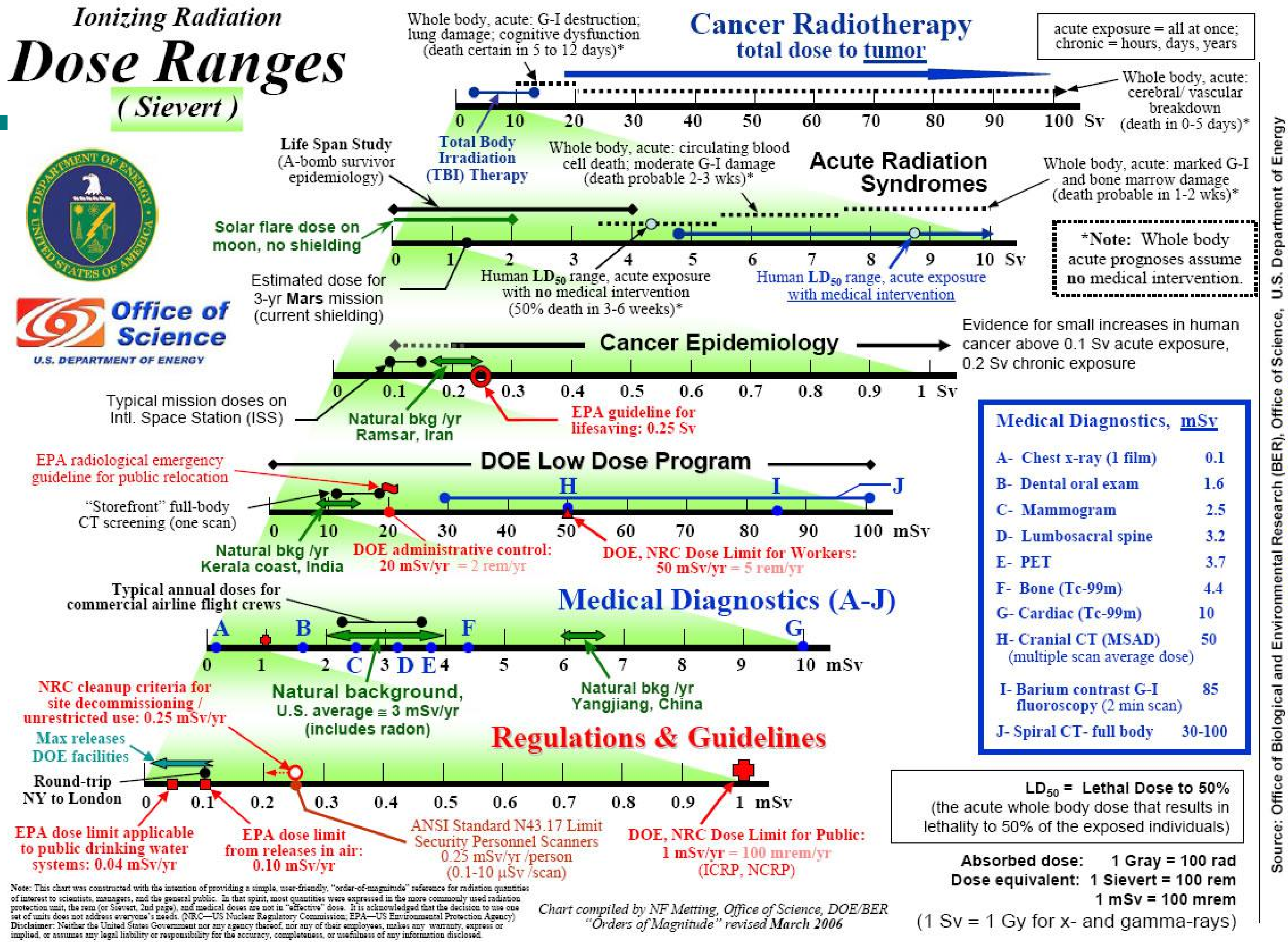
[d] Department of Energy facilities, smelters, transportation, etc.

* Source: National Council on Radiation Protection and Measurements, Report No. 93.
(Total adds up to more than 100% due to rounding off of percentages.)

Radiation Protection

- Gama (hν): 1.5 in. lead, 3 in .steel, 7 in. thick concrete can reduce the effect of gamma rays to permissible levels.
- Beta (electron) can be shielded by several inches of plastic, thin plywood and sheet metal. Can penetrate up to 1/4 in. into the tissue




Ionizing Radiation
Dose Ranges
(Sievert)
Whole body, acute: G-I destruction; lung damage; cognitive dysfunction (death certain in 5 to 12 days)*
Cancer Radiotherapy
total dose to tumor
acute exposure = all at once; chronic = hours, days, years
Whole body, acute: cerebral/ vascular breakdown (death in 0-5 days)*
0 10 20 30 40 50 60 70 80 90 100 Sv
Life Span Study (A-bomb survivor epidemiology)
Total Body Irradiation (TBI) Therapy
Whole body, acute: circulating blood cell death; moderate G-I damage (death probable 2-3 wks)*
Acute Radiation Syndromes
Whole body, acute: marked G-I and bone marrow damage (death probable in 1-2 wks)*
Solar flare dose on moon, no shielding
0 1 2 3 4 5 6 7 8 9 10 Sv
*Note: Whole body acute prognoses assume no medical intervention.
Estimated dose for 3-yr Mars mission (current shielding)
Human LD50 range, acute exposure with no medical intervention (50% death in 3-6 weeks)*
Human LD50 range, acute exposure with medical intervention
Office of Science
U.S. DEPARTMENT OF ENERGY
Cancer Epidemiology
Evidence for small increases in human cancer above 0.1 Sv acute exposure, 0.2 Sv chronic exposure
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1 Sv
Typical mission doses on Intl. Space Station (ISS)
Natural bkg /yr Ramsar, Iran
EPA guideline for lifesaving: 0.25 Sv
Medical Diagnostics, mSv
A- Chest x-ray (1 film) 0.1
B- Dental oral exam 1.6
C- Mammogram 2.5
D- Lumbosacral spine 3.2
E- PET 3.7
F- Bone (Tc-99m) 4.4
G- Cardiac (Tc-99m) 10
H- Cranial CT (MSAD) (multiple scan average dose) 50
I- Barium contrast G-I fluoroscopy (2 min scan) 85
J- Spiral CT- full body 30-100
EPA radiological emergency guideline for public relocation
DOE Low Dose Program
H I J
"Storefront" full-body CT screening (one scan)
0 10 20 30 40 50 60 70 80 90 100 mSv
Natural bkg /yr Kerala coast, India
DOE administrative control: 20 mSv/yr = 2 rem/yr
DOE, NRC Dose Limit for Workers: 50 mSv/yr = 5 rem/yr
Typical annual doses for commercial airline flight crews
Medical Diagnostics (A-J)
A B F G
0 1 2 C 3 D E 4 5 6 7 8 9 10 mSv
NRC cleanup criteria for site decommissioning / unrestricted use: 0.25 mSv/yr
Natural background, U.S. average ≅ 3 mSv/yr (includes radon)
Natural bkg /yr Yangjiang, China
Max releases DOE facilities
Regulations & Guidelines
Round-trip NY to London
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1 mSv
LD50 = Lethal Dose to 50% (the acute whole body dose that results in lethality to 50% of the exposed individuals)
EPA dose limit applicable to public drinking water systems: 0.04 mSv/yr
EPA dose limit from releases in air: 0.10 mSv/yr
ANSI Standard N43.17 Limit Security Personnel Scanners 0.25 mSv/yr /person (0.1-10 μSv /scan)
DOE, NRC Dose Limit for Public: 1 mSv/yr = 100 mrem/yr (ICRP, NCRP)
Absorbed dose: 1 Gray = 100 rad
Dose equivalent: 1 Sievert = 100 rem
1 mSv = 100 mrem
(1 Sv = 1 Gy for x- and gamma-rays)
Note: This chart was constructed with the intention of providing a simple, user-friendly, "order-of-magnitude" reference for radiation quantities of interest to scientists, managers, and the general public. In that spirit, most quantities were expressed in the more commonly used radiation protection unit, the rem (or Sievert, 2nd page), and medical doses are not in "effective" dose. It is acknowledged that the decision to use one set of units does not address everyone's needs. (NRC—US Nuclear Regulatory Commission; EPA—US Environmental Protection Agency)
Disclaimer: Neither the United States Government nor any agency thereof, nor any of their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of any information disclosed.
Chart compiled by NF Metting, Office of Science, DOE/BER
"Orders of Magnitude" revised March 2006
Source: Office of Biological and Environmental Research (BER), Office of Science, U.S. Department of Energy
http://www.science.doe.gov/ober/

Radiation Measurements

- Units of exposure: The coulomb per kilogram (C/kg) is the SI unit of ionizing radiation exposure, and measures the amount of radiation required to create 1 coulomb of charge of each polarity in 1 kilogram of matter. (1 Roentgen = $2.58 \times 10-4$ C/kg)
- Absorbed Dose: more diagnostic of damage. The gray (Gy), with units J/kg, is the SI unit of absorbed dose, which represents the amount of radiation required to deposit 1 joule of energy in 1 kilogram of any kind of matter. 100 rad = 1 Gy.

Equivalent Doses

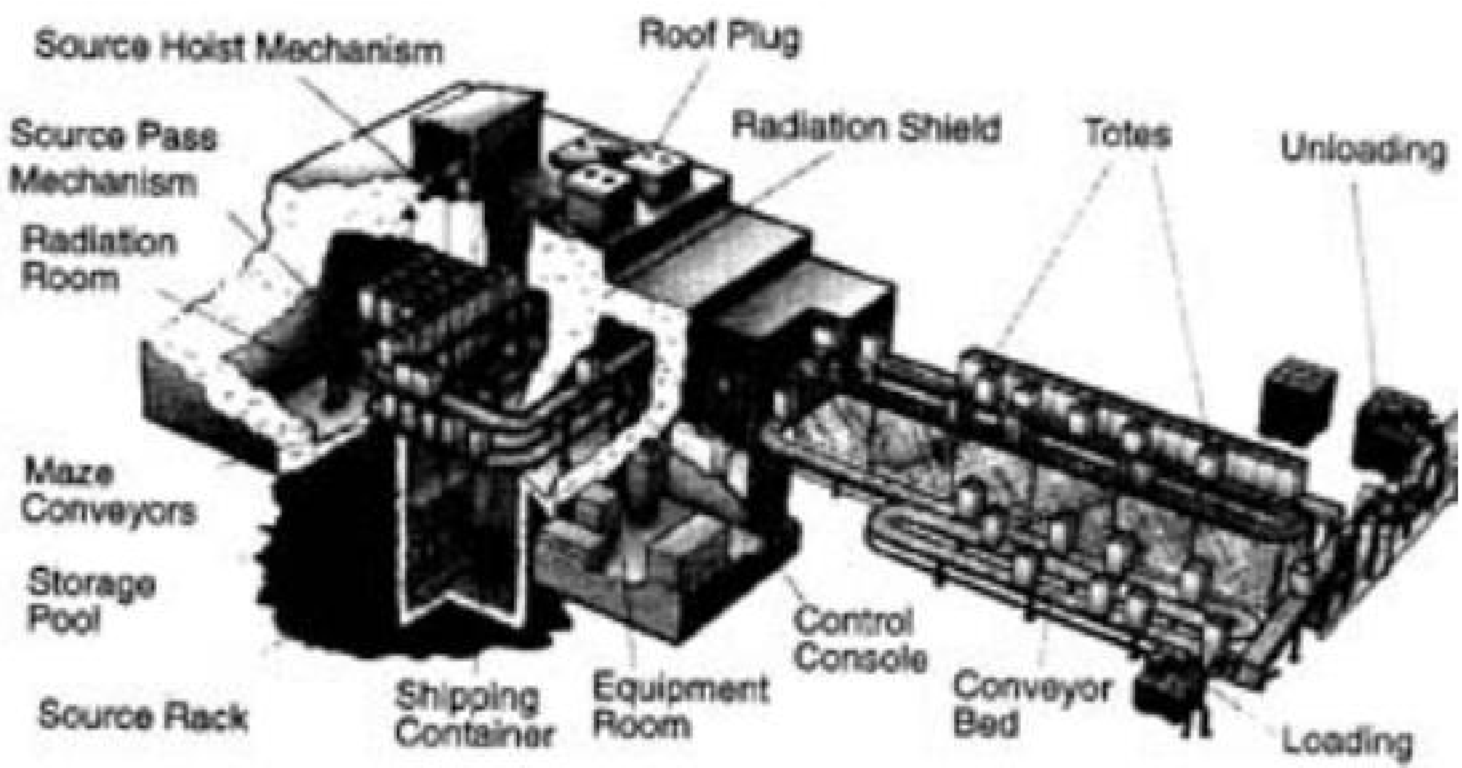
* The sievert (Sv) is the SI unit of equivalent dose. Although it has the same units as grays, J/kg, it measures something different. It is the dose of any type of radiation in Gy that has the same biological effect on a human as 1 Gy of x-rays or gamma radiation.

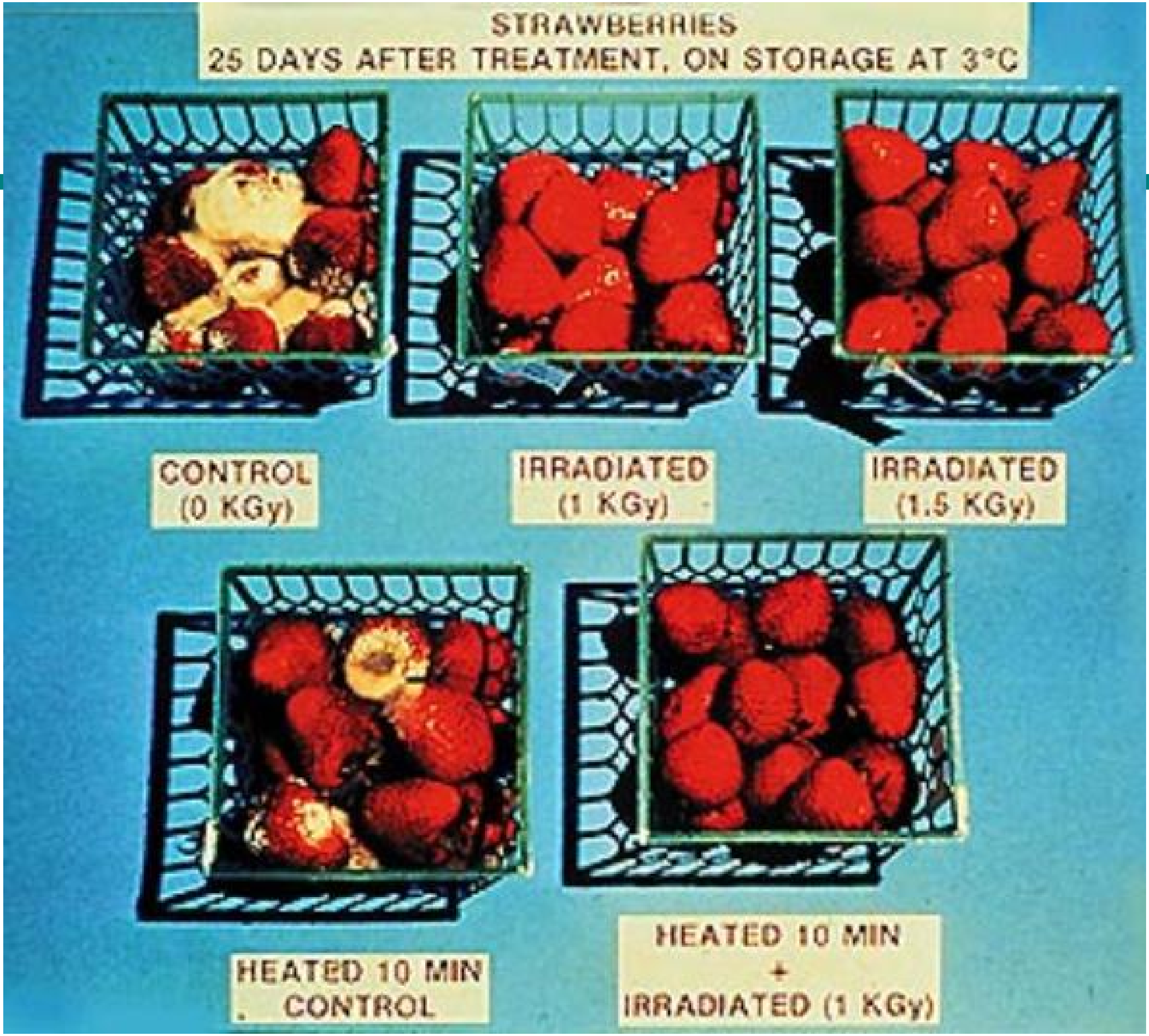
* The rem (Roentgen equivalent man) is the traditional unit of equivalent dose. 1 sievert = 100 rem. Because the rem is a relatively large unit, typical equivalent dose is measured in millirem (mrem), 10^{-3} rem, or in microsievert (μSv), 10^{-6} Sv. 1 mrem = 10 μSv.

'background' dose = 3 mSv = GI x-ray
Lethal dose = 4-5 Sv = 4-5 Gy of x-rays

Radiation Sterilization

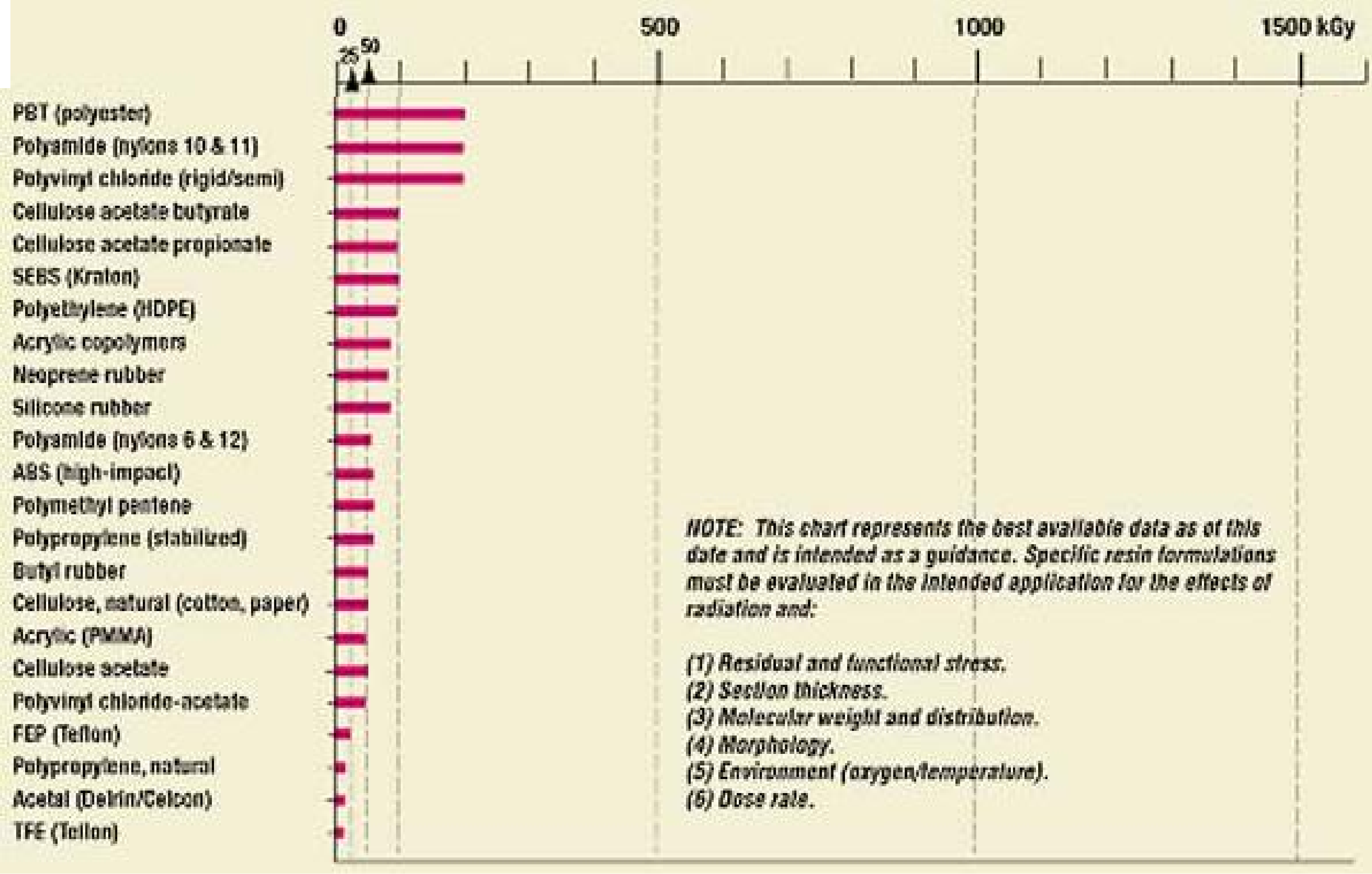
- Co-60 or Cs-137 gama sources
- radiation dose limit of 30 kilograys allowed for herbs and spices (10^9 chest x-rays)
- other foods: 10 kilogray limit




STRAWBERRIES
25 DAYS AFTER TREATMENT, ON STORAGE AT 3°C
CONTROL
(0 KGy)
IRRADIATED
(1 KGy)
IRRADIATED
(1.5 KGy)
HEATED 10 MIN
CONTROL
HEATED 10 MIN
+
IRRADIATED (1 KGy)

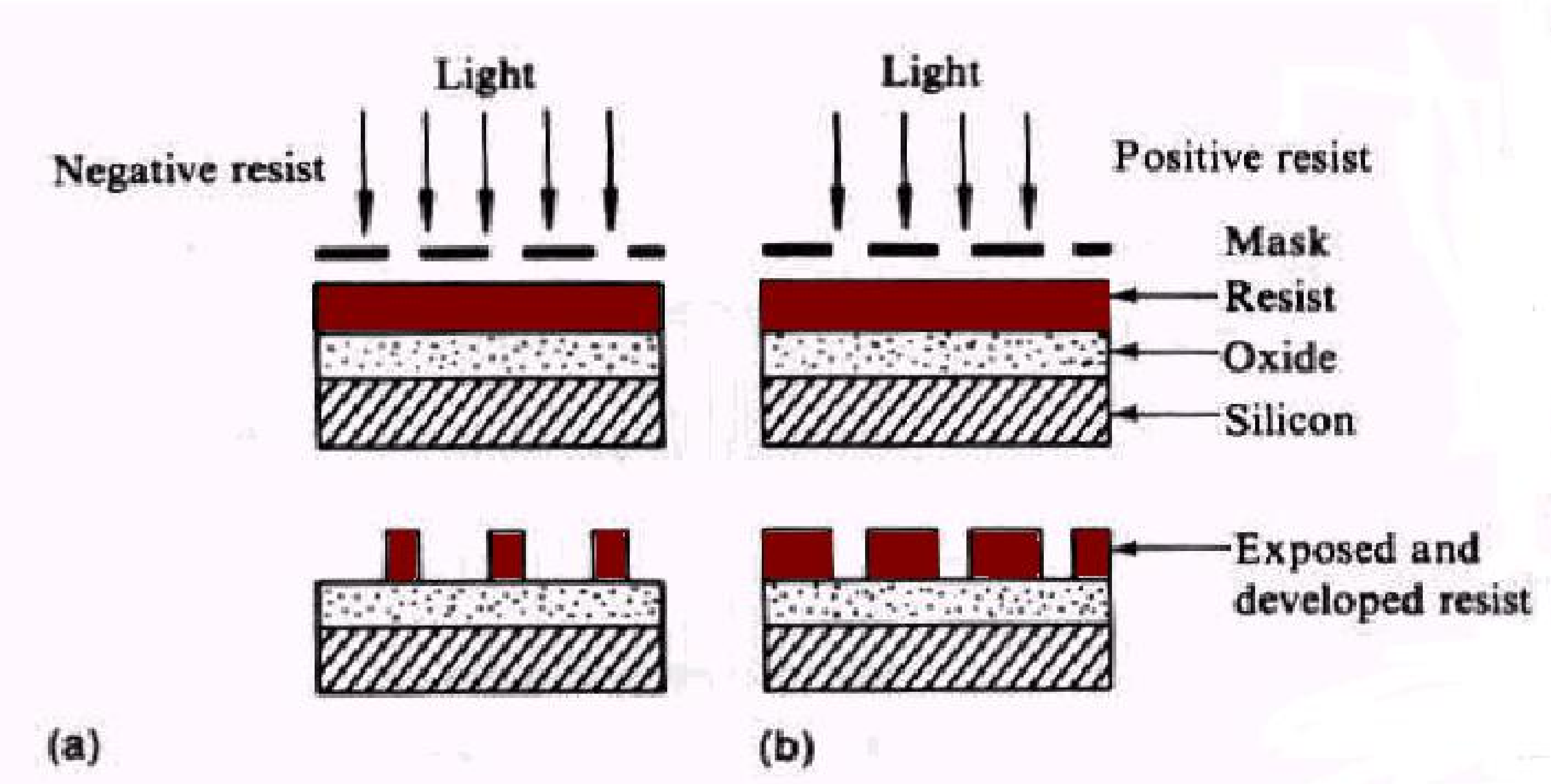
radioactive isotopes (gamma rays from Co-60 or rarely, from Cs-137), particles accelerators (high-energy electrons).

0
25
50
500
1000
1500 kGy
Epoxy (aromatic)
Liquid crystal polymer
Phenolic
Polyester (thermoset)
Polystyrene
Polyurethane (thermoset)
Polyvinylidene fluoride (Kynar)
Polyethylene (low/med.)
PET (polyester, rigid)
ETG (polyester, flexible)
Polyimide
ECTFE
ABS
High-impact styrene (HIPS)
Polycarbonate
Polysulfone Udel
PVC, flexible
Polyvinylidene chloride (Saran)
Natural rubber (latex)
Styrene-butadiene rubber
Chlorinated polyvinyl chloride
Polyethylene (UHMW)
EPDM

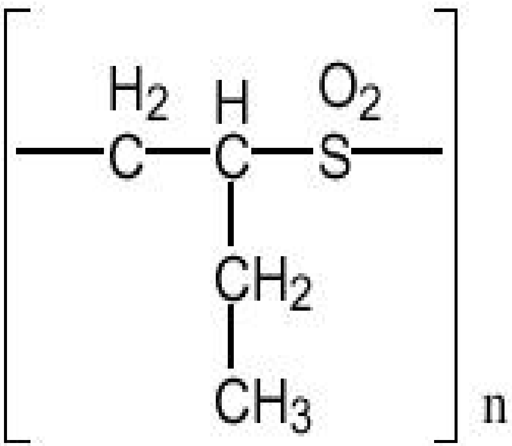


0
25
50
500
1000
1500 kGy
PBT (polyester)
Polyamide (nylons 10 & 11)
Polyvinyl chloride (rigid/semi)
Cellulose acetate butyrate
Cellulose acetate propionate
SEBS (Kraton)
Polyethylene (HDPE)
Acrylic copolymers
Neoprene rubber
Silicone rubber
Polyamide (nylons 6 & 12)
ABS (high-impact)
Polymethyl pentene
Polypropylene (stabilized)
Butyl rubber
Cellulose, natural (cotton, paper)
Acrylic (PMMA)
Cellulose acetate
Polyvinyl chloride-acetate
FEP (Teflon)
Polypropylene, natural
Acetal (Delrin/Celcon)
TFE (Teflon)
NOTE: This chart represents the best available data as of this date and is intended as a guidance. Specific resin formulations must be evaluated in the intended application for the effects of radiation and:
(1) Residual and functional stress.
(2) Section thickness.
(3) Molecular weight and distribution.
(4) Morphology.
(5) Environment (oxygen/temperature).
(6) Dose rate.

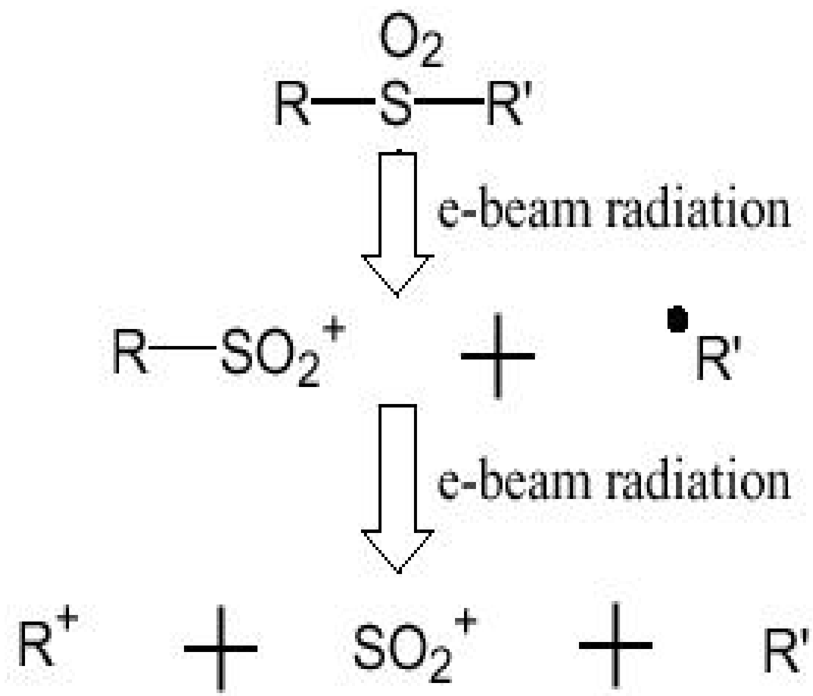
Negative and Positive Photoresist

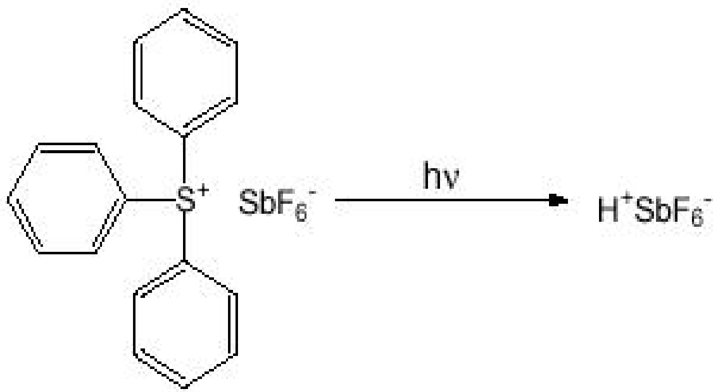


Positive Tone Photoresists



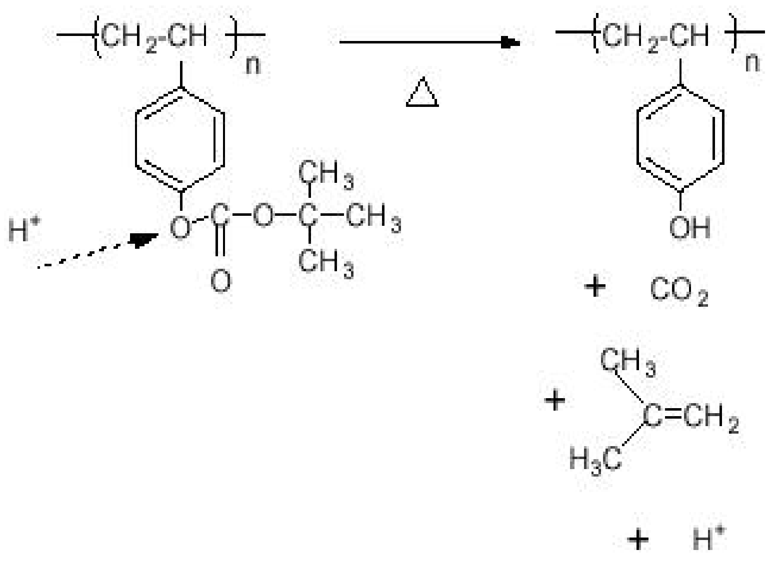
polybutene-1-sulfone

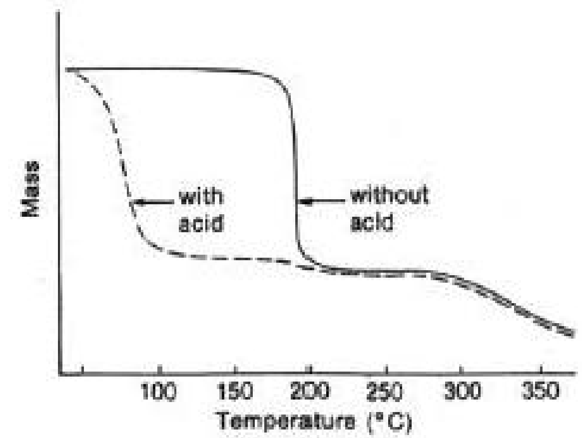




Resist Components

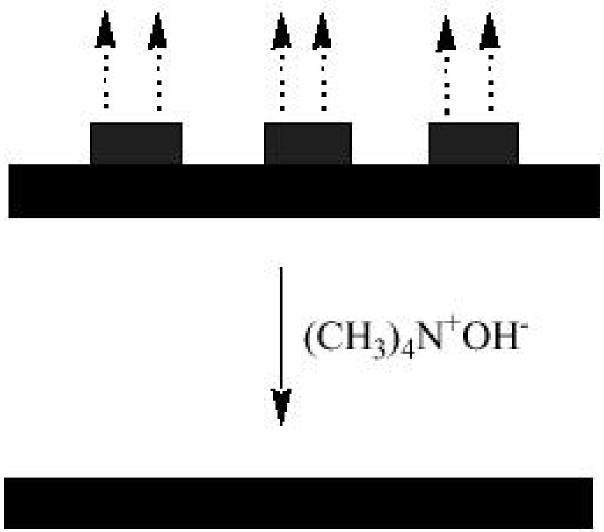
- Polymer
- Solvent
- Photoacid Generator (PAG)
- Additives (e.g. DI,plasticizer)

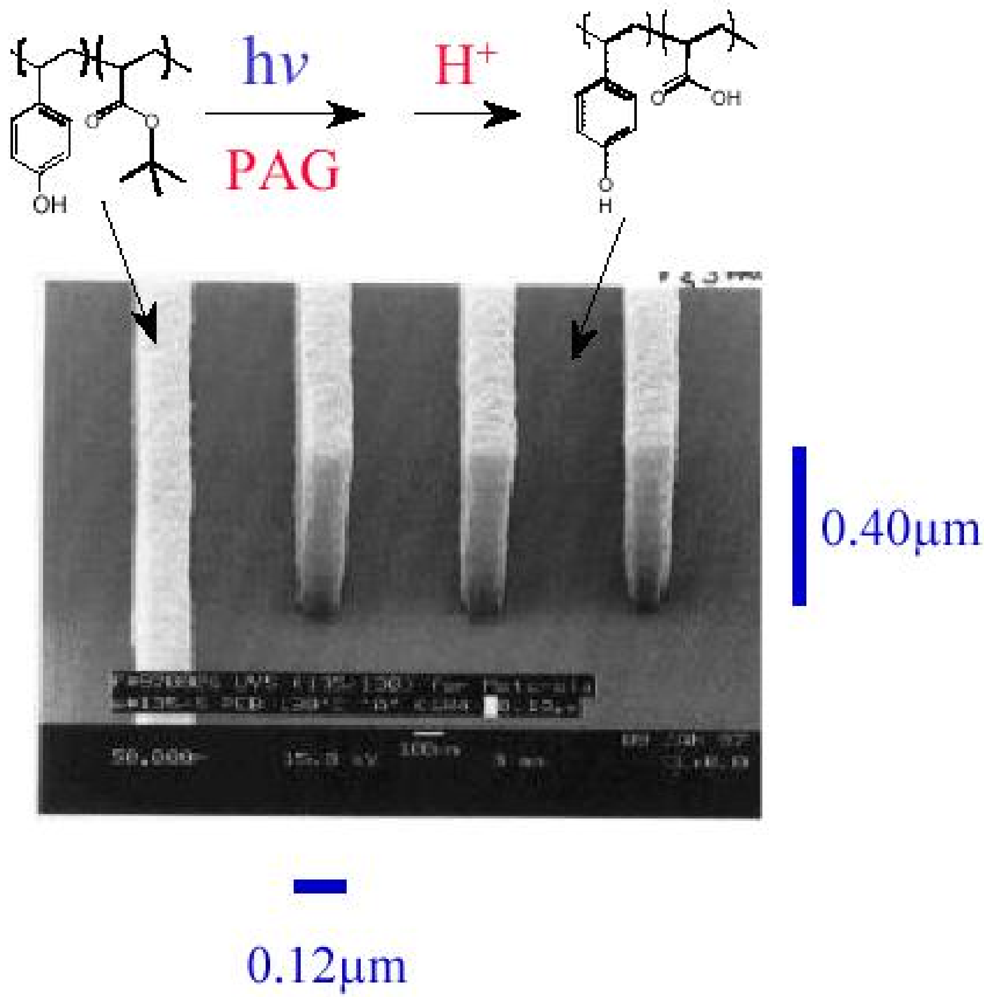


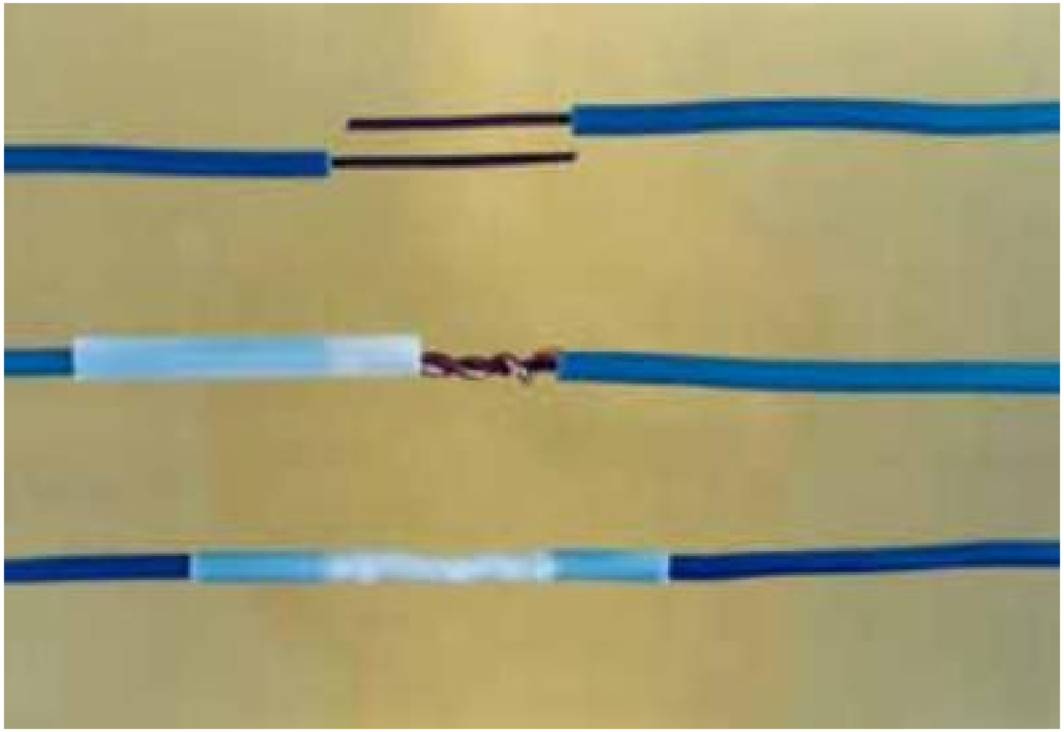


Positive CA Photoresist Chemistry

Aqueous Base Dissolution



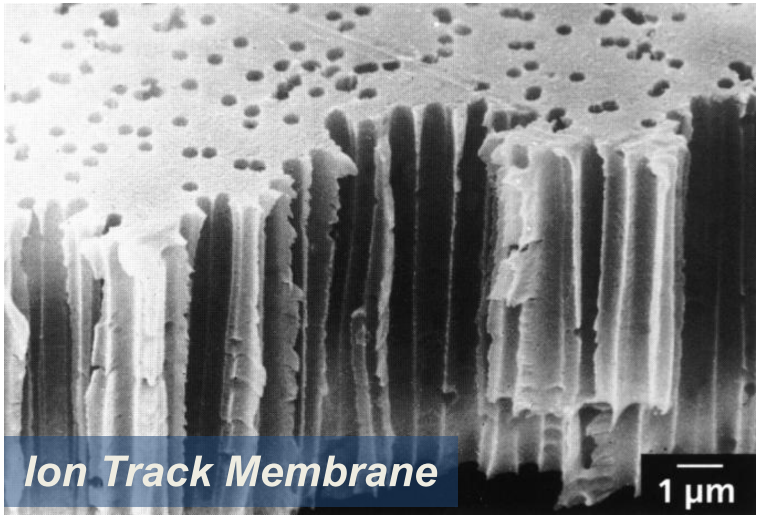




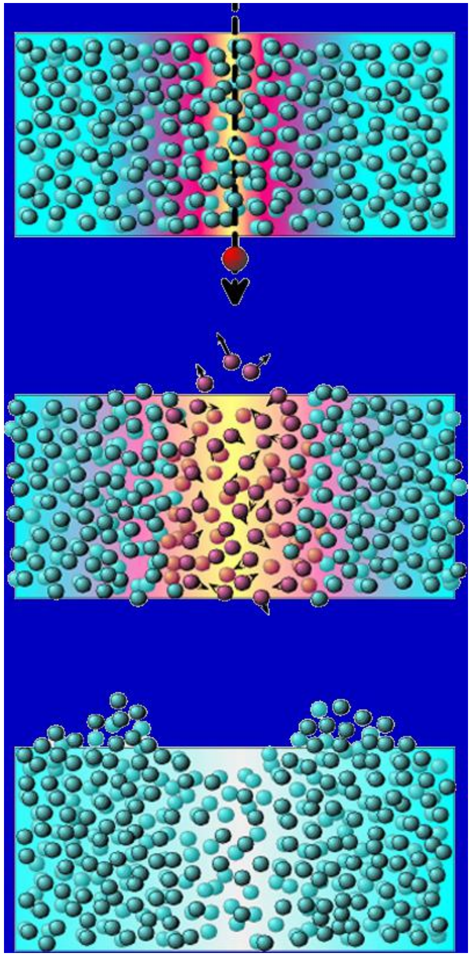
polyethylene heating to 120-200ºC they reduce their inside diameter up to half.

irradiation with high-energy ions

breaking of chemical bonds, formation of free radicals, amorphisation

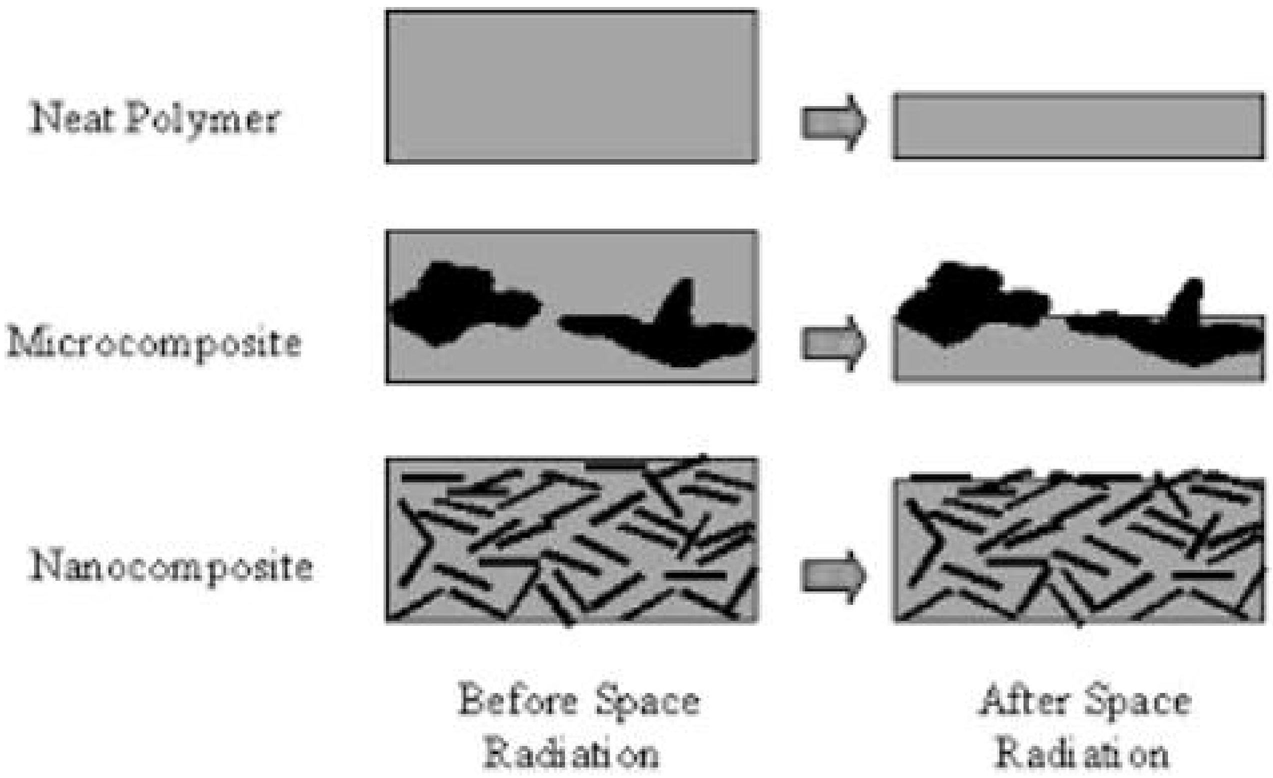


pore size: 0.05 –15 µm



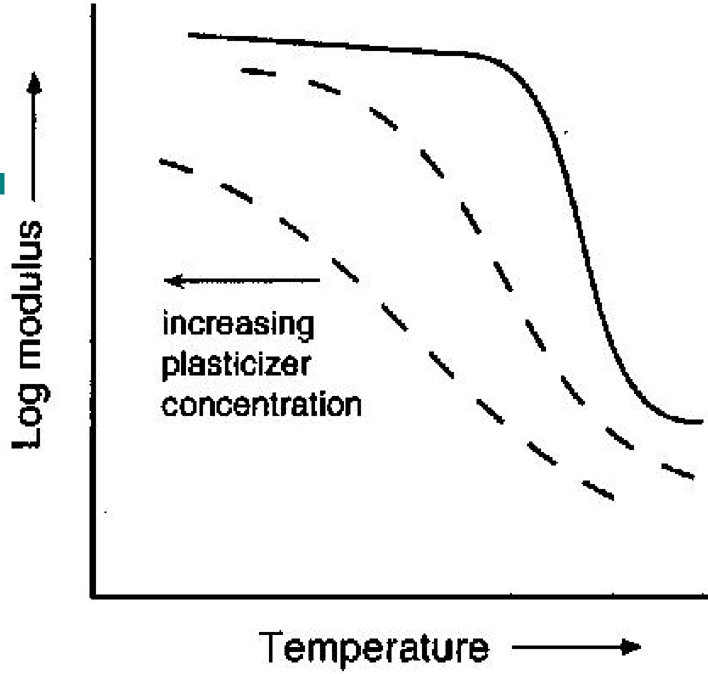
ION TRACK MEMBRANES

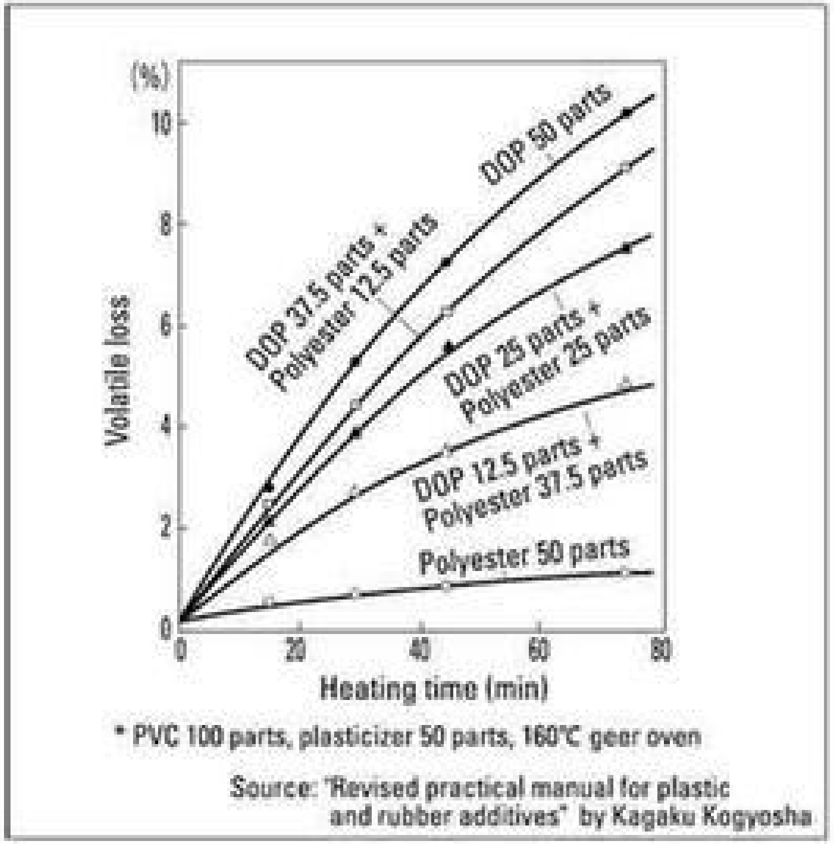
Protect polymeric materials against aggressive space environment as nanocomposites



Schematic drawing of changes to neat polymer, microcomposite and nanocomposite produced by space radiation

Loss of Plasticizer





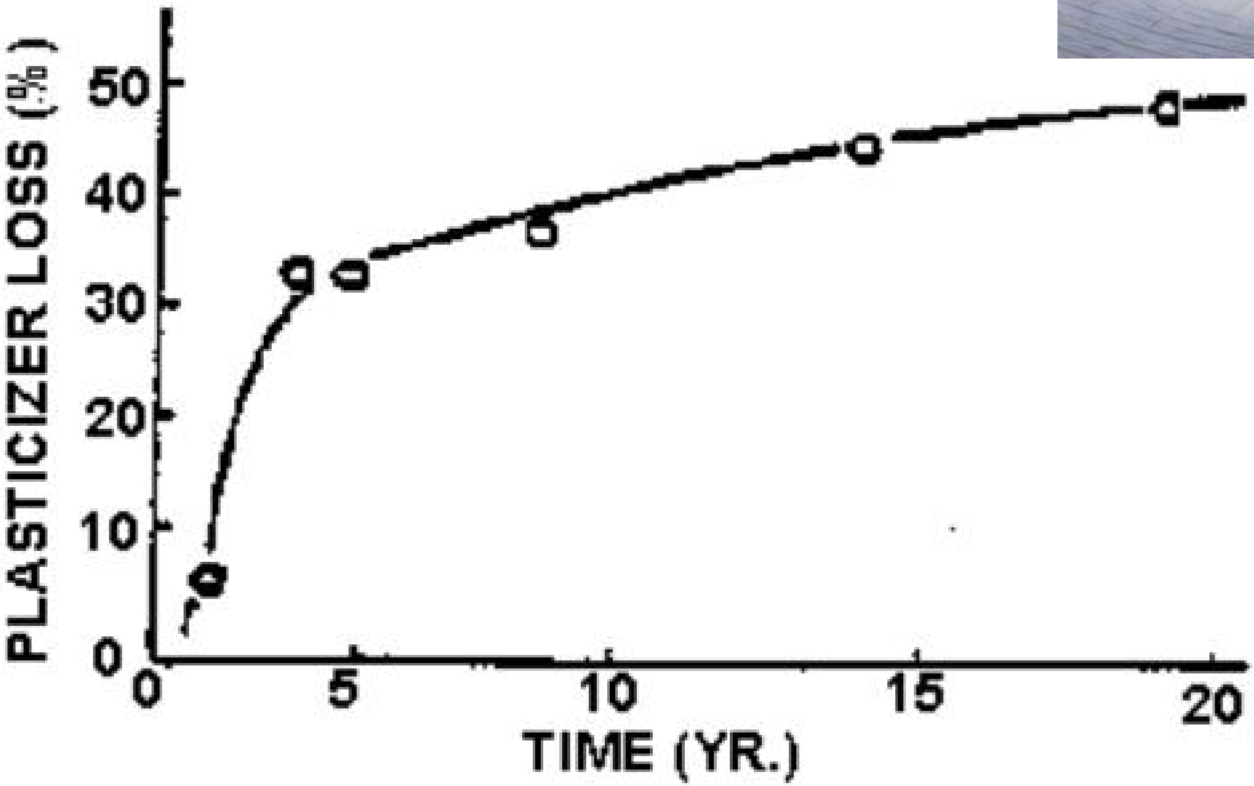
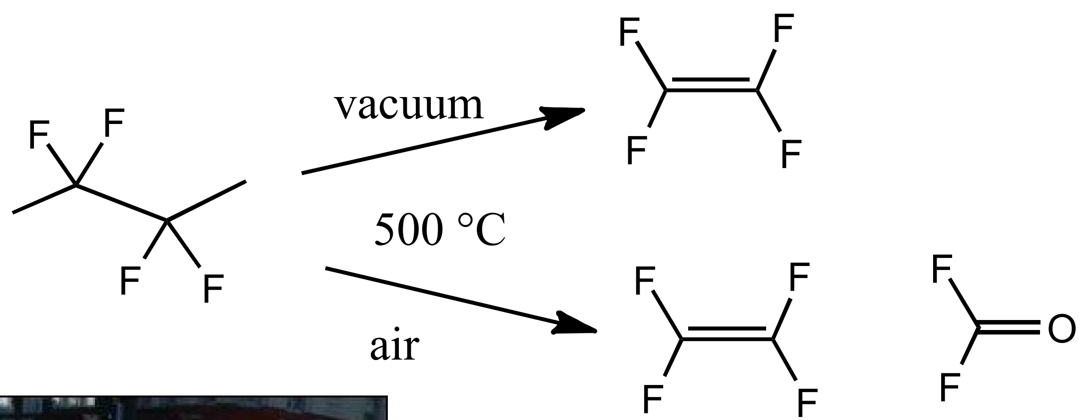
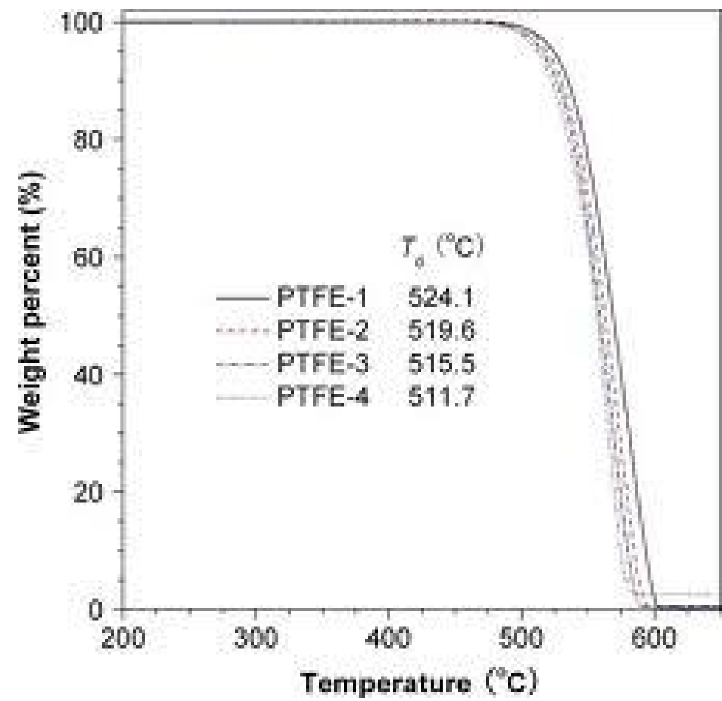


Figure 4. Loss of plasticizer from 10 mil PVC canal liner

Heating Teflon

Don’t heat above 260 ° C (500 F, actually 375 ° C or 700 F).





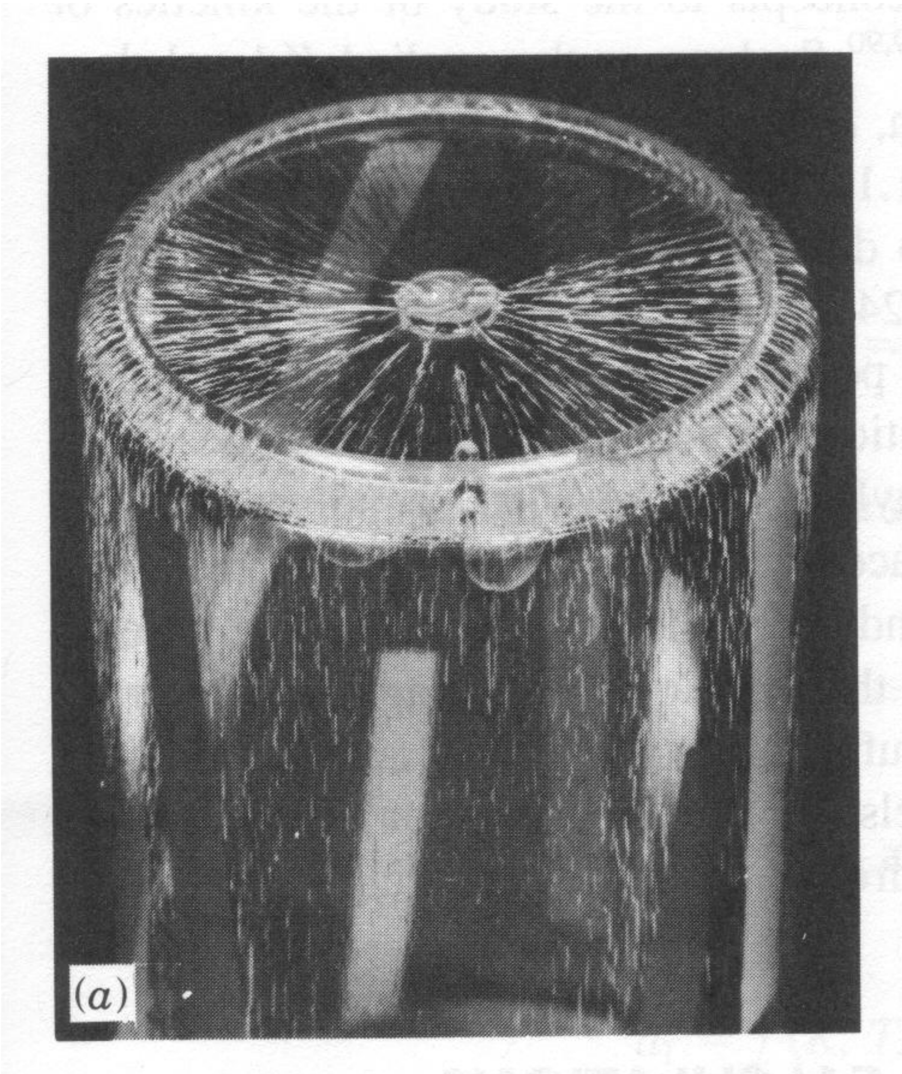
Environment Assisted Cracking

•EAC caused by
- –Stress
- –Environment
- –Susceptible material

•**Examples:**

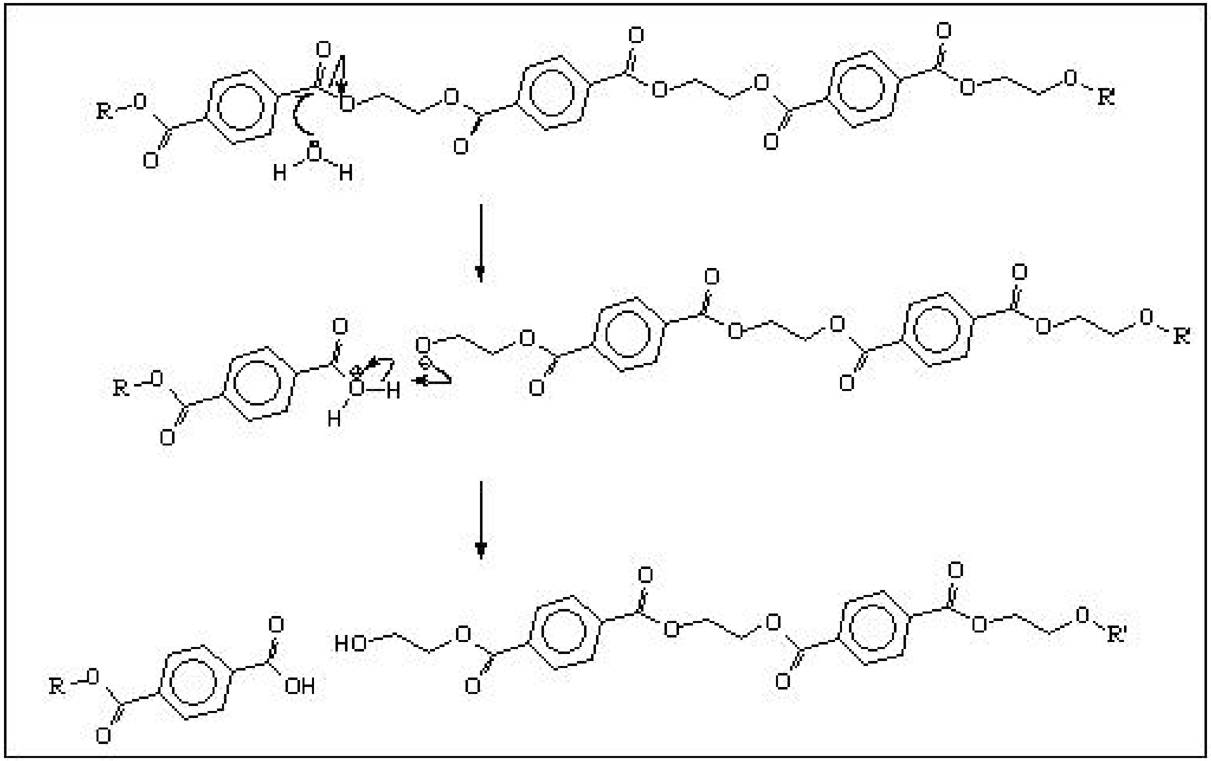
•Acrylic exposed to solvent

•PVC and PE water and gas pipelines (water and detergents at high temperature are the degrading medium)



Hydrolytic Degradation of Polymers

Reduction in chain length



Linear polyesters, polyamides (Nylons), polyurethanes, polyureas & polycarbonates, Catalyzed by acid or base

Hydrolysis during processing due to residual water

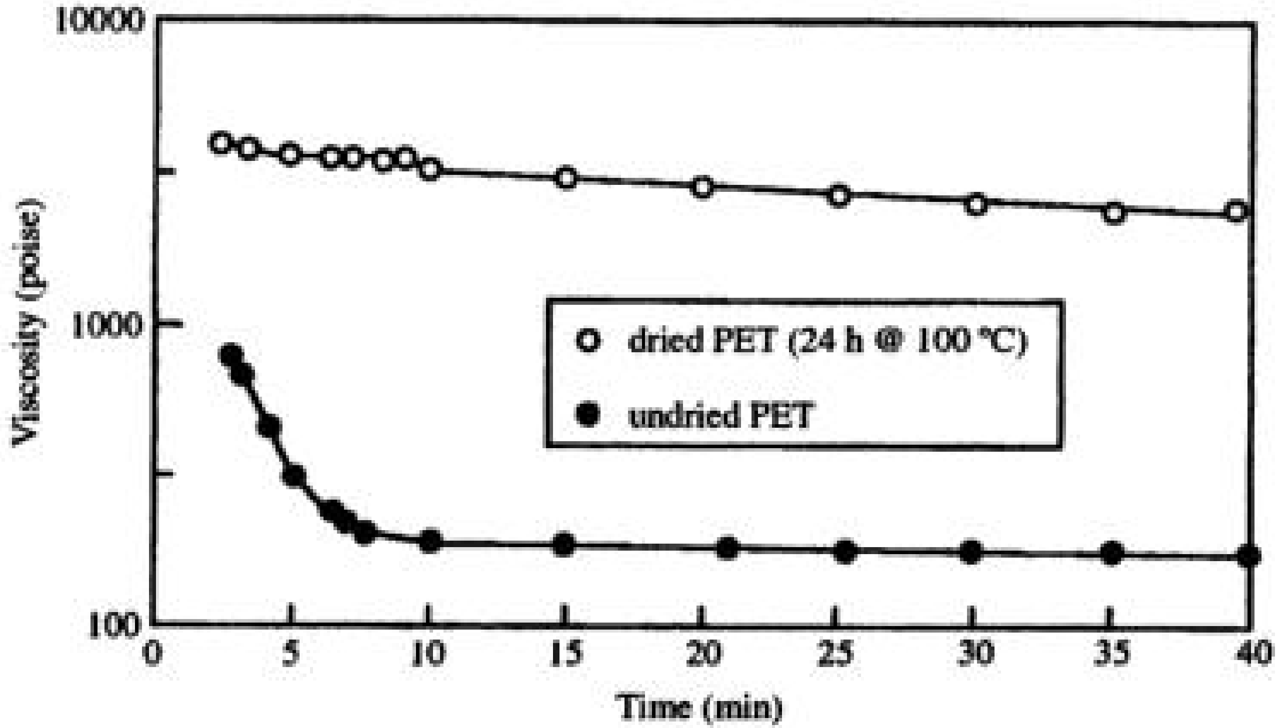
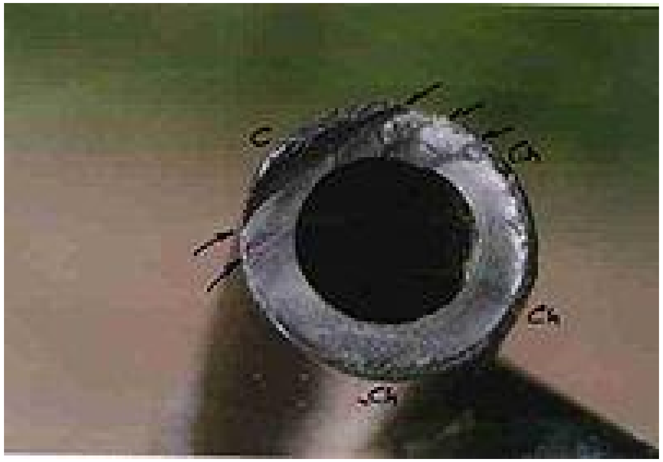


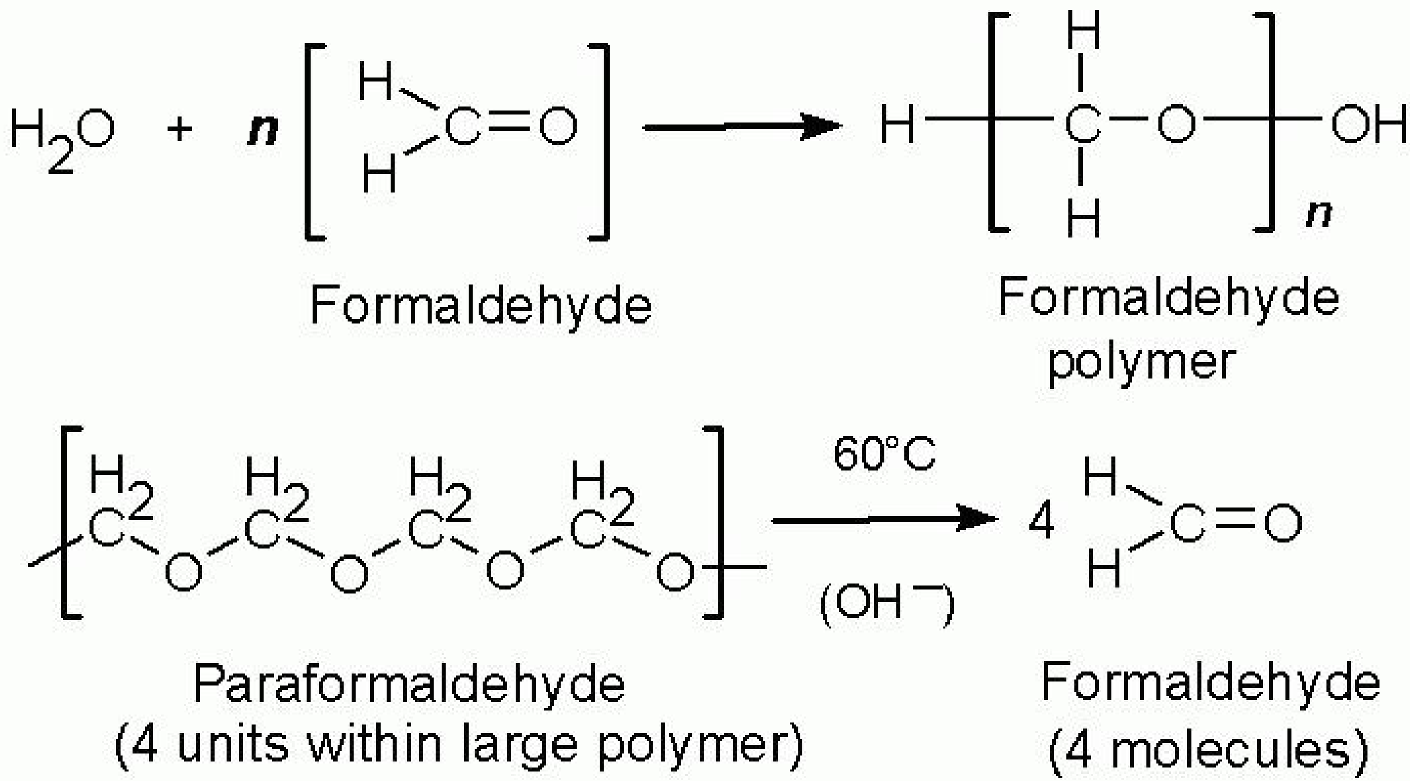
Figure 13.2 Melt viscosity of PET at 285 °C showing the effect of drying the resin before processing. The sample dried for 24 h at 100 °C under vacuum shows good melt stability as indicated by little change in its viscosity over time. In contrast, the undried PET exhibits a considerable drop in MW due to main-chain hydrolysis. (Adapted from Seo, K. S., Cloyd, J. D. *J. Appl. Polym. Sci.* (1991) 42, p. 845)

Nylon Hydrolysis

One drop of sulfuric acid from battery caused Nylon fuel line to fail



Polyacetal


H
H
C=O
H2O + n
H
C
O
OH
n
Formaldehyde
Formaldehyde
polymer
H2 H2 H2 H2
C C C C
O O O O
60°C
(OH−)
4
Paraformaldehyde
(4 units within large polymer)
Formaldehyde
(4 molecules)

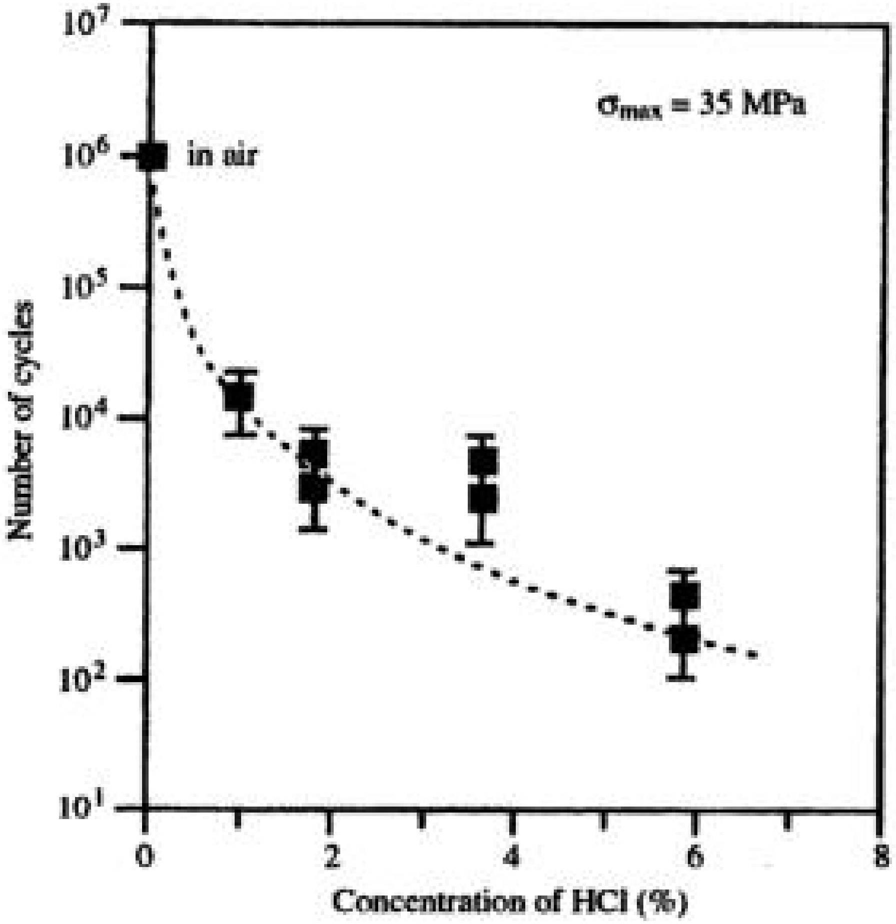
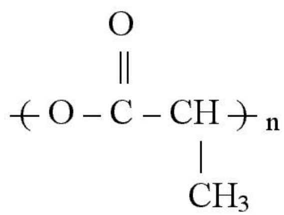
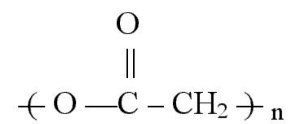


Figure 13.10 Effect of fatigue loading on crack formation in polyacetal in the presence of acid. The plot shows the number of cycles necessary to cause crack initiation in polyacetal as a function of HCl concentration. (Adapted from Chudnovsky, A., Zhou, Z. W., Stivala, S. S. and Patel, S. H., *Proc. ANTEC'94*, 3284 (1994))

HYDROLYZABLE POLYMERS: INTENTIONAL DEGRADATION



Poly(lactic acid)



Poly(glycolic)acid

Degradation of polyethylene glycol block polymer

Day zero Days 10 Days 30

- Trimethylene carbonate polymers
- Polyethylene glycol block polymers
- Dioxanone polymers
- Lactone polymers
- Lactic acid polymers
- Glycolic acid polymers

Polyhydroxybutryate

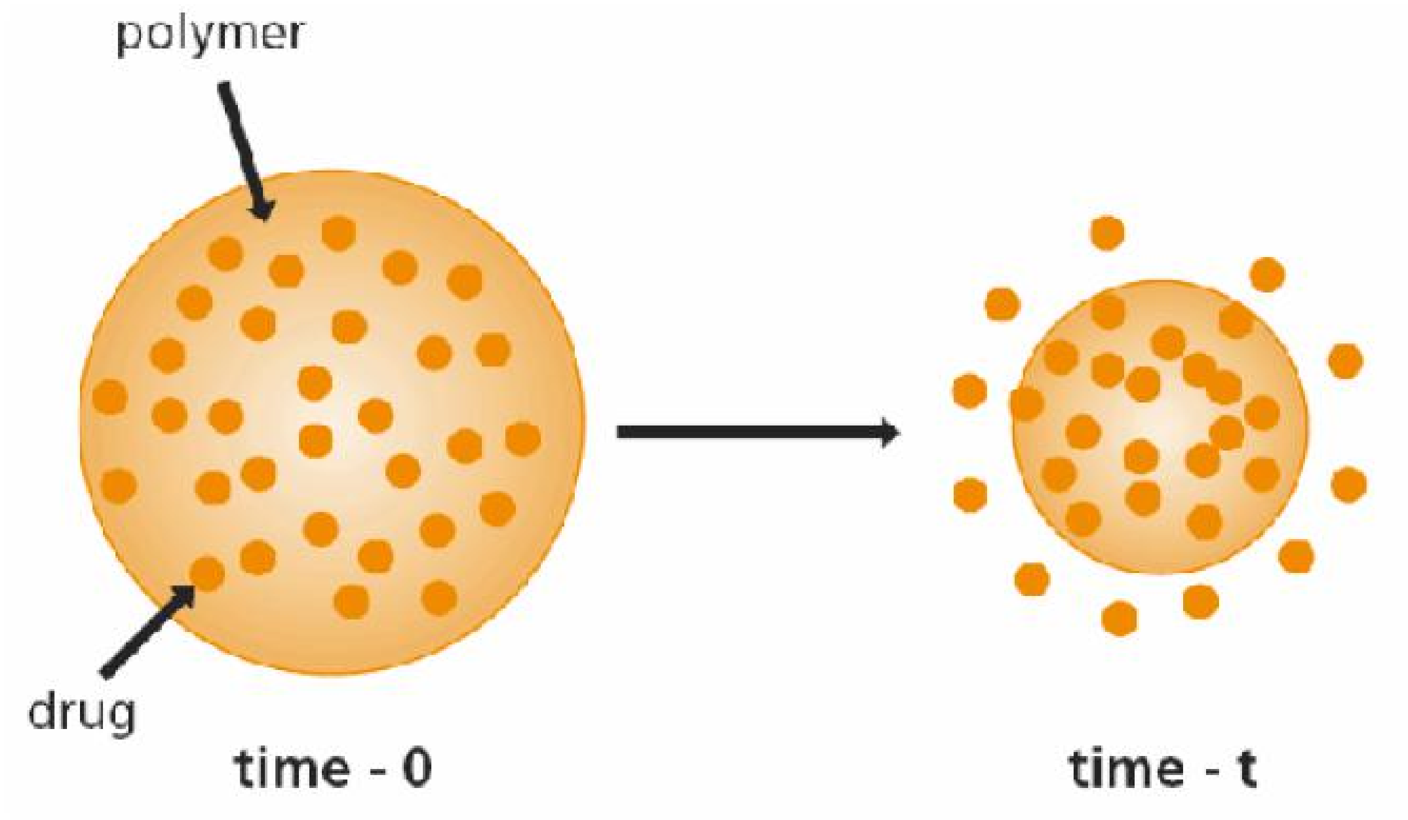
Polyesters

Polymer	Melting Point (°C)	Glass-Transition Temp (°C)	Modulus (Gpa)[a]	Degradation Time (months)[b]
PGA	225—230	35—40	7.0	6 to 12
LPLA	173—178	60—65	2.7	>24
DLPLA	Amorphous	55—60	1.9	12 to 16
PCL	58—63	(—65)—(—60)	0.4	>24
PDO	N/A	(—10)— 0	1.5	6 to 12
PGA-TMC	N/A	N/A	2.4	6 to 12
85/15 DLPLG	Amorphous	50—55	2.0	5 to 6
75/25 DLPLG	Amorphous	50—55	2.0	4 to 5
65/35 DLPLG	Amorphous	45—50	2.0	3 to 4
50/50 DLPLG	Amorphous	45—50	2.0	1 to 2

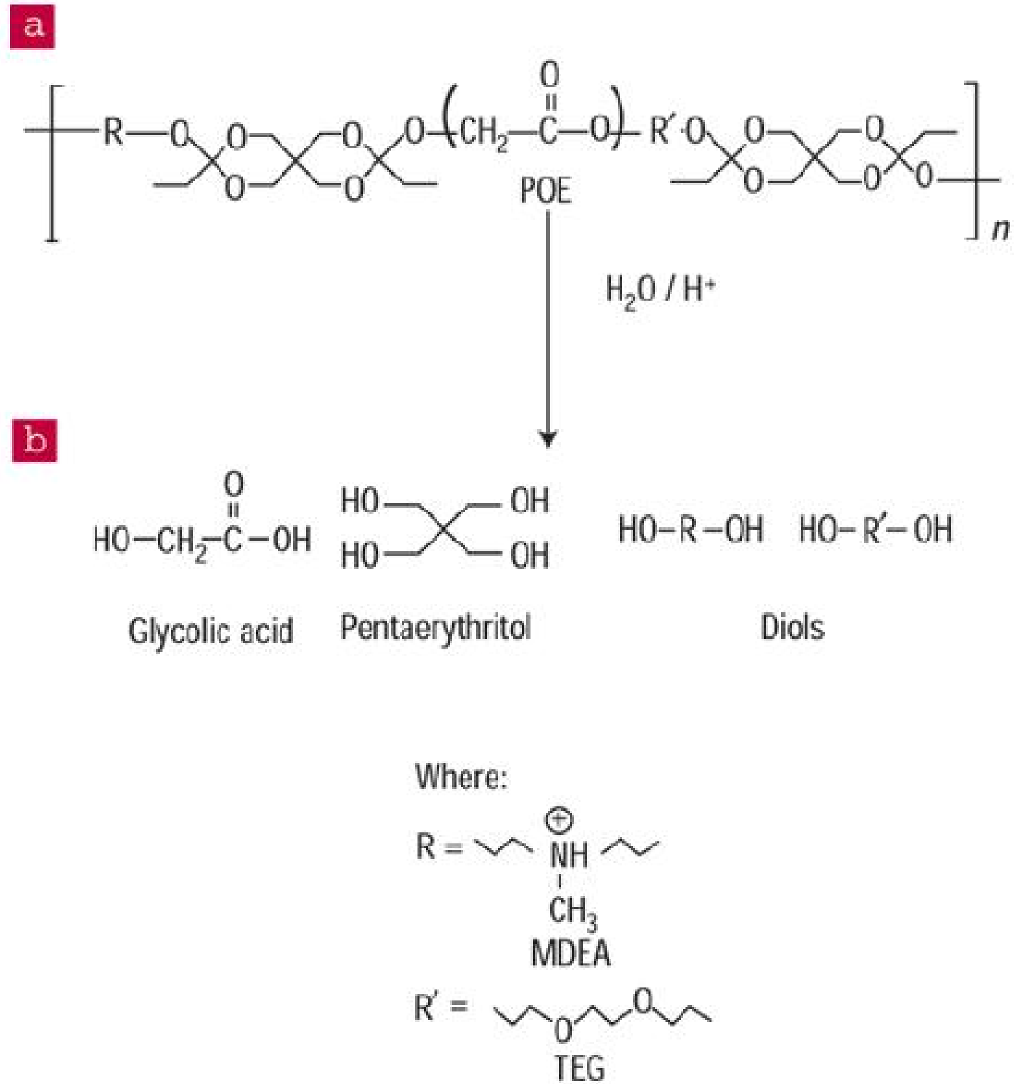
a Tensile or flexural modulus.

b Time to complete mass loss. Rate also depends on part geometry.

Drug Delivery Systems



DRUG DELIVERY SYSTEMS



Mechanical Degradation of Polymers

- High shear during extrusion (with heat)
- Grinding polymers
- Shear in solutions
- Ultrasonic
  - Break carbon-carbon bonds (radicals form)
  - Use to make graft polymers

HOMEWORK

- Well, no written HW this week.
- Read through all the slides including those I have not been able to cover in class.
- Mid-term next Thursday